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Renormalization Group

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Contents

1 What is the RG

The renormalization group is a general tool for understanding the behaviour of complex, spatially extended systems with many degrees of freedom, like for example, lattice models of classical spins, field theories, quantum many body systems (eg interacting particles etc).

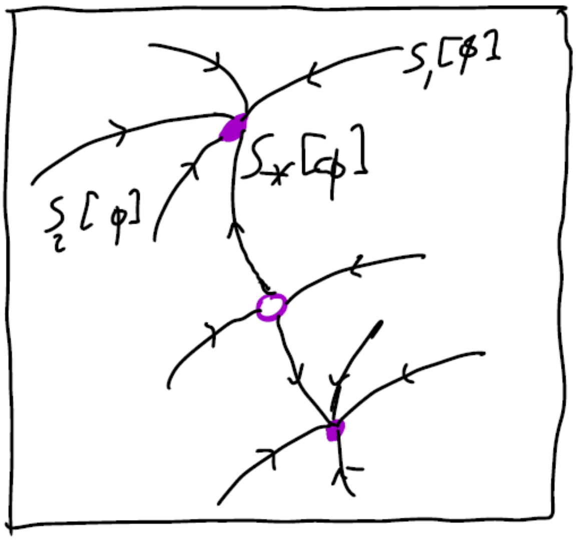
The RG group operates at a more abstract level than the physical theories we have met so far. It is not to be viewed as a set of formulas that lead to a single result, but rather a general framework of thinking about theories and the description of the system, at different length scales. For example, two systems that are microscopically different, may be governed by the same equations when viewed from larger scales. The key idea is that the appropriate and useful theoretical description of the system is different at different length scales. This means that "zooming out", an idea to be made precise through the RG, allows us to explore the universal ingredients of a microscopic model. A simple example regards classical liquids which may be different at the small scales being made up from different microscopic constituents, but at the larger scales, they can be understood by the universal theory provided by the Navier-Stokes equation, through macroscopic quantities like the density and the viscosity of the fluid.

To qualitatively understand the renormalization group we imagine an abstract construct that we refer to as a *theory space*, which contains a QFT described by some action,

$$S[\phi] = \int d^d x \ (\nabla\phi)^2 + \sum_i \lambda_i \phi^2 + \dots,$$

where the different couplings $\{\lambda_i\}$ correspond to different points in the theory space. For example the usual ϕ^4 and ϕ^3 theories correspond to two such points in the infinite dimensional theory space, while lattice models with different couplings correspond to different ones. We can draw a cartoon picture that looks like the following one,

We now want to make sense of the idea that starting from a microscopic model, when we zoom out and look at the model from larger scales, we are sitting at a different model on the theory space. This is referred as *flowing in theory space through coarse graining.* At some point, generically, the flow stops because we hit an RG fixed point, which means that there is no further change under zooming out. The same fixed point can be reached by coarse graining transformations if we started from a different microscopic model (eg a theory with more than one field).



Fixed point theories are special because they are invariant under coarse graining and they are *attractors.* The fact that any model which starts within the basin of attraction of this fixed point ends up at the same fixed point is the idea of *universality.* However, if we make the fields or coupling sufficiently different, we may find that the flow ends up at a different fixed point, which has its own basin of attraction.

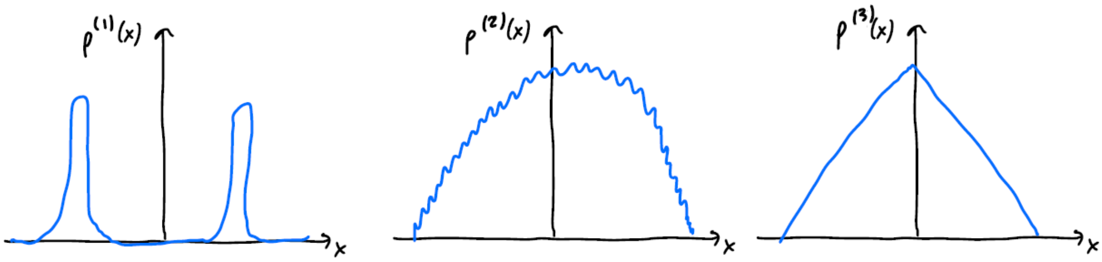
From a condensed matter physics point of view, we are interested in phase transitions. In the RG language, one reason for having different fixed points is having different phases, for example ordered vs disorder, solid vs liquid. The presence of different phases on the theory space means that there are also phase transitions, which correspond also to fixed points in the theory space. The difference here is that such fixed points are *unstable* and have lines flowing both into and out from the phase.

1.1 Toy Example for motivation

We take a large number N of independent and identically distributed random variables, $x_1, x_2, \ldots, x_N$. Those are the microscopic degrees of freedom, and we may imagine that x_i corresponds to the magnetisation of the ith grain of magnet. The microscopic model is defined by the probability distribution of the x_i, $P(x)$. For simplicity we assume that $< x >= 0$. We now define a coarse-grained variable which contains less information and is given by,

$$X = \frac{1}{N} \sum_{i=1}^{N} x_i,$$

corresponding to the magnetisation averaged over a region of volume proportional to N.



The central limit theorem states that as $N \to \infty$, $P_N(x) \to$ Gaussian, under the weak assumption that the initial random variable has finite variance. In this toy model, the theory space is infinite, corresponding to the different probability distributions that correspond to the picture described above. However, the Gaussian distribution is a fixed point, since as we average over larger and larger distances, the distribution flows to the Gaussian. Note that is only in the limit of $N \to \infty$ that we reach the fixed point. This fact can be proved by studying the cumulants of X and provides a simple example of universality.

However, in other parts of theory space we may find other fixed points. This can happen for example for sufficiently broad initial probability distributions, $P(x)$, with non-finite variance, $\int p(x)x^2 dx = \infty$ (e.g. Levy's stable distributions). Introducing correlations or interactions between the x_i will also make the distribution flow to a different fixed point. These provide usual examples of statistical mechanics and field theories in the continuum limit.

2 The Ising and O(N) models

The $O(N)$ model is a historically important family of models that is rich enough to display most of the key features of RG flows. Recall that in classical statistical mechanics ($\hbar = 0$), the fundamental object is the partition function,

$$Z = \int_{\text{configs}} e^{-\beta E}$$

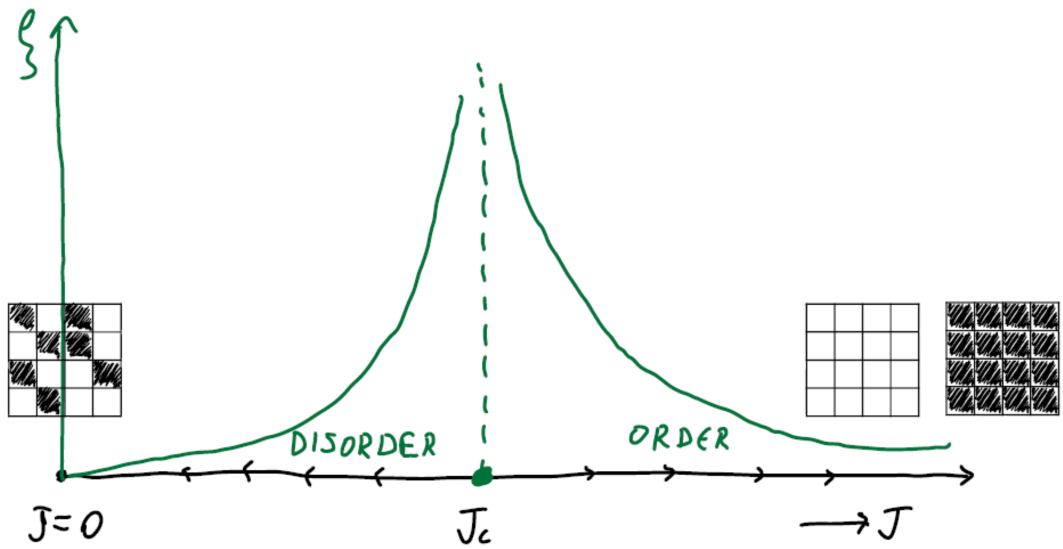
where E is the energy of the configuration and $\beta = 1/k_B T$. The degrees of freedom are classical *spins* which are N-component unit vectors, $\vec{s} = (s^1, \ldots, s^N)$. $O(N)$ models exist on hypercubic lattices where each spin is placed on each site and nearest neighbours interact ferromagnetically, so that,

$$e^{-\beta E} = e^{J \sum_{<ij>} \vec{s}_i \cdot \vec{s}_j}.$$

As can be seen, the Boltzmann weight is largest when the spins are aligned, and thus the lowest energy configuration is the one with all the spins are aligned (ferromagnetically). Generally, as is the case for classical systems, there is competition between the energy and the entropy at all T. Also, note that we have absorbed the temperature dependence into the coupling J.

2.1 A first look at the Ising model

We first start with a rather qualitative discussion on the Ising model in $d = 2$ dimensions, which is a very simple model with a phase transition between order (low T, thus large J) and disorder (large T, low J). A flow diagram for the Ising model is shown below.



At $J \to \infty$, the spins are so strongly interacting which essentially means that they are all aligned with each other. There are two configurations that satisfy this, either all spins point up or down, and each configuration occurs with probability $p = 1/2$. On the other side, when $J \to 0$, the spins are equally likely to be pointing up or down, and the phase is disordered. There is a point in between at some critical value J_c where there is a crossover from order to disorder.

Note that one-point functions, $\langle s_i \rangle = 0$ for all i by symmetry, when there is no symmetry breaking term (like an external magnetic field). To see this, we note that $E(\{s\}) = E(\{-s\})$ and

$$\langle s_i \rangle = \frac{1}{Z} \sum_{\{s\}} s_i e^{-\beta E(\{s\})} = \frac{1}{Z} \sum_{\{-s\}} (-s_i) e^{-\beta E(\{s\})} = - \langle s_i \rangle = 0$$

We can look however at the correlation between two spins. For $J < J_c$ we have,

$$\langle s_i s_j \rangle \sim e^{-r/\xi},$$

for $r \gg 1$, the lattice spacing, and ξ a characteristic length. Note that in general there is no reason to assume that ξ is constant but rather a $\xi(\theta)$ dependent function. This dependence in fact becomes weak close to J_c. Note also the exponential decay - this is a landmark of weakly interacting degrees of freedom, and it means that we can understand its functional form by perturbing around the limit of no interactions, $J = 0$ (we will do that later).

For $J > J_c$, two spins are more likely to be pointing in the same direction than in a different one, even when they are asymptotically very far away. Thus,

$$\lim_{r \to \infty} \langle s_i s_j \rangle = M^2,$$

where M is the spontaneous magnetisation density and $M > 0$.

Note. What is large in a finite system of L sites? We define an average spin for the system as,

$$\bar{S} = \frac{1}{L^2} \sum_i s_i$$

When the system size becomes large, the average $\bar{S} \approx \pm M$, with equal probabilities for $+$ and $-$. This information is shared by the whole system, and fluctuations around the ordered phase are weakly correlated (in the sense of the exponential decay).

So more properly, the two-point function for $J > J_c$ takes the form,

$$\langle s_i s_j \rangle = M^2 + A e^{-r/\xi}$$

As we approach J_c, the correlation length ξ diverges in a power-law fashion that we will understand later. At $J = J_c$ we cannot perturb around any of the two extreme limits, and the two-point function takes the form,

$$\langle s_i s_j \rangle \sim \frac{1}{r^{2x}},$$

with $x = 1/8$. We will understand this with the machinery of the renormalisation group.

The alternative way of thinking about the distinction between the two phases is more directly related to the ideas of RG. We first consider an infinite sized system and successively coarse-grain by using the “majority rule” to replace a big square of spins (which are up or down) with a single square that points to the direction of the majority of the spins. This requires the approximation that under coarse-graining we reach the same Ising model with different coupling values. We first note that the perfectly ordered state at $J = \infty$ is invariant under this transformation. The $J = 0$ configurations are also invariant, or more properly, their probability distributions are invariant under the transformation, since they are essentially non-correlated random variables.

Away from the extreme values of J (the fixed points), the state is not invariant under coarse-graining, since there are spins pointing in both directions. In the disorder phase, the absence of long-range order means that when we average over a sufficiently large region (larger than the correlation length), these are uncorrelated. Under the coarse graining transformation the characteristic length scale ξ decreases and thus J decreases. What happens eventually is that the correlation length becomes much smaller than the lattice spacing, which means that the spins are fully uncorrelated and we have reached the $J = 0$ point.

If we start exactly at $J = J_c$ neither of the above will happen and we will never reach any of the probability distributions at the trivial fixed points. What happens is that we reach a non-trivial and scale invariant probability distribution where the spins have non-zero correlations and unlike the $J = \infty$ case they do not have perfect correlations.

2.2 High T Expansion: Making correlations geometrical

There is a clever geometrical way of obtaining corrections to the partition function at all orders by expanding around the extreme phases. At very high T we are dealing with independent spins, and note that $s_i s_j = \pm 1$ for all i, j, which allows us to write the identity,

$$\begin{aligned} e^{J s_i s_j} &= \cosh J \left[1 + \tanh J s_i s_j\right] \\ e^{\pm J} &= \cosh J (1 \pm \tanh J) \end{aligned}$$

Writing, $x = \tanh J$ and using the identity above, we can write the partition function as,

$$Z = (\cosh J)^{N_b} \sum_{\{s\}} \prod_{\langle ij \rangle} (1 + x s_i s_j),$$

where N_b is the total number of bonds, and including this prefactor has the effect of changing the free energy by a constant value. We will now expand the product diagrammatically by denoting the $x s_i s_j$ term by a solid line, and the 1 term by a dotted line. Therefore the partition function takes the form,

$$Z = (\cosh J)^{N_b} \sum_{\text{graphs } G} \sum_{\{s\}} \prod_{\langle ij \rangle \in G} (x s_i s_j).$$

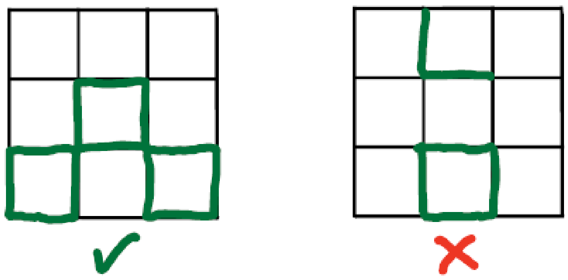
For example, the following graph corresponds to a the term,

$$Z = (\cosh J)^4 \sum_{s_i, s_j, s_k, s_l} (1 + x s_i s_j)(1 + x s_j s_k)(1 + x s_k s_l)(1 + x s_l s_i)$$

Note now that summing over a single spin, gives us only a power of s_i,

$$\sum_{s_i = \pm 1} (s_i)^P = \begin{cases} 2 & \text{if P is even} \\ 0 & \text{if P is odd} \end{cases}$$

Therefore, performing the whole summing will give 0 unless every site is adjacent to an even number of solid bonds. We call the graphs that obey this *closed graphs*. Some examples and non-examples of closed graphs are the following,



Going back to the original partition function, we find that,

$$Z = (\cosh J)^{N_b} 2^{N_{\text{spins}}} \sum_{\text{G closed}} x^{\text{links in } G}$$

2.2.1 Free energy

We can now write a power series expansion for the free energy, $F = -\frac{1}{\beta}\ln Z$, which we call the high temperature expanson since $x = \tanh J = \tanh \beta\tilde{J}$ is small when T is large. Note also that this works in any number of dimensions, on any lattice.

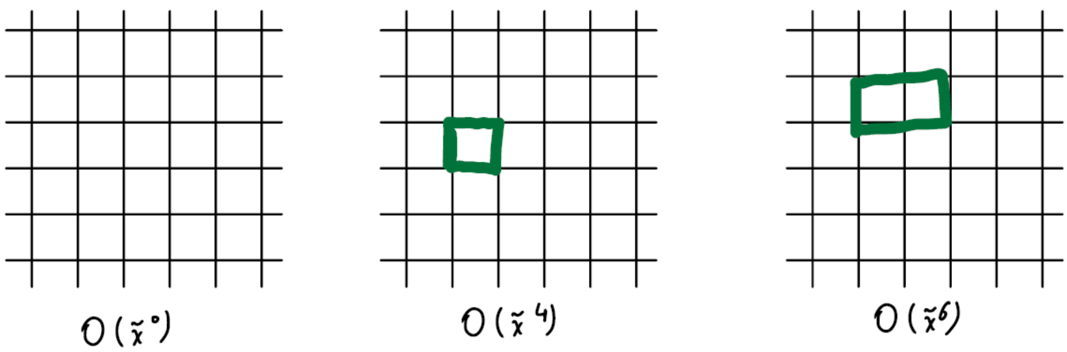
In $d = 2$ with PBC we have $N = N_{\text{spins}}$ and $N_b = 2N$. Thus,

$$-\beta F = \ln Z = N \ln\left(2\cosh^2 J\right) + \ln \sum_{\text{G closed}} x^{|G|}$$

Thus,

$$\sum_{\text{G closed}} x^{|G|} = 1 + Nx^4 + 2Nx^6 + \dots$$

which is diagrammatically represented by the following diagrams,



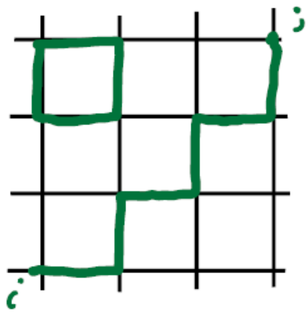
The prefactors are justified because there are N ways to place the square in the first case, but $2N$ ways to put the 6-vertex rectangle, since we can also rotate it by 90-degrees. Therefore, to this order, the free energy takes the form,

$$-\beta F = N\left(\ln\left(2\cosh^2 J\right) + x^4 + 2x^6 + O(x^8)\right).$$

Notice that these first terms are proportional to N which is expected since the free energy is extensive. It is now obvious that all higher order terms contribute only with terms proportional to N, however it turns out that all higher order in N terms like N^2 cancel when we take the logarithm. (**Prove this!**)

2.2.2 Correlation functions

For correlation functions, we insert extra factors of s_i in the evaluation of the sums and find that the graphs that contribute are those with an odd number of adjacent bonds at site i. An example of an allowed graph is the following,



The two-point function is then given by,

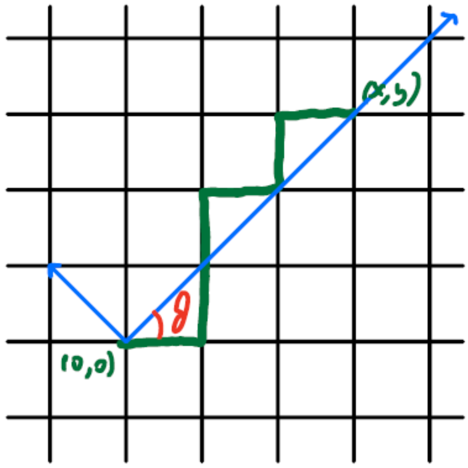
$$\langle s_i s_j \rangle = \frac{1}{Z}(2\cosh^2 J)^N \sum_{\substack{\text{G closed} \\ \text{except at i,j}}} x^{|G|}$$

Let us consider only the simple limit $x \to 0$, and therefore the sum in the expression above is dominated by those graphs with the smallest number of occupied bonds. Labeling each spin by its (x, y) coordinates, the shortest path that we can draw is a directed path from $(0,0) \to (x,y)$, without any disconnected clusters to contribute with extra factors of $\tilde{x}$[1]. The correlation function is then,

$$\langle s(0,0)s(x,y)\rangle = \sum_{\substack{\text{directed paths} \\ (0,0)\to(x,y)}} \tilde{x}^{|x|+|y|}$$

For fixed x, y, the weight of each path is the same, so it remains to count the number of paths, which is the same as the number of ways of choosing right and up steps from the $x + y$ total steps. This is given by $\binom{x+y}{x}$, and thus,

$$\langle s(0,0)s(x,y)\rangle = \frac{(x+y)!}{x!y!}\tilde{x}^{|x|+|y|}$$



In a classical statistical mechanics language, we find that the number of configurations counted above is given by an entropic factor e^S whereas the energy of the path in this effective partition function is given by $e^{-\beta E}$.

Exercise 1. *Use Sterling's approximation, $N! = exp(N \ln N - N + \frac{1}{2}\ln(2\pi N) + O(1/N)$, to show that the correlation function takes the form,*

$$\langle s(0,0)s(x,y)\rangle = \frac{A(\theta)e^{r/\xi(\theta)}}{\sqrt{r}}$$

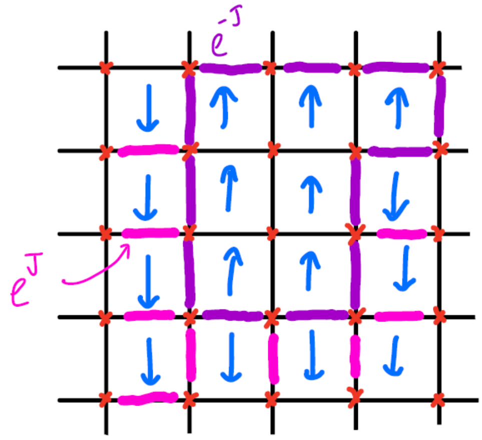
where $r = \sqrt{x^2+y^2}$, $A(\theta) = \sqrt{\frac{|\cos\theta|+|\sin\theta|}{2\pi|\cos\theta \; \sin\theta|}}$ and $\xi(\theta)^{-1} = (\cos\theta+\sin\theta)\ln\frac{1}{\tilde{x}} + \cos\theta\ln\cos\theta + \sin\theta\ln\sin\theta - (\cos\theta+\sin\theta)\ln(\cos\theta+\sin\theta)$. This shows that correlations decay faster on the diagonal versus along the axis, which makes sense since the paths that connect points along a diagonal are simply bigger. The $1/\sqrt{r}$ is related to the probability of a 1d random walker to return to the origin. To see this, we can think of one of the directions (eg the diagonal that connects $(0,0)\to(x,y)$ as time, and the normal to it as space, this is precisely the path of a 1d random walker.

Despite the crude approximations, we get the characteristic functional form of the disordered phase.

2.3 Low Temperature Expansion

We now check the low temperature expansion of the 2d Ising model, through which we will see that the model is in fact self-dual. We will understand this by thinking about domain walls. We first call the lattice, the one on which the spins live, and the dual lattice is defined to be the one formed by the centres of the plaquettes of the lattice. Given a configuration with up and down spins, a domain wall is defined to be the graph on the lattice that separates spins of different orientation, as in the figure below. A spin configuration thus corresponds to a unique graph on the dual lattice, as is explained in the figure below,

[1]Here $\tilde{x} = \tanh J$ from before, since we are using x to label the horizontal coordinate.



The weight of a configuration is given by the total length of the domain walls: if a bond doesn't have a domain wall on it, its weight is e^J and if it does have a domain wall on it, its weight is e^{-J}. Therefore, the partition function takes the form,

$$Z = 2(e^J)^{N_{\text{bonds}}} \sum_{\substack{\text{G closed on} \\ \text{dual lattice}}} (e^{-2J})^{\text{links in } |G|}$$

where the factor of 2 that comes from the fact that every configuration of domain walls corresponds to two spin configurations related by a global spin-flip, $S \to -S$.

Note. Remarkably we see that this partition function has the same form as the high-temperature expansion, a fact known as the *Kramers-Wannier duality.* with a different weight for each bond. In the high temperature expansion,

$$\tilde{x} = \tanh J,$$

while in the low temperature expansion,

$$\tilde{x}' = e^{-2J}$$

This allows us to write a formula for the exact relationship for the free energies in the high and low temperature phases, and obtain the exact value of the location of the critical point. This is one of the few exceptions that an exact value can be obtained for the critical point. Assuming that there is a single phase transition from order to disorder, which is of course the case here, the transition must happen at the only self-dual point which is given by $J = J'$, a little bit of algebra shows that the critical point is given by,

$$\sinh(2J)^2 = 1 \iff J_c = \frac{1}{2}\log\left(\sqrt{2}+1\right)$$

2.4 Scale Invariance

The crucial feature of the Ising model at $J = J_c$ is scale invariance: under coarse-graining, the theory approaches a fixed point that is invariant. We have seen this by using the block-spin transformation.

We will now study a simpler example of scale invariance with the example of a 1-dimensional random walk. The walker is displaced by $\delta r = \pm 1$ each time interval $\delta t = 1$. The key result of the random walk is that the average of the square of the displace is,

$$\langle r^2 \rangle = t.$$

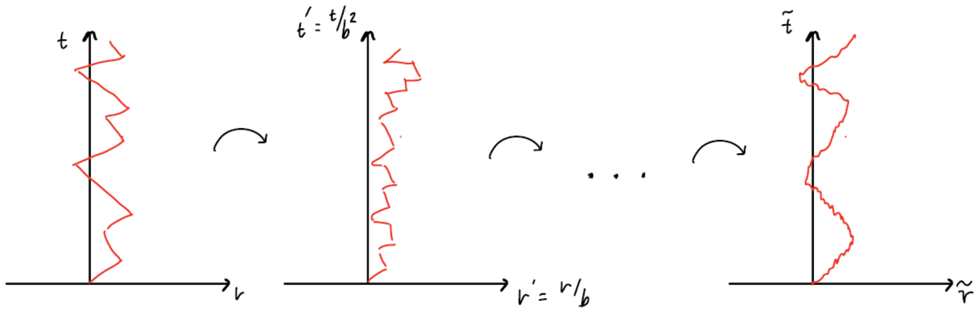
We say that $r \sim t^{\frac{1}{z}}$, where $z = 2$ is the *scaling exponent* that tells us how one variable changes as a function of another. Here z is a *dynamical exponent* since it relates the scaling of space and time.

At large scales, the problem is scale invariant under the rescaling,

$$r' = \frac{r}{b} \qquad t' = \frac{t}{b^z}.$$

This is the meaning of the dynamical exponent: it tells us how we need to change time in relation to space if we want to make a rescaling that leaves the problem unchanged. Here the scaling is *anisotropic* in t, r.

Qualitatively, what this means is that if we draw a configuration in the $r - t$ plane and make the rescaling to draw the configuration in the $r' - t'$ plane iteratively and ad infinatum, we will converge to what we call the *continuum brownian motion*. The latter **is** invariant under changing the lengths since we have reduced the step size to an infinitesimal value which is invariant under scaling.



Suppose now that we ask what is the probability that the walk passes through a small interval δr at some time t,

$$p_t(r) = \binom{t}{\frac{t+r}{2}}.$$

Taking t and r to be large, we find (using Stirling's formula) that the solution is,

$$p_t(r)\delta r = \frac{1}{2\pi t} e^{-r^2/2t} \delta r,$$

where δr here is such that $1 \ll \delta r \ll r$, so that it is much larger than the lattice spacing and we can use the continuum approximation, but it is much smaller than r. The precise meaning of scale invariance is then that under the scale transformation 2.4, we have,

$$p_t(r)\delta r = p_{t'}(r')\delta r'$$

2.5 Transfer Matrix Method

A very useful method of solving classical lattice models with short range interactions in one dimension is that of the transfer matrix. The idea is to build the partition function by choosing a point x_0 on the lattice and evolving the system successively from $x_0 \to x_1 \to x_2 \to \dots$ using the transfer matrix. As an example, we consider the 1D Ising model and assume periodic boundary conditions, so we can write the partition function as,

$$\begin{aligned} Z &= \sum_{\{s\}} e^{Js_1 s_2} e^{Js_2 s_3} \dots e^{Js_L s_1} \\ &= \sum_{\{s\}} M_{s_1,s_2} M_{s_2,s_3} \dots M_{s_L,s_1} \end{aligned}$$

where the second step follows because the partition function has the same structure as matrix multiplication with a matrix whose rows and columns are labeled by the spin index s_i that takes values ± 1. Thus, M is given by,

$$M_{s_1,s_2} = e^{Js_1 s_2} = \begin{pmatrix} e^J & e^{-J} \\ e^{-J} & e^J \end{pmatrix}_{s_1,s_2}$$

Performing the internal sums, we are left with a simpler expression,

$$Z = \sum_{s_1} (M^L)_{s_1,s_1} = \operatorname{Tr} M^L$$

when periodic boundary conditions are used. However, the transfer matrix method does not rely on PBC. We can also find correlation functions using the transfer matrix. Consider the two-point function,

$$\langle s_i s_j \rangle = \frac{1}{Z} \sum_{\{s\}} e^{Js_1 s_2} \dots e^{Js_{i-1} s_i} s_i e^{Js_i s_{i+1}} \dots e^{Js_{j-1} s_j} s_j e^{Js_j s_{j+1}}.$$

Now it is useful to insert the diagonal matrix $\Lambda = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ next to the s_i, so that the correlation function takes the form,

$$\langle s_1 s_{1+r} \rangle = \frac{\operatorname{Tr} \Lambda M^r \Lambda M^{L-r}}{\operatorname{Tr} M^L}$$

The great thing about the transfer matrix is that we can simply compute both the partition function and the correlation functions just by computing the eigenvalues of the transfer matrix M. Suppose that the eigenvalues of M are $\lambda_1 \geq \lambda_2 \geq \dots$ (for a more general transfer matrix), the partition function takes the form,

$$Z = \sum_\alpha \lambda_\alpha^L = e^{-Lf} + \dots,$$

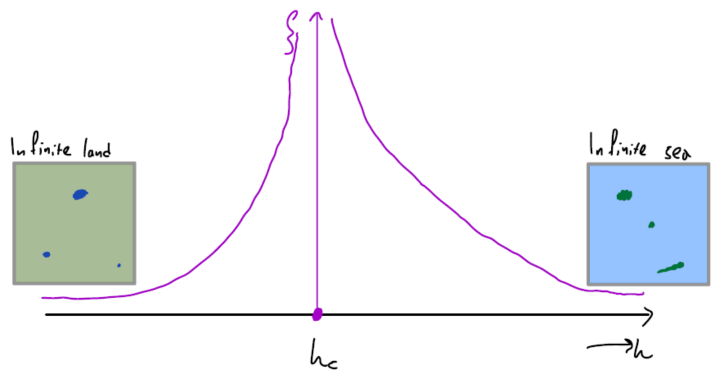
to leading order and for large L, and with $f = -\ln \lambda_1$. Similarly, it can be shown that the correlation function is $\sim e^{-\xi} = \lambda_1/\lambda_2$.

2.6 Percolation

We now look at another important example of scale invariance: percolation. Consider a infinite random landscape with hills and valleys, which is statistically homogeneous (eg every region looks the same as every other region) and which has short-range correlations in space. The characteristic lengthscale of the hills is a. Imagine now that we start to flood this landscape up to some level in the vertical direction of height h. The question we are asking is what does the landscape look like as a function of h.

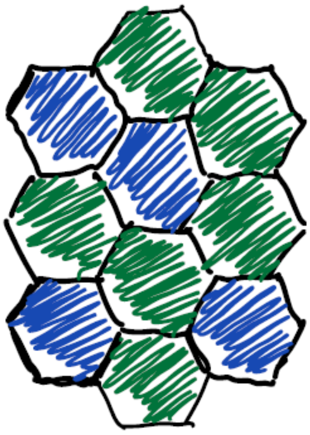
Qualitatively we know that for small h, almost all the landscape is land apart from some small lakes at the bottom of deep lakes. On the other extreme, the opposite situation happens: almost everything is water apart from small islands which are the tops of high mountains. It turns out that there is a sharp phase transition at some critical height h_c, called the *percolation threshold*, which separates the "land phase" (where land percolates) and the "sea phase" (where sea percolates). The phase transition is continuous phase transition in the sense that there is a length scale that diverges at the phase transition.

Denoting the length scale by ξ, the qualitative plot of the length scale is given by the following figure, In the land phase, ξ is giving us the typical size of the lakes so that the density of lakes of



size $\gg \xi$ is exponentially small. In the sea phase, it is the density of islands that is exponentially small. The scaling of ξ at h_c follows a power law fashion,

$$\xi \sim |h - h_c|^{-4/3}$$



We can model (eg numerically) the landscape by hexagonal tiles and generate the random landscape by the rule,

$$\text{Tile} = \begin{cases} \text{land} & \text{w/ probability } p \\ \text{sea} & \text{w/ probability } 1-p \end{cases}$$

An example of such a landscape is given in the figure to the left. The phase diagram in the parameter p has a single critical point on which the land phase transition to sea phase. There is a symmetry of land $\Longleftrightarrow$ sea, by the mapping $p \Longleftrightarrow 1-p$ and therefore the critical p is $p_c = 1/2$. We let the probability that two tiles O and R are in the same (connected) land mass be given by $G(R)$.

In the land phase, there is a finite probability that two points, even if being infinitely far-away from each other, will like at the same land mass, so we expect that,

$$G(R) \longrightarrow \text{const.} \quad \text{as } R \to \infty$$

while in the sea phase the same probability is exponentially small,

$$G(R) \sim e^{-R/\xi}.$$

At exactly the critical point, $p_c = 1/2$, the probability is again a power law,

$$G(R) \sim \frac{1}{R^{\frac{5}{24}}}$$

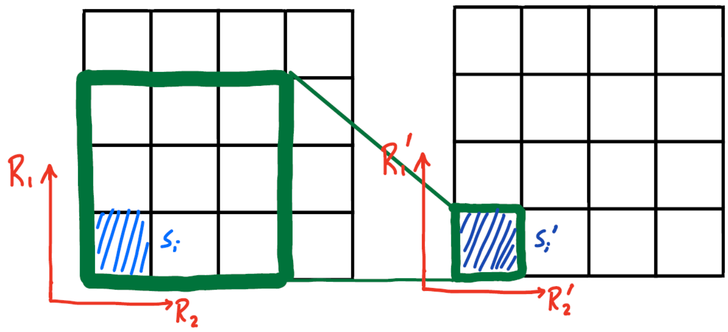
What is the phase at p_c though? It is neither land nor sea, but rather a phase with both lakes and islands nested together inside each other on all length scales. Since $\xi \to \infty$ at that point, there is no characteristic length scale for islands.

The phase at p_c is scale invariant in the sense that if we are to zoom in at some small region of the big landscape, it will look statistically the same, no matter how many times we iterate that process, so long as we haven't reached the level of the hexagons, which are the microscopic constituents of the model. At that point, scale invariance breaks down. However, as long as we are studying the problem at length scales much larger than the lattice spacing, the model is scale invariant. These clusters are called random fractals with a fractal dimension d_f that gives the scaling of the "mass", the total area within the cluster (or total number of connected lattice sites), is related to the size R of the cluster, (mass of cluster) $\sim R^{d_f}$. For the 2-dimensional percolation, $d_f = 91/48 \approx 1.9$.

Note. We have chosen to work with a lattice made of hexagons because it is such that only land or sea percolates for all values of $p \neq p_c$: it is impossible for both to percolate, in the sense of having infinite sized clusters. This is not necessarily true for other lattices. For example we could use a square lattice by saying that two tiles are connected if they share one corner (but not necessarily an edge). Therefore, it is possible for some values of p that both land and sea percolates. However, based on universality, the critical exponents will be the same independent of the two-dimensional lattice used.

3 Real Space RG *(Cardy Chapter 3)*

The key idea of real space RG is the *block spin* idea. We start with a configuration on a lattice indexed by (R_1, R_2). According to the block-spin idea, we choose a 3×3 sublattice to replace it with a single lattice site, and rescaling its size so that the lattice spacing is the same as the starting one. We thus reach a coarse-grained lattice.



Let us call the original spins by $S = \pm 1$ and the coarse-grained ones, referred to as the *block spins* by $S' = \pm 1$, and similarly for the lattice spacing. The idea is that we are discarding the short wavelength degrees of freedom, retaining the long wavelength ones. More properly, we need to define an effective Hamiltonian for the block spins. Given a Hamiltonian of the original spins $\mathcal{H}[S]$, we need a protocol that relates the two,

$$\mathcal{H}[s] \longrightarrow \mathcal{H}'[s'].$$

In other words, what $\mathcal{H}'$ do we have to choose so that the probability distribution of $\{s'\}$ is preserved? This means that if we calculate, for example, expectation values of s' using $\mathcal{H}'$, it has to coincide with the expectation values of s using $\mathcal{H}$.

Note that the length is also rescaled according to,

$$\underline{R}' = \frac{1}{b}\underline{R},$$

where b is the rescaling factor. In this case $b = 3$. This is in order to get physics of the same "resolution", since we want $\underline{R}$ and $\underline{R}'$ to be measured in lattice spacings, and it is simply a good way to do book-keeping.

Recall now the schematic behaviour of the 2D Ising model which has two trivial fixed points, and a non-trivial fixed point that separates the order and disordered phases. In the ordered phase there is a characteristic length scale over which fluctuation about the ordered state are correlated. On the other hand, in the disordered phase, there is a characteristic length scale over which the spins themselves are correlated. What we have done here is to project the behaviour of the model in the abstract theory space onto a one-dimensional subspace controlled by a single parameter, under reasonable approximations. Within the approximation, the fixed point corresponds to a fixed point Hamiltonian $\mathcal{H}_*$ which is invariant under the block spin transformation.

Before studying the 2D Ising model with real space RG, it is simpler to consider the flows in the 1D Ising model. For $J = \infty$ there is stable ordered phase, but for any other value, the model will flow to the disorderd phase fixed point $J = 0$. Writing $t_i = s_i s_{i+1}$ to represent a domain wall, we can write the partition function for the 1D Ising model as,

$$Z = \sum_{\{t\}} \prod_i e^{Jt_i}.$$

Since the t_i are non-interacting, the probability (and thus the density) for the domain wall to attain each value is simply given by,

$$p(t_i = -1) = \frac{e^{-J}}{e^J + e^{-J}} \sim e^{-2J} \text{for large } J \gg 1.$$

Thus there are connected regions of size $\xi \sim e^{2J}$, which is the correlation length. This means that looking at the system at scales much less than ξ, it looks like we are sitting at the $J = \infty$ fixed point, but on scales much larger than ξ, there are many domain walls in between, and the correlations are very weak, which means that we are sitting at the trivial fixed point. Therefore, we may think of the correlation length as a measure that tells us how much we have to coarse grain by to escape from the unstable fixed point.

Majority Rule

Consider again nine microscopic spins within a single block, $s_1, \ldots s_9$, which we want to use in order to define the single block-spin, s'. We are thus after a function that relates the two, and the simplest function to use is,

$$s' = \text{sgn} \sum_{i=1}^{9} s_i$$

Given a definition for the block spin, we can then write a *renormalised Hamiltonian*,

$$e^{-\mathcal{H}'[s']} = \sum_{\substack{s \\ (\mapsto s')}} e^{-\mathcal{H}[s]}$$

where we have absorbed the β factor into the definition of $\mathcal{H}$ and the notation for the sum means that we sum over all s configurations that are consistent with a given s' configuration. We can think of that as a partial partition function which sums over a subset of the original configurations. More explicitly,

$$\sum_{\substack{s \\ (\mapsto s')}} = \sum_s \prod_{\text{blocks}} \delta_{\text{sgn}(\sum s), s'}$$

From the definition above, it follows that the RG tranformation preserves the partition function and thus the free energy,

$$Z' = \sum_{s'} e^{-\mathcal{H}'[s']} = \sum_{s'} \sum_{\substack{s \\ (\mapsto s')}} e^{-\mathcal{H}[s]} = Z.$$

More importantly, the probability distribution of the coarse-grained degrees of freedom, $P(\{s'\})$, is also preserved,

$$
\begin{aligned}
P(\{s'\}) &= \frac{1}{Z}\sum_{s} e^{-\mathcal{H}[s]} \prod_{\underline{R}} \delta_{s'_{\underline{R}},\mathrm{sgn}(\dots)} \\
&= \frac{1}{Z'} \sum_{\substack{s \\ (\mapsto s')}} e^{-\mathcal{H}[s]} \\
&= \frac{1}{Z'} e^{-\mathcal{H}'[s']}
\end{aligned}
$$

Parametrising the Hamiltonian

To parametrise the Hamiltonian, we consider listing all the possible couplings in a vector $\underline{K} = (K_1, K_2, K_3, \dots)$, which is a point in theory space. For example, a model Hamiltonian for an Ising model plus next nearest neighbour interactions is,

$$
-\mathcal{H}[s] = K_1 \sum_{\langle ij \rangle} s_i s_j + K_2 \sum_{\langle\langle ij \rangle\rangle}
$$

In general, starting from a point $(K_1, 0, 0, \dots)$, the RG transformation that we defined above will produce more complicated couplings,

$$
\underline{K}' = \mathcal{R}(\underline{K}),
$$

which in components reads,

$$
K'_a = \mathcal{R}_a(K_1, K_2, \dots).
$$

This is what we cal the "RG transformation". It i a map that takes a point in theory space, to another point in theory space. A cartoon representation of these in the hyperplace of couplings is shown in the following figure. This map can be iterated as many times as we want, which results in finite jumps in the plane. Once we get closed to a fixed point, these jumps become very small, and they will become continuous flows being described by a differential equation (as is expected from a field-theoretical point of view).

RSRG for 1D Ising model

Even though we can solve the 1D Ising model directly, through the transfer matrix method, it is worth going through it from the RG point of view. The Hamiltonian is,

$$
-\mathcal{H}[s] = K \sum_{i} s_i s_{i+1},
$$

and we will define block spins by using a scaling factor of 3 the simple rule for defining them we will use the *decimation* rule, so that,

$$
s'_i = f(s_j, s_{j+1}, s_{j+2}) = s_{j+1},
$$

i.e. we aways take the middle spin out of the three. It turns out that this idea works only in 1-dimension. The partition function splits into two sums, made from the middle of the blocks, and all the other ones,

$$Z = \sum_{\dots,s_1',s_2',\dots} \sum_{\dots,s_1,s_3,s_4\dots} \left(e^{Ks_1s_1'} e^{Ks_1's_3} e^{Ks_3s_4} e^{Ks_4s_2'} \dots \right)$$

so that,

$$e^{Ks_1's_2'+A} = \sum_{s_3,s_4} e^{Ks_1's_3} e^{Ks_3s_4} e^{Ks_4s_2'}$$

for some constant A that we will calculate later. To perform the sum, we will use the high temperature expansion with $\mu = \tanh K$ and $e^{Ks_1s_2} = \cosh K(1 + \mu s_1 s_2)$. Thus,

$$\sum_{s_3,s_4} e^{Ks_1's_3} e^{Ks_3s_4} e^{Ks_4s_2'} = \sum_{s_3,s_4} (\cosh K)^3 (1 + \mu s_1' s_3)(1 + \mu s_3 s_4)(1 + \mu s_4 s_2')$$

From the discussion before, only the closed graphs (in s_3, s_4) are going to contribute and the only possibilities are with all the links empty (contributing with a factor of 4), or all the links occupied (contributing with a factor of $4\mu^3 s_1' s_2'$). Therefore,

$$\begin{aligned} \sum_{s_3,s_4} e^{Ks_1's_3} e^{Ks_3s_4} e^{Ks_4s_2'} &= 4(\cosh K)^3 (1 + \mu^3 s_1' s_2') \\ &\equiv 4(\cosh K)^3 (1 + \mu' s_1' s_2') \end{aligned}$$

We now define a new coupling $\mu' = \tanh K' = \mu^3$, and thus $\tanh K' = (\tanh K)^3$. We can rewrite the answer above in the form,

$$\frac{4(\cosh K)^3}{\cosh K'} \cdot (\cosh K')(1 + \mu' s_1' s_2') = \frac{4(\cosh K)^3}{\cosh K'} e^{K' s_1' s_2'} \equiv e^{A + K' s_1' s_2'}.$$

The constant A has only an effect on the total free energy but not on any correlation functions. The RG equation in the variable μ is thus,

$$\boxed{\mu' = \mathcal{R}(\mu) = \mu^3}$$

From the RG equation we can immediately see that the only fixed points are $\mu = 0$ and $\mu = 1$. Additionally, the $\mu = 1$ fixed point is the unstable zero temperature fixed point since any number $(\mu)^3 < \mu$ for all $\mu < 1$. The $\mu = 0$ on the other hand is the stable disordered fixed point, that describes the disordered phase. Therefore, there is no long range ordered in the 1-D ising model except at $T = 0$.

Correlation length

We saw that by studying the Ising model using domain walls, the correlation length grows exponentially with the temperature. Here we will try to understand that in two ways.

Consider $\xi = \xi(\mu)$

Under the RG transformation, the new correlation is $\xi(\mu')$, in units of the new lattice spacing. But since all we did was to rescale all the lengths by a factor of b, we must have $\xi(\mu') = \xi(\mu)/b$. This is merely an equation that describes how the new units relate to each other. But bringing in the RG equation from above, $\mu' = \mu^3$, is enough to determine the functional equation of ξ,

$$\xi(\mu^3) = \frac{\xi(\mu)}{3}.$$

The solution to the equation above is logarithmic, which we write in the form,

$$\xi = \frac{\text{const.}}{\ln(1/\mu)}.$$

Iterating the RG equation

Consider again the 1D chain of spins, and we iterate using the decimation rule, to obtain a coarse-grained chain, and every time the lengths, and thus the correlation length, gets divided by 3. Iterating k times, the correlation length becomes $\xi/3^k$. At the same time, the coupling becomes μ^{3^k}. The idea now is to reach a region of the parameter space where we know what the physics are, and can understand what the correlation length is.

Assume that the initial μ is close to 1 so that $\xi \approx 1$. We now iterate until the renormalised μ is of the order of magnitude of the lattice spacing (the precise value doesn't matter, only that its difference from $\mu = 1$ is of order $\mathcal{O}(1)$). At that point, the correlation length is also of order $\mathcal{O}(1)$, even if the initial coupling before the coarse-graining was very large. Thus,

$$\mu_{\text{renorm}} = \mu^{3^k} \sim \mathcal{O}(1) \iff 3^k \sim \frac{\mathcal{O}(1)}{\log(1/\mu)}$$

and for the correlation length,

$$\xi_{\text{renorm}} = \frac{\xi}{3^k} \sim \mathcal{O}(1) \iff \xi \sim \frac{\mathcal{O}(1)}{\log(1/\mu)}$$

Previously we found that the correlation length grows exponentially as the temperature is decreased. Let us check this here also. We have $\mu = \tanh K \approx 1 - 2e^{-2K}$ for $K \gg 1$ and $\log(1/\mu) = -\log\mu \approx 2e^{-2K}$, which gives $\xi \propto e^{2K} = e^{2\tilde{K}/T}$.

3.1 RG with a nontrivial fixed point

We consider a very simple case with just one coupling, K, and a non-trivial fixed point given by $K_* = \mathcal{R}(K_*)$. This implies that the correlation length is $\xi = \infty$, which can be seen by

$$\xi(K') = \frac{1}{b}\xi(K) \implies \xi(K_*) = \frac{1}{b}\xi(K_*) \to \begin{cases} 0 & \text{only if } K = 0 \\ \infty & \text{else} \end{cases}$$

Near the $K = 0$ fixed point, the correlation length is small $\xi \lesssim 1$, and we can study this perturbatively as we have done with the high temperature expansion. Similarly, if we are close to the $K = \infty$, the correlation length is again similar to the lattice spacing, and again we can understand this with perturbative physics as we did with a low temperature expansion. However, near the non-trivial fixed point, this is a region of emergent physics where the correlation length is $\xi \gg 1$, and can be understood by a fixed point theory and perturbations of that. The difference is that the zeroth order of the perturbation theory is always non-trivial. With the correlation length being much larger than the lattice spacing means to effective descriptions in terms of field theory is likely to be useful.

Linearize RG EQ near the fixed point

We will understand much of the universal physics by linearising the RG equations by writing $\delta \underline{K} = \underline{K} - \underline{K}_*$. With one parameter only, we have $\delta K' = \mathcal{R}'(K_*)\delta K$. It is useful to define a y "RG eigenvalue" by $\mathcal{R}'(K_*) = b^y$ so that,

$$\delta K' = b^y \delta K$$

Note here that if $y > 0$, $b^y > 1$ and $|\delta K|$ increases under the RG transformation.

Also note that, y is an example of a **universal** quantity. Even though we have freedom on how we define the model, arbitrariness on how we define the block spins, and arbitrariness in the choice of b, but y is robust. It is in a sense a fundamental constant.

As a consistency check, we consider rescaling by a factor of b^2, which we can achieve by rescaling by a factor of b twice (on blocks of size b) ,

$$\delta K \to b^y b^y \delta K$$

or by using one block of size b^2,

$$\delta K \to (b^2)^y \delta K$$

The *correlation length exponent* is define by $\nu = \frac{1}{y}$. We have,

$$\xi(\delta K') = \xi(b^y \delta K) = \frac{1}{b}\xi(\delta K)$$

which is solved by $\xi(\delta K) = \frac{\text{const}}{|\delta K|^{1/y}}$, or,

$$\xi(K) = \frac{\text{const}}{|K - K_*|^\nu}$$

The constant may depend on the sign of δK.

With more couplings we can generalise this by writing $\underline{K}' = \underline{\mathcal{R}}(\underline{K})$ or in components, $K_a = \mathcal{R}_a(K_1, K_2, \dots)$. The linearised version of the RG equation with many couplings reads,

$$\delta K'_a = \sum_b T_{ab} \delta K_b + \mathcal{O}(\delta K^2),$$

where $T_{ab} = \frac{\partial R_a}{\partial K_b}\Big|_{K_*}$. We now define the *scaling variables* $\{u_i\}$, which can be chosen so that the matrix T is replaced by a diagonal matrix, and the RG equation has the form,

$$u_i' = b^{y_i} u_i.$$

The choice we have made is in writing the eigenvalue in the form b^{y_i}, but the *RG eigenvalue* is referring to the y_i (even though b^{y_i} is the eigenvalue of $\underline{T}$). This will make sense later when we will write a differentital equation instead of an iterative scheme.

Now, T doesn't have to be symmetric and therefore it will not have the same left and right eigenvectors. We can write $u_i = \underline{e}^{(i)} \cdot \underline{\delta K} = e_a^{(i)} \delta K_a$, so that the renormalised variables are,

$$u_i' = \underline{e'}^{(i)} \cdot \underline{\delta K'} = \underline{e'}^{(i)} \cdot \underline{T} \cdot \underline{\delta K'}$$

and we choose $\underline{e^{(i)}}$ to be a left eigenvector of $\underline{T}$, so that $\underline{e'}^{(i)} \cdot \underline{T} = \lambda^{(i)} \underline{e^{(i)}}$

Symmetry

Taking the symmetries of the Hamiltonian into account restrict the form of the matrix $\underline{T}$. For example, let us take the Ising model with a nearest neighbour coupling K and a magnetic field h and assume that there is a non trivial fixed point at $(K_*, 0)$. The model has symmetry $\delta h \to -\delta h'$ and $\delta h' \to -\delta h'$ while K remains fixed. This corresponds to making the change of variable $S \to -S$ and $S' \to -S'$. This is the $\mathbb{Z}_2$ symmetry of the Ising model. Because the RG equations have to be invariant under this symmetry, this restricts their form to,

$$\begin{aligned} \delta K' &= A \delta K + \mathcal{O}(\delta K^2, \delta h^2) \\ \delta h' &= B \delta h + \mathcal{O}(\delta K \delta h) \end{aligned}$$

where A, B are some numerical values. In terms of the matrix T the RG equations take the form,

$$\begin{pmatrix} \delta K' \\ \delta h' \end{pmatrix} = \begin{pmatrix} b^{y_t} & 0 \\ 0 & b^{y_h} \end{pmatrix} \begin{pmatrix} \delta K \\ \delta h \end{pmatrix}$$

so that in general we only have to diagonalise T within a symmetry block.

Classification of couplings and universality

We can now classify couplings into three types, based on the sign of the corresponding RG eigenvalues. The following table summarises this.

$y_i > 0$	Relevant	$b^{y_i} > 1$: u_i grows under the RG
$y_i < 0$	Irrelevant	$b^{y_i} < 1$: u_i shrinks under the RG
$y_i = 0$	Marginal	Have to look at higher orders to see what happens

This is a crucial classification that allows us to understand universality and answer the question: can we get the same long-distance behaviour for different microscopic models? This is a question of how many relevant ($y_i > 0$) couplings there are at a given fixed point. The important fixed points have a small number of relevant/marginal couplings and an infinite number of irrelevant ones.

As an example, the topology of flow diagram for the 2D Ising model is shown on the following figure. In the plot we assume that the $\mathbb{Z}_2$ symmetry is preserved and thus magnetic fields are not included. The usual 1D flow diagram that we have been drawing so far really sits in a higher dimensional space which has a lot of directions in which the fixed point is stable. Of course, with a 2D plot we can only show one such direction.

To relate this to a microscopic model we assume that the microscopic model is controlled by a single parameter (such as the temperature). As we change the temperature, the microscopic model moves along a path through theory space (dotted line on plot). This is not an RG trajectory, but rather a trajectory of the microscopic model as the temperature is varied. Where is the critical point (which is the microscopic model at the critical temperature) of the microscopic model? The critical point of the microscopic model is the value of the temperature where it flows towards the non-trivial fixed point, shown as big dot in the figure. Notice that the critical point is different from the fixed point (purple dot), which represents the point towards all critical points flow to. This platonic object is characterised by a set of universal numbers given by the RG eigenvalues. More precisely, the y_i are properties of the linearised RG flows **near** the fixed point.

We again stress out that the fixed point governs te long distance behaviour (as $L \to \infty$) of many microscopic models at T_c (an infinite dimensional continuum to be precise). The following cartoon explains this. The previous "plane" is a subspace in this higher dimensional space and the space of models that flow to the previous fixed point looks like a plane. There is one relevant direction that comes out of this plane, described by the "thermal scaling variable" u_t with eigenvalue y_t. The plane is a "critical surface" of models that flow to the fixed point, which is in fact an infinite dimensional subspace with many couplings that do not contribute to the long-distance behaviour. It is therefore parametrised by the irrelevant scaling variables ($y_i < 0$).

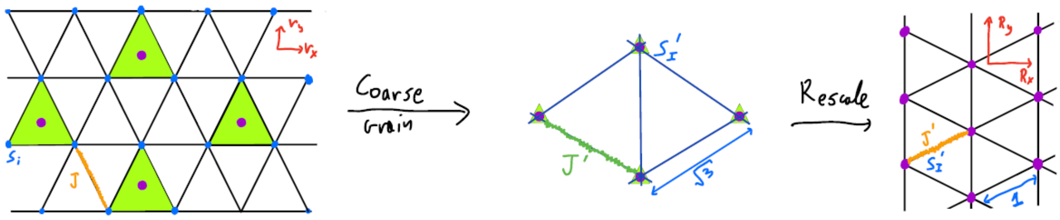
3.2 Scaling near a non-trivial fixed point

We will now study the scaling near a non-trivial fixed point for the case of the 2D Ising model. Recall that we define the block spins using the majority rule and the RG transformation that related the new parameters to the old ones is $\underline{K}' = \mathcal{R}(\underline{K})$. We will use the *Niemeijer–van Leeuwen* method (see Kardar Ch. 6) on a 2D lattice in the cases where there is only an Ising coupling, and when we introduce a symmetry breaking field. We will also explore the consequences of the non-trivial fixed point on the thermodynamics and correlation functions (see Cardy Ch. 3.4).

Near the fixed point, we linearise the RG equation by $\underline{K} = \underline{K}_* + \underline{\delta K}$, which in components becomes $\delta K'_a = T_{ab}\delta K_b + \mathcal{O}(\delta K^2)$. We make a change of basis by introducing the scaling variables u_α which are linear combinations of the δK_α and they obey the simple RG equation $u'_\alpha = b^{y_\alpha} u_\alpha$. In the 1D Ising model there was a single relevant scaling variable, whose flows are directed away from the fixed and end up in two different phases. In addition there are in general an infinite number of irrelevant scaling variables whose flows are directed towards the fixed point.

3.2.1 Niemeijer-van Leeuwen

For convenience we will use a triangular lattice instead of a square lattice. On the triangular lattice, the spins are grouped into cells (triangles) of three spins, which makes the use of the majority rule



for a renormalised Hamiltonian apparent. The partition function of the 2D Ising model is,

$$Z = \sum_{\{s\}} e^{J\sum_{\langle ij\rangle} s_i s_j}$$

The block spin is defined (without any ambiguity) as,

$$s_I' = \text{sgn}(\sum_{i\in I} s_i)$$

and the lattice geometry remains the same after the coarse-graining, so that iteration is possible. The rescaling factor can be found by a trigonometric argument, or by noting that rescaling areas by b results in the rescaling of areas by b^2. But since a unit of area of the initial lattice which corresponded to three spins, now corresponds to one spin, we get that the rescaling factor is $b = \sqrt{3}$.

By definition, the RG transformation on the Hamiltonian is,

$$e^{-\mathcal{H}'[s']} = \sum_{\substack{s \\ (\mapsto s')}} e^{-\mathcal{H}[s]}$$

The idea of Niemeijer and van Leeuwen was to split up the Hamiltonian into two terms: one which contains the bonds of microscopic spins that lie inside the block spin ($\in \triangle$) and those which are outside ($\notin \triangle$),

$$-\mathcal{H}_0 = J\sum_{\substack{\langle ij\rangle \\ \in\triangle}} s_i s_j, \qquad -\mathcal{H}_1 = J\sum_{\substack{\langle ij\rangle \\ \notin\triangle}} s_i s_j.$$

We now want to treat $\mathcal{H}_1$ as a perturbation, but we do not have a small parameter to control this. Instead we will do an uncontrolled perturbation, where we consider adding higher and higher order corrections to what are treating as small. The algebra is quite complicated for higher order terms, but it is in principle possible to perform.

When $\mathcal{H}_1 \to 0$, this is a problem of decoupled blocks, and is therefore exactly solvable. The partition function for $\mathcal{H}_0$ is then,

$$Z_0[s'] = \sum_{\substack{s \\ (\mapsto s')}} e^{-\mathcal{H}_{\prime}[s]}$$

which allows us to interpret the renormalised Hamiltonian, as an expectation of $\mathcal{H}_1$ in the Z_0 ensemble,

$$e^{-\mathcal{H}'[s']} = \sum_{\substack{s \\ (\mapsto s')}} e^{-\mathcal{H}_0[s]}e^{-\mathcal{H}_1[s]} = Z_0[s']\langle e^{-\mathcal{H}_1[s]}\rangle_{J=0;\text{Spin}=s'}$$

which is still an exact expression.

Now, the partition function for the intra block spins describes uncoupled blocks, and can therefore be split into a product of partition functions for single blocks,

$$Z_0[s'] = \prod_{\text{blocks}I} Z_\triangle[s'_I].$$

and by symmetry, the partition function for the blocks of spin $+1$ and those with spin -1 are the same, denoted by $Z_\triangle$.

> **Note.** Cumulant Expansion
>
> $$\begin{aligned}\langle e^x\rangle &= 1 + \langle x\rangle + \frac{1}{2!}\langle x^2\rangle + \frac{1}{3!}\langle x^3\rangle + \dots \\ &= e^{\langle x\rangle + \frac{1}{2}(\langle x^2\rangle - \langle x\rangle^2) + \frac{1}{3!}(\langle x^3\rangle - 3\langle x\rangle^2\langle x\rangle + 2\langle x\rangle^3) + \dots}\end{aligned}$$
>
> where the terms on the exponential on the second line are called the *cumulants*. Note that taking $e^{\langle x\rangle}$ as an approximation is not the same as $1+\langle x\rangle$, and the cumulants will vanish only if the fluctuations in x vanish. Therefore, $e^{\langle x\rangle}$ can be a good approximation even in the case when $1+\langle x\rangle$ is not. This will be a good approximation at very low temperatures, for example.
>
> As an example, if $x = \text{const.}$, then $e^{\langle x\rangle}$ is exact since all the higher order **cumulants** vanish, but $1+\langle x\rangle$ is not exact, because the higher **moments** do not vanish.

We will work to first order, keeping only the first cumulants here, so that,

$$\begin{aligned}e^{-\mathcal{H}'} &\approx Z_0[s']e^{\langle \mathcal{H}_1[s]\rangle_{0,s'}} \\ &= Z_\triangle^{N'} \exp\left(-\langle\mathcal{H}_1\rangle_{0,s'}\right)\end{aligned}$$

where N' is the number of block spins. Now,

$$\begin{aligned}-\langle\mathcal{H}_1\rangle_{0;s'} &= J \sum_{\substack{\langle ij\rangle \\ i,j\text{on diff. blocks}}} \langle s_i s_j\rangle_{0,s'} \\ &= 2J\sum_{\langle IJ\rangle} < s_i >_{0,s'_I} < s_j >_{0,s'_J}\end{aligned}$$

where on the second sum we have grouped terms according to which blocks they couple and the factor of two comes from the fact that two bonds contribute with the same contribution. We have split the expectation values into a product of expectation since there is no coupling in the Z_0 ensemble. By symmetry now,

$$\langle s_i\rangle_{0,s_I} = \gamma s'_I$$

for some constant γ, and $s'_I = \pm 1$. Also, $\gamma < 1$ since there are fluctuations in the microscopic spin s_i for each value of s'_I. We can calculate γ explicitly by thinking about the configurations of a single block. We thus get,

$$-\mathcal{H}'[s'] = N'\ln Z_\triangle + 2J\gamma^2\sum_{\langle IJ\rangle} s'_I s'_J,$$

and note that at this order in the approximation, the renormalised Hamiltonian has only nearest neighbour coupling.

Single block expectation value

We now perform the explicit calculations. Each ferromagnetic bond has Boltzmann weight e^J while an antiferromagnetic one has e^{-J}, and therefore the partition function is given as a sum of the following configurations,

$$\begin{aligned} Z_\triangle &= Z_{\triangle,+} = Z_{+++} + Z_{++-} + Z_{+-+} + Z_{-++} \\ &= e^{3J} + 3e^{-J} \end{aligned}$$

The expectation values are thus,

$$\begin{aligned} \langle s_i \rangle_{0,s'_I=+1} = \gamma &= \frac{\sum_{4\text{ configs}} e^{-\mathcal{H}_0} s_i}{\sum_{4\text{ configs}} e^{-\mathcal{H}_0}} \\ &= \frac{e^{3J} + e^{-J} + e^{-J} - e^{-J}}{e^{3J} + 3e^{-J}} \\ \iff \gamma &= \frac{e^{4J}+1}{e^{4J}+3} \end{aligned}$$

This gives us now the RG transformation for J, $J' = \mathcal{R}(J)$,

$$J' = 2J \left(\frac{e^{4J}+1}{e^{4J}+3} \right)^2$$

Fixed points

- $\underline{J=0}$ This a stable disordered fixed point. To see this, plug in the RG equation a small $J \ll 1$, to get $J' = J/2$.
- $\underline{J=\infty}$ This is a stable ordered fixed point. For $J \gg 1$, the RG equation reads, $J' = 2J$.

In order to have two stable fixed point at the extrema, the simplest scenario is to have an unstable fixed point at some finite J_*. To find it, we plug $J' = J$ in the RG equation and find,

$$e^{4J_*} = \frac{3-\sqrt{2}}{\sqrt{2}-1} \implies \boxed{J_* \approx 0.336}$$

The 2D Ising model on the triangular lattice is in fact exactly solvable with $J_c = 0.275$. The approximation is therefore not exact, but this was expected.

To get the universal exponents, we linearise the RG equation around the fixed point, $J = J_* + \delta J$ and $J' = J_* + \delta J'$. Plugging in the RG equation, we get,

$$\delta J' \equiv b^{y_t} \delta J = (1.624\dots)\delta J$$

where y_t is the RG eigenvalue of J. The subscript t is short for "thermal" and refers to the fact that it can be used to go from one side of the diagram to the other by eg changing the temperature. With $b = \sqrt{3}$ we find $y_t \approx 0.882$.

Recall that the correlation length exponent is given by $\nu = 1/y_t \approx 1.13$. The true result is $y_t = \nu = 1$, which shows that the approximation is surprisingly good.

Adding a magnetic field

We now consider adding an infinitesimal field $h \ll 1$, and include this term as to what we considered as a perturbation,

$$e^{-\mathcal{H}'[s']} = \sum_{\substack{s \\ (\mapsto s')}} e^{-\mathcal{H}_0[s]} e^{-\mathcal{H}_1[s] + h\sum_i s_i}$$

The new Hamiltonian is thus,

$$\mathcal{H}'[s'] = N' \ln Z_\triangle + J \sum_{\langle ij \rangle} \langle s_i s_j \rangle_{0,s'} + h \sum_i \langle s_i \rangle_{0,s'}$$

It remains to calculate the last term in a similar way as before,

$$\begin{aligned} h \sum_i \langle s_i \rangle_{0,s'} &= h \sum_I \sum_{i \in I} \langle s_i \rangle_{0,s'} \\ &= 3h \sum_I \langle s_i \rangle_{0,s'_I} \\ &= 3h\gamma \sum_I s'_I \end{aligned}$$

We thus have the RG transformation of the magnetic field as,

$$h' = 3\gamma(J)h$$

with the same fixed point as before $J = J_*$, $h = 0$. Linearising around that,

$$h' = 3\gamma(J_*)h \implies \boxed{h' = \frac{3}{\sqrt{2}}h}$$

The RG eigenvalue for the magnetic field is $y_h \approx 1.37$, and the true value is $y_h = 1.88$, which shows that this is not a good approximation. However $y_h > y_t > 0$.

Note. When is our approximation good? The approximation that we did is good when $J \gg 1$ and $J \ll 1$. To see this, recall that what we did was to replace the term $\langle e^{J\sum_{\text{inter}} s_i s_j} \rangle_{0;s'}$ with the lowest order term in the cumulant expansion, $e^{J\sum_{\text{inter}} <s_i s_j>_{0;s'}}$. For $J \ll 1$, this is a good approximation by usual perturbation theory. If $J \gg 1$, the spins are well ordered and thus $s_i s_j$ has very weak fluctuations, and thus higher cumulants are small

3.3 Symmetry

We now consider the $\mathbb{Z}_2$ symmetry of the above model. We only considered a $\mathbb{Z}_2$ even coupling of nearest neighbours, and a $\mathbb{Z}_2$ odd magnetic term, but we could have included more terms like $\sim \sum s_i s_j s_k$. In fact, a higher order approximation would require keeping track of more couplings.

The linearised RG equation we found was a diagonal matrix since the off diagonal elements vanish by symmetry. More generally we would get a block-diagonal matrix with a $\mathbb{Z}_2$ even and a

$\mathbb{Z}_2$ odd block. We would then diagnalise each block and find say the eigenvalues $b^{y_t}, b^{y_t'}, b^{y_t''}, \ldots$ for the even block. However, we would find that the first eigenvalue is positive $y_t > 0$ and the other ones are negative and successively more negative, $0 > y_t' \leq y_t'' \leq \ldots$. But because we only have only one relevant eigenvalue, we only need to tune one coupling to access the critical point.

Emergent symmetries

There are also symmetries that are not present in the microscopic model but *emerge* under the RG. The RG preserves all the symmetries of $\mathcal{H}$ but it can also generate new symmetries asymptotically (i.e. at the fixed point). The simplest example is rotational invariance. For example, the square lattice has only a finite-angle rotational invariance. The same is true for a triangular lattice (even though the symmetry group is larger). But the Ising fixed point has continuous rotations in space, and this is an emergent symmetry hidden in RSRG since we work with a lattice. We will revisit this later in the context of field theory.

Another example is provided by scale invariance, as a consequence of fixed point. Microscopically, since the Hamiltonian changes there is no scale invariance. But under the RG flow when we get aymptotically close to the non-trivial fixed point, the Hamiltonian is the same which is exactly scale invariance. Scale invariance is in fact often part of a larger emergent symmetry: *conformal invariance.* A conformal map is one that preserves the angles on a local orthonormal set of axes, and they look locally like rescaling transformations. See Cardy, final chapter for more information on conformal transformations.

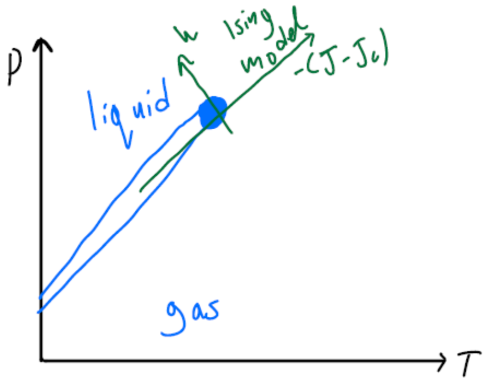
Another famous example is the $\mathbb{Z}_2$ symmetry of the Ising model, not as a fundamental symmetry of the microscopic model, but as an emergent symmetry. Suppose that we are in 3D, and in that case there is only one relevant $\mathbb{Z}_2$ odd scaling variable, the magnetic field, at the Ising fixed point. Thus $y_h > 0$ and $0 > 0 \leq y_h' \leq y_h'' \ldots$ and similarly for the $\mathbb{Z}_2$ even ones. The RG flow diagram thus looks like,

Consider now a microscopic model $\mathcal{H}$ with no $\mathbb{Z}_2$ symmetry and with 2 parameters (couplings). We can draw the parameter space in the original theory space. Since it doesn't have a $\mathbb{Z}_2$ symmetry, we cannot say anything about this 2-dimensional surface: it sits at a non-zero value of the irrelevant odd coupling, h. Nevertheless, it is possible that the $\mathbb{Z}_2$ symmetry emerges at one point in the parameter space of the model. This point is shown at the diagram. An example of such a model would be the Hamiltonian,

$$\mathcal{H} = J \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i + h' \sum_{\triangle} s_i s_j s_k$$

There are now 2 odd couplings, u_h and u_h', and each would be a different combination of h and h'. One combination is relevant but any linearly independent combination is irrelevant. Therefore in order to get to the Ising critical point, we only have to make the "correct" linear combination vanish, as is given by the RG eigenvectors.

The emergent $\mathbb{Z}_2$ symmetry is what happens at the critical endpoint of the liquid-gas transition (as we will see later in the field theory context). As can be seen from the PT phase diagram below, there is a first order phase transition that terminates at a point which is governed by a non-trivial RG fixed point, which is the Ising fixed point. This means that all universal properties of vapour at the critical endpoint coincide with those of the Ising magnet at T_c. The key difference is that at the Ising model (in the presence of symmetry) we have to tune one single parameter to access the critical point whereas in the liquid-gas transition we need to tune two parameters (in the absence of symmetry).



We can rationalise the above by thinking about the *lattice gas*. Take a cubic lattice and at each site sits a variable which is an occupation number that takes the value $n_i = 0, 1$. For example a gas with low density would have only a few occupied sites, but a liquid state would have a high density with many occupied sites. The mapping to Ising model is then trivially,

$$s_i = 2n_i - 1$$

To access the two phases we can tune a magnetic field: a positive field in the Ising model would correspond to a high density. Going back to the PT phase diagram, the direction parallel to the coexistence line is the $-\delta J = -(J - J_c)$ and the one normal to it is the δh in the Ising model. The first order transition is the ordered state in the Ising model: the two coexisting states with the same free energy, gas and liquid, map to the magnetised states of the Ising model.

3.4 Scaling Operators & Scaling Variables

Consider a Hamiltonian of the form,

$$\mathcal{H}[s] = K_1 \sum_{\langle ij \rangle} s_i s_j + K_2 \sum_{\langle\langle ij \rangle\rangle} s_i s_j + \dots$$

Close to the fixed point there is a preferred basis of couplings, $\{u_\alpha\}$, given by $u_\alpha = \underline{e}^{(\alpha)} \cdot \underline{\delta K}$. We can then rewrite $\mathcal{H}$ in terms of the u_α instead of the $K's$,

$$\begin{aligned}
\mathcal{H} &= \mathcal{H}_*[s] + \sum_\alpha u_\alpha(\dots) \\
&= \mathcal{H}_*[s] + \sum_\alpha u_\alpha \sum_r O_\alpha(r)
\end{aligned}$$

where $(\dots)$ represents a perturbation that involves all the spin interactions, and is different for each scaling variable. We denote these terms by O_α, which are local operators, and the expression is translationally invariants. Notice also that we now use the position r of the site rather than the label i for a spin.

We call $O_\alpha(r)$ a *scaling operator*[2] where "operator" refers to some function of spins in a neighbourhood close to r, so it is local. Let $O'_\alpha(R)$, for $R = r/b$, be the same function but now for the block spin s'. For example, if $O_\alpha(r) = s_i$, then $O'_\alpha(R) = s'_I$.

We now imagine coarse-graining correlation functions that involve $O_\alpha(r)$. As is the case for the scaling variables which have a simple behaviour under the RG, similarly for the scaling operators, their correlation function behave in a simple way under the RG transformation. The claim is that there is a simple relationship between the scaling operators of the microscopic and coarse-grained model, given by,

$$\boxed{O_\alpha(r) \sim b^{-X_\alpha} O'_\alpha(R)}$$

where X_α is the *scaling dimension* of O_α, and "$\sim$" means equivalent if we consider correlation functions at separations much larger than the lattice spacing.

We will look at one example. In our previous calculation in the triangular lattice, within the accuracy of our approximation, the magnetic field h was a scaling variable and the spin s was the corresponding scaling operator. We found $\langle s_i \rangle_{0,s_I} = \gamma s'_I$, which we now interpret as $s_i \sim \gamma s'_I$, in correlation functions. This means that if we calculate $\langle s_i s_j \rangle$, for i, j in distinct blocks, we can apply the same approximation to find,

$$\langle s_i s_j \rangle = \gamma^2 \langle s_I s_J \rangle .$$

(**Check this!**. To do this, work at the lowest non-trivial order as was done in the derivation of the RG equations.)

Relation to RG

We will use a consistency argument by calculating the renormalised Hamiltonian in two different ways.

1. $\mathcal{H}_*[s] + u_\alpha \sum_r O_\alpha(r) \to \mathcal{H}_*[s'] + (b^{y_\alpha}) u_\alpha \sum_R O'_\alpha(R)$, to linear order in perturbation in u_α

2. $$\begin{aligned} \mathcal{H}_*[s] + u_\alpha \sum_r O_\alpha(r) &= \mathcal{H}_*[s] + u_\alpha \sum_{R\text{blocks}} \sum_{r \in R} O_\alpha(r) \\ &= \mathcal{H}_*[s] + u_\alpha \sum_{R\text{blocks}} \sum_{r \in R} b^{-X_\alpha} O'_\alpha(R) \\ &\sim \mathcal{H}_*[s] + u_\alpha b^{-X_\alpha} b^d \sum_R O'_\alpha(R) \end{aligned}$$

   since $\sum_{r\in R} = b^d$, the number of spins in the block. In the last step, when we performed the RG transformation, we have used the fact that $O'_\alpha(R)$ is already expressed in terms of block spins, so the RG step leaves it unchanged.

By consistency, both expressions should be equal, which means that.

$$\boxed{y_\alpha = d - X_\alpha}$$

[2] Nothing to do with quantum mechanics here!

which is a crucial relationship that related RG flows (y_α) to correlation functions. In the previous example, $O_\alpha(r) = s$ and $u_\alpha = h$, and we have found $b^{y_h} = 3/\sqrt{2}$ and $b^{-X_h} = \gamma = 1/\sqrt{2}$, which is in agreement with the relation just derived!

Scaling of Correlation functions

Let $G_\alpha(r) = \langle O_\alpha(r) O_\alpha(0) \rangle_*$, the correlator at the critical point, which is the correlator at the fixed point at long distances. We use $O_\alpha(r) = b^{-X_\alpha} O'_\alpha(R)$ to get,

$$G_\alpha(r) = b^{-2X_\alpha} \langle O'_\alpha(\frac{r}{b}) O'_\alpha(0) \rangle_*.$$

But since this is evaluated at the fixed point, the correlation functions of the coarse-grained (primed) spins are exactly the same as the correlation functions of the microscopic spins: s and s' have the same $\mathcal{H}_*$. Therefore,

$$\boxed{G_\alpha(r) = b^{-2X_\alpha} G_\alpha(r/b)}$$

and $X_\alpha > 0$. This is solved by,

$$\boxed{G_\alpha(r) = \frac{A}{r^{2X_\alpha}}}$$

More relevant scaling variables u_α (larger y and thus smaller X) are conjugate to scaling operators with more slowly decaying correlators. Thus, the most important operators are those which are most strongly correlated at larger distances. This generalises to multipoint correlation functions also,

$$\langle O_\alpha(r_1) O_\alpha(r_2) \ldots O_\alpha(r_N) \rangle_* \sim b^{-2NX_\alpha} \langle O'_\alpha(\frac{r_1}{b}) \ldots O'_\alpha(\frac{r_N}{b}) \rangle_*$$

Off-critical correlators

We now consider what happens to the correlation functions not at the critical exactly but slightly away from it, which will give rise to the notion of a "scaling function". Consider $T \geqslant T_c$ which means that u_t is small and positive since $u_t \sim T - T_c$. We evaluate the two-point function ($\langle s_i s_j \rangle$),

$$G(r, T - T_c) = b^{-2X_h} G(r/b, (T - T_c) b^{y_t})$$

since at lowest order in the perturbation expansion, u_t rescales according to the RG equation b^{y_t}. The solution is now,

$$\boxed{G(r, T - T_c) = r^{-2X_h} g(r/\xi)}$$

for $\xi = \frac{1}{(T-T_c)^\nu}$, $\nu = 1/y_t$ and g is the "scaling function". To see this, we use the relation above with $b = r$ to get $G(r, T - T_c) = r^{-2X_h} G(1, (T - T_c) r^{y_t})$ (which only makes sense with the linearised RG equation). Now $G(1, (T - T_c) r^{y_t}) = G(1, r/\xi) \equiv g(r/\xi)$. When $r \ll \xi$, $g = \text{const}$ and the relation reduces to a power law, as above. When $r \gg \xi$, $g \sim e^{-r/\xi}$.

3.4.1 Example: 2-point function on Cylinder (2D Ising model)

Consider a cylinder of circumference L, and place an Ising model on it. We want to find the 2-point function between spins at points r_1 and r_2, and for simplicity, we will do this at the critical point,

$$G(r_1, r_2; L) = \langle s(r_1)s(r_2)\rangle_{*,L}$$

The important point here is that the cylinder has no intrinsic curvature so that locally it does look like an Ising model on the plane. If $1 \ll |r_1 - r_2| \ll L$, we expect that G is the same as in the ∞ plane case, the usual Ising model. So $G \sim \frac{A}{|r_1-r_2|^{2X_h}}$. On the other limit, $|r_1 - r_2| \gg L \gg 1$, this model will look like a one-dimensional model. To see this formally, we can think of writing a $2^L \times 2^L$ transfer matrix for the system and the correlation function is then given by the ratio of the two largest eigenvalues and decays exponentially with L.

We can make this more precise with the RG. We first zoom out by a factor of $b \sim L$,

$$\begin{gathered} G(r_1, r_2, L) \longrightarrow G(r_1/b, r_2/b, L/b)b^{-2X_h} \\ G(r_1/L, r_2/L, 1)L^{-2X_h} \end{gathered}$$

and now this effective 1D model is not parametrically large in L, all length scales and the coupling is $\mathcal{O}(1)$. Only $|r_1 - r_2|$ can become large. The correlation length of the effective model, ξ_{1D}, is some $\mathcal{O}(1)$ constant. We now that this then,

$$G(R, R', 1) \sim e{-}|R - R'|/\xi_{1D}$$

and therefore,

$$\boxed{G(r_1, r_2, L) \sim L^{-2X_h} \exp\left(-\frac{|r_1 - r_2|}{L\xi_{1D}}\right)}$$

To get the prefactor we have used our knowledge of the scaling at the 2D fixed point and our expectation on the endpoint of the RG flow had to be, using what we know about $1D$ models and their correlation functions.

More generally, not assuming any relation between r and L, we get that,

$$G(r_1, r_2, L) = L^{-2X_h} F\left(\frac{r_1 - r_2}{L}\right)$$

for some universal scaling function $F(z)$, with $z = \frac{|r_1-r_2|}{L}$. For z large, we have, $F(z) \sim Ae^{-z/\xi_{1D}}$. For small z we want to reproduce the power law behaviour, so $F(z) \sim Az^{-2X_h}$ and thus $G \sim L^{-2X_h}\left(\frac{|r_1-r_2|}{L}\right) = A|r_1 - r_2|^{-2X_h}$.

We can assume further conformal invariance of the 2D fixed information to find that $\xi_{1D} = \frac{1}{2\pi X_h}$. The idea is that the plane can be conformally mapped to the plane using the exponential map.

3.5 Scaling for Free Energy

We now discuss the scaling predictions for the free energy and other thermodynamic quantities. We know that we can understand universality in terms of RG fixed points. We will now see how the critical exponents that we found actually encode information about phase transitions also, in the context of the 2D Ising model. Recall that the free energy is an extensive quantity defined by $Z = e^{-\beta F}$ and $F = \langle E \rangle - TS$. We first define the reduced free energy per site,

$$f(\underline{K}) = -\frac{1}{N} \ln Z$$

which is an intensive quantity that is finite as $N \to \infty$. Close to the critical point, it is a function of only the relevant scaling variables $u_t \propto t = \frac{T-T_c}{T_c}$ and $u_h \propto h$, for small t, h.

We first need to understand the functional form of $f(u_t, u_h)$ and we start with considering the RG transformation of the partition function Z. So far, in the renormalization of the Hamiltonian, we discarded the constant piece, $N' \ln Z_\triangle$, only thinking about the coupling term. With N spins, we have,

$$Z(\underline{K}; N) = e^{-Ng(\underline{K})} Z(\underline{K}'; N/b^d)$$

where in our case $b^d = 3$ and $e^{-Ng(\underline{K})} = Z_\triangle^{N/3}$. Equivalently,

$$f(\underline{K}) = g(\underline{K}) + \frac{1}{b^d} f(\underline{K}')$$

and we assume that $g(\underline{K})$ is some analytic function in $\underline{\delta K}$. This is in agreement with what we found in our example where $Z_\triangle \sim e^{3J} + 3e^{-J}$, resulting in an analytic expansion of $\ln Z$ in δJ. This is actually an important assumption because the free energy $f(\underline{K})$ can be a non-analytic function of the couplings that comes from the fact that it involves contributions from modes at all length scales. It is this non-analyticity that results in singular behaviour of thermodynamic quantities near T_c. However, $g(\underline{K})$ only involves modes from a finite range of length scales, in that case length scales between the lattice spacing and the size of the block, preventing any non-analyticity. These singularities in the free energy are governed by the RG eigenvalues y_t and y_h (or equivalently X_t and X_h).

Historically it was a breakthrough to reduce the critical singularities of specific heat, magnetisation, magnetic susceptibility etc to just two critical exponents. The critical exponents serve as fingerprints that can be measured in experiments or simulations, which reveal the underlying theory of the fixed point controlling the physics. Given the free energy as a function of the couplings, we can take derivatives to obtain the *standard critical exponents.*

- <u>Specific heat:</u> $C = \frac{\partial \langle E \rangle}{\partial T} \sim |T - T_c|^\alpha$
- <u>Magnetisation:</u> $M = \langle s \rangle = \frac{\partial f}{\partial h}\Big|_{h=0} \sim |T_c - T|^\beta$ (for $T < T_c$).
- <u>Magnetic susceptibility:</u> $\chi = \frac{\partial M}{\partial h}\Big|_{h=0} \sim |T - T_c|^{-\gamma}$
- Right at the $T = T_c$, $M = 0$ but the susceptibility diverges. The scaling is given by, $M \sim |h|^{1/\delta}$
- <u>Correlation length:</u> $\xi \sim |T - T_c|^{-\nu}$ with $\nu = 1/y_t$.
- <u>Correlation functions:</u> $\langle s(r) s(r') \rangle \sim \frac{1}{r^{2X_h}} = \frac{1}{r^{d-2+\eta}}$, with $\eta = 0$ in MFT (mean field theory).

All these critical exponents reduce to expression that involve the RG eigenvalues.

The reduced free energy transforms under RG according to Eq. 3.5. However, its singular part is obeying a simpler relation, $f_s(\underline{K}) = \frac{1}{b^d} f_s(\underline{K}')$. To see this, imagine writing f as a generalised Taylor series including non-integer powers of the couplings u_t, u_h. The first step of the RG transformation is thus,

$$f_s(u_t, u_h) = b^{-d} f_s(b^{y_t} u_t, b^{y_h} u_h)$$

which we can iterate n times to get,

$$f_s(u_t, u_h) = b^{-nd} f_s(b^{ny_t} u_t, b^{ny_h} u_h)$$

and we choose n such that $b^n = \ell$. This means that we coarse-grain up to a scale ℓ. Now, since $y_t, y_h > 0$, the arguments on the RHS are growings. We can get rid of ℓ from the RG equation by iterating RG until $|\ell^{y_t} u_t| = u_{t_0}$, an arbitrary fixed reference point, which we restrict to a small value so that the linearised RG is valid. Thus, $\ell = \left|\frac{u_{t_0}}{u_t}\right|^{1/y_t}$. Inserting this into the RG equation gives,

$$f_s(u_t, u_h) = \left|\frac{u_{t_0}}{u_t}\right|^{-d/y_t} f_s\left(u_{t_0}, \left|\frac{u_{t_0}}{u_t}\right|^{y_h/y_t} u_h\right)$$

The upshot of this is that now the free energy is a function of a single argument instead of two, since u_{t_0} is a fixed value chosen by us. The only unknown variable is the second one in f_s.

In terms of the RG parameters, the initial values $u_t = \text{const.}t$ and $u_h = \text{const.}h$. We further define $\Phi(z) = f_s(u_{t_0}, z)$, so that,

$$f_s(t, h) = \left|\frac{t}{t_0}\right|^{d/y_t} \Phi\left(\frac{h/h_0}{|t/t_0|^{y_h/y_t}}\right)$$

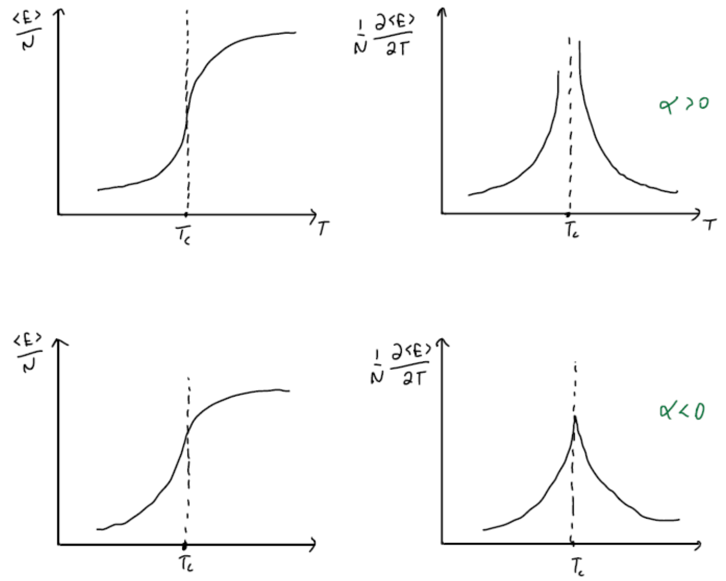
The parameters t_0 and h_0 are not universal constants but depend on the definition of the lattice model etc. However, the function Φ is a universal scaling function that encodes all the large length scales information from the coarse graining. We neglected the irrelevant couplings which would only contribute through subleading corrections to the above form, when t, h are small.

Heat Capacity and Magnetisation

We are talking here about a continuous phase transition which are characterised by a diverging correlation length ξ and scale invariance at the critical point. Let us first recall what a first order phase transition. In the RG language, this corresponds to a trivial unstable fixed point (see Cardy). First order transition have continuous free energy across the transition, $f = \frac{\langle E\rangle}{N} - \frac{TS}{N}$, but the internal energy is discontinuous. The disordered phase has higher $\langle E\rangle$ and higher S and the ordered phase has lower values for both. The jump corresponds to the latent heat: the heat that needs to be supplied to the system to “excite” it from the low energy ordered phase to the higher energy disordered phase. At $T = T_c$, the free energy is equal for both phase.

On the other hand, a continuous phase transition has continuous internal energy but non in general analytic at T_c. What we see from both plots is that the heat capacity scales as,

$$C = \frac{1}{N}\frac{\partial \langle E\rangle}{\partial T} \sim |T - T_c|^{-\alpha}$$



So when the specific heat diverges, $\alpha > 0$, and when there is a cusp, $\alpha < 0$.

Let us consider the simple case $h = 0$ where the formula from above reduces to,

$$f_s(t, 0) = \Phi(0) \left| \frac{t}{t_0} \right|^{d/y_t}$$

In that case, the heat capacity is,

$$C \sim \frac{\partial^2 f}{\partial t^2}\Big|_{h=0} \sim |t|^{-(2-d/y_t)}$$

so that $\alpha = 2 - d/y_t = 2 - d\nu$. Therefore, the divergence of the heat capacity at the critical point will depend only on the value of ν. For the 2D Ising model, $d = 2$ and $\nu = 1$, thus $\alpha = 0$ which leads to a logarithmic divergence.

Similarly, for the magnetisation we calculate,

$$M \sim \frac{\partial f}{\partial h}\Big|_{h \to 0^+} \sim \left| \frac{t}{t_0} \right|^{\frac{d-y_h}{y_t}}$$

for $t < 0$. For $t > 0$ we have $M = 0$. Thus the critical exponent from $M \sim (T_c - T)^\beta$ gives $\beta = \frac{d-y_h}{y_t} = \nu X_h$, where X_h is the scaling dimension of the spin.

We can also check that the scaling of the magnetisation, $M \sim |t|^{\nu X_h}$ is consistent with the scaling of the spin operator, by deriving the above reuslt in a different way. We have that the magnetisation at some fixed temperature $t < 0$ is,

$$M = \lim_{h \to 0^+} \langle s \rangle_t$$

where we included a symmetry breaking field to select one of the two ordered phases. On the RG flow diagram, we are near the fixed point, at a small negative δt, and since we are close to the critical point, the magnetisation $\langle s \rangle$ is small. We can thus use RG to renormalise "enough" until we reach some $t = -\mathcal{O}(1)$ value, where $\langle s \rangle = \mathcal{O}(1)$ also. The length scale that we need to coarse grain up to is the correlation length ξ. When we are no longer close to the fixed point, the value of $\langle s \rangle$ is no longer small. So let's say we coarse grain up to a length scale ℓ.

$$M = \lim_{h \to 0^+} \ell^{-X_h} \langle s' \rangle_{t\ell^{y_t}}$$

As always, the rescaling can be done in steps of b so that $b^k = \ell$. Now we choose $\ell = |t|^{-\nu} = |t|^{-1/y_t} = \xi$, which gives,

$$M = \lim_{h \to 0^+} \xi^{-X_h} \langle s' \rangle_{-1}$$

so that $\lim_{h \to 0^+} \langle s' \rangle_{-1}$, the expectation value of the spin deep in the ordered phase, is some $\mathcal{O}(1)$ constant. Finally the scaling of the magnetisation is,

$$M \sim \xi^{-X_h} \sim |t|^{\nu X_h}$$

Thus, instead of taking derivatives of the free energy, we can find the critical exponents by thinking about the scaling of the relevant operator (or correlation function). For the specific heat it is not just the spin as above, but rather a more complicated correlation function that involves 4 spins (2 links $\langle ij \rangle$ and $\langle kl \rangle$). This can be seen by expanding everything out: $C = \frac{1}{N} \frac{\partial \langle \mathcal{H} \rangle}{\partial T} \propto \langle \mathcal{H}^2 \rangle - \langle \mathcal{H} \rangle^2$, with $\mathcal{H} = \sum_{\langle ij \rangle} s_i s_j$. Thus the correlation function of the "energy" scaling operator.

4 Landau Theory

The basic idea of Landau theory is universality: many models flow to the same fixed point. The Landau Ginzburg theory (LG) gives us a canonical model with a given symmetry which is a useful point of reference. Continuum field theory allows more powerful RG tools and in some cases (including many useful cases), the continuum theory allows an exact description of the fixed point and its correlation functions. We start by writing the L.G. theory for the Ising model.

4.1 Ising field theory: ϕ^4 theory

The partition function for the Ising model is given by,

$$Z = \int \mathcal{D}\phi(\underline{r})\ e^{-\mathcal{H}}$$

with the Hamiltonian being,

$$\mathcal{H} = \int d^d r\ \left[\frac{1}{2}(\nabla\phi)^2 + \frac{t}{2}\phi^2 + \frac{u}{4}\phi^4\right]$$

We can relate this to the lattice sites we used before by regularizing the integral such that $\mathcal{D}\phi \to \prod_i d\phi_i$, where i runs over all the lattice sites, or more often the product is over k, the momentum modes. In eiher case, there is a **UV cutoff** introduced because the model has a smallest length scale given by the lattice spacing a or equivalently as the largest momentum $|k| \leq \frac{1}{a}$. Then, the partition function is guaranteed to make sense.

There is a $\mathbb{Z}_2$ symmetry in the field theory, which is the analogue of the spin flip symmetry on the lattice: $\mathcal{H}$ is invariant under $\phi(\underline{r}) \to -\phi(\underline{r})$. The vague idea is that we can think of the field $\phi(\underline{r})$ is related to the spin in the lattice site, averaged over a spatial region, and therefore it no longer just takes the values ± 1. The true justification for using the LG theory however is universality: the lattice and the ϕ^4 theories are actually different, but at long length scales, they flow to the same fixed point. Finally, the $O(n)$ model exists also as a field theory. When $S \mapsto \vec{S}$, $\phi \mapsto (\phi^1, \phi^2, \ldots, \phi^n)$ and the Hamiltonian is,

$$\mathcal{H} = \int d^d\underline{r}\ \left[\frac{1}{2}(\nabla\vec{\phi})^2 + \frac{t}{2}\vec{\phi}^2 + \frac{u}{4}\vec{\phi}^4\right]$$

4.2 Mean Field Theory

The mean field theory is useful to guess the phase diagram of the given model. It gives us exact universal exponents in high enough dimensions and remains a starting point for approximation in lower dimensions. By MFT here we refer to the saddle point treatment of the field theory. We thus find the field configuration that minimises the Hamiltonian. Evidently, such a configuration is constant in space, $\nabla\phi = 0$, and thus,

$$\mathcal{H} = L^d V(\phi), \qquad V(\phi) = \frac{t}{2}\phi^2 + \frac{u}{4}\phi^4$$

We can understand this through the plots of V for the cases where t takes either sign. When $t > 0$, near $\phi = 0$, $V \sim \phi^2$, when $t = 0$, $V \sim \phi^4$, and when $t < 0$ the potential has two minima. Thus $t > 0$ corresponds to disorder with $\langle\phi\rangle = 0$, and $t < 0$ corresponds to order with $\langle\phi\rangle \neq 0$. The case $t = 0$ is the critical point. To find the minimum value of $\langle\phi\rangle$ we minimise the potential,

$$\frac{\partial V(\phi)}{\partial\phi}\phi(t + u\phi^2) = 0 \iff \langle\phi\rangle = \sqrt{-\frac{t}{u}}$$

The above value gives the MF estimate for the order parameter exponent. $M \sim |T_c - T|^\beta \sim (-t)^\beta$ and therefore $\beta_{MF} = \frac{1}{2}$. We can compare that with what we found in the case of the 2D Ising model, where we had $\beta = X_h\nu$, and $X_h = \frac{1}{8}$, $\nu = 1$. This is strongly not mean field like! However in sufficiently high dimensions, the square root singularity becomes exact. For the free energy, we can read off the value of the energy from the approximation we made. $f = \mathcal{H}$ at saddle point, neglecting fluctuations. Thus,

$$\mathcal{H} = \begin{cases} 0 & t > 0 \\ -\frac{t^2}{4u} & t < 0 \end{cases}$$

For the specific heat, one can show that there is a jump at $t = 0$, since C is a constant on either side of t, and this corresponds to a critical exponent $\alpha_{MF} = 0$.

Correlation Functions

We will now go to first non-trivial order of the saddle point approximation by expanding around the saddle point solution to quadratic order, $\phi(\underline{r}) = \phi + \delta\phi(\underline{r})$. At $T = T_c$ ($t = 0$), $\phi = 0$. Thus, expanding to quadratic order,

$$\mathcal{H} = \frac{1}{2}\int d^d\underline{r}\ (\nabla\phi)^2$$

which is the usual *free, massless, field theory.* The Free field theory (FFT) is perhaps the most important RG fixed point. It describes the exact properties of

- Critical points in sufficiently high dimensions.
- Goldstone modes in ordered phases.
- Long-distance power law correlations of the XY model ($O(2)$ model) in d=2.
- Many 1D quantum models.

and many more. It is also a rare example where we can *solve* the fixed point exactly. It is thus a starting point for more complex fixed points, not only as a pedagogical example, but concretely, we can understand many fixed points by perturbing around the free fixed point.

Note. Fourier transform conventions:

$$\phi(\underline{r}) = \int d^d\underline{k}\ e^{-i\underline{k}\cdot\underline{r}}\phi(\underline{r})\phi(\underline{r}) = \int \frac{d^d\underline{k}}{(2\pi)^d}\ e^{i\underline{k}\cdot\underline{r}}\phi(\underline{k})$$

For a real $\phi(\underline{r})$, $\phi(\underline{k}) = \phi(\underline{-k})^*$. The Dirac delta function is,

$$\delta^d(\underline{k}+\underline{k}') = \int \frac{d^d\underline{k}}{(2\pi)^d}\ e^{i(\underline{k}+\underline{k}')\cdot\underline{r}}$$

The partition function in Fourier transform is then given by,

$$Z = \int \prod_k d\phi_k\ \exp\left[\int \frac{d^dk}{(2\pi)^d} k^2 |\phi(k)|^2\right]$$

and in the functional integral we integrate over the real and imaginary part of each ϕ_k avoiding double counting of $\phi(k)$ and $\phi(-k)$. The two-point function is then given by,

$$\boxed{\langle \phi(\underline{k}\phi(\underline{q})\rangle = (2\pi)^d \delta^d(\underline{k}+\underline{q})\frac{1}{k^2}}$$

A simple way to show this is by introducing the generating functional with a soure term J,

$$Z[J] = \int \mathcal{D}\phi(\underline{e})^{-\frac{1}{2}\int \frac{d^d}{(2\pi)^d}k^2|\phi(k)|^2 + \int \frac{d^dJ}{(2\pi)^d}(k)\phi(k)} = Z\langle e^{\int \frac{d^dJ}{(2\pi)^d}(k)\phi(k)}\rangle$$

Now, ϕ is a gaussian variable (it has quadratic weight) and for any gaussian variable ϕ, if A is a linear combination of components of ϕ,

$$\langle e^A\rangle = e^{\langle A\rangle + \frac{1}{2}\langle A^2\rangle}$$

In our case, $\langle\phi\rangle = 0$ by symmetry, and thus by applying the above formula we arrive at,

$$Z[J] = Ze^{\frac{1}{2}\int \frac{d^dk}{(2\pi)^d}\int \frac{d^dk'}{(2\pi)^d} J_k J_{k'}\langle\phi_k\phi_{k'}\rangle}$$

On the other hand we can alternatively compute the functional integral by completing the square,

$$Z[J] = Ze^{\frac{1}{2}\int \frac{d^dk}{(2\pi)^d}\frac{J(k)J(-k)}{k^2}}$$

so comparing the two results gives the two-point function as above. In real space, we find,

$$\langle\phi(0)\phi(\underline{r})\rangle = 2\int_{1/L}^{1/a} \frac{d^dq}{(2\pi)^d} e^{i\vec{u}\cdot\underline{r}}\frac{1}{q^2}$$

where the minimum momentum is set by the system size and the maximum momentum is set by the UV cutoff.

The behaviour of the two-point function depends on the dimension. If $d > 2$, the integral is finite as $L \to \infty$ and the result can be shown to be,

$$\langle\phi(0)\phi(\underline{r})\rangle \sim \begin{cases} r^{-(d-2)} & t = 0 \\ r^{-\frac{d-2}{2}}e^{-r/\xi} & t > 0, r \gg \xi \text{ with } \xi \sim t^{-1/2} \end{cases}$$

The mean field exponent is then $\nu_{MF} = \frac{1}{2}$. Expanding now around ϕ in the disordered phase,

$$\mathcal{H} = \frac{1}{2}\int \frac{d^dk}{(2\pi)^d}\,(k^2+t)\phi(k)\phi(-k)$$

so that the characteristic length scale is set by $1/\sqrt{t}$. In free field theory, the scaling dimension of ϕ is then,

$$\boxed{X_\phi = \frac{d-2}{2}}$$

We contrast with 2D Ising model where $X_s = \frac{1}{y}$. The FFT is a simple RG fixed point as we will explain later.

Note. Mean Field Theory does not obey the relationships between exponents that we found previously using the RG (eg $\beta_{MF} \neq \nu_{MF}X_\phi$). This is because MFT is pathological from the point of view of the RG. There are some assumptions about scaling that are made which are not true in the RG.

4.3 MFT examples

Liquid-gas transition

Imagine that we write a LG "Hamiltonian' for the liquid gas transition', and the quotes are due to the fact that the description is on a coarse-grained level. The Hamiltonian that we write really contains contributions from both energy and entropy, and it is therefore a free energy. We will stick to $\mathcal{H}$ though as a symbol for this action. This action is a function of the density ρ or more precisely, $\phi = \rho - \rho_c$. There is no symmetry for ρ, so the most general form that it takes is,

$$\mathcal{H} = \int (\nabla\phi)^2 + m\phi + t\phi^2 + h\phi^3 + w\phi^4 + \ldots$$

However, we can remove the linear term $m \mapsto 0$ by shifting ϕ by a constant. Coupling constant which can be eliminated by redefining the variables are called "redundant" couplings. Thus the potential is, $V(\phi) = t\phi^2 + h\phi^3 + w\phi^4$, and the phenomenology is now different. The behaviour of V is summarised by the plots below, where starting from the upper left plot, the temperature t is increased in an anticlockwise fashion. Generically, when there is a cubic term in the action, then we have a discontinuous first order phase transition of the order parameter.

In that case there are no power law correlation at t_c since the system will be sitting in one of the two minima. Expanding $\phi(r) = \phi_0 + \delta\phi$, we find that $\delta\phi$ has nonzero mass, leading to non-power law correlations. This is to be contrasted with the ϕ^4 case.

If we tune two parameters in the action, $t \to 0$ and $h \to 0$, the action becomes,

$$\mathcal{H} = \int (\nabla\phi)^2 + t\phi^2 + u\phi^4 + O(\phi^5)$$

In MFT, $O(\phi^5)$ terms are unimportant at small t since then ϕ is small and those terms are subleading and negligible. Thus higher order terms do not change the ϕ^4 theory **close to** $h = 0, t = 0$. The transition as we vary t has the same exponents as the MFT of the Ising transition. This is a version of universality: higher powers of ϕ are irrelevant to asymptotic scaling behaviour at MF Ising critical point. Thus, the MF treatment leads to the guess that the liquid-gas critical endpoint lies in the Ising universality class. It is only guess for us, but it can be confirmed by more careful RG treatment.

Tricritical Ising Model

In the Landau theory, we consider a Landau functional with potential,

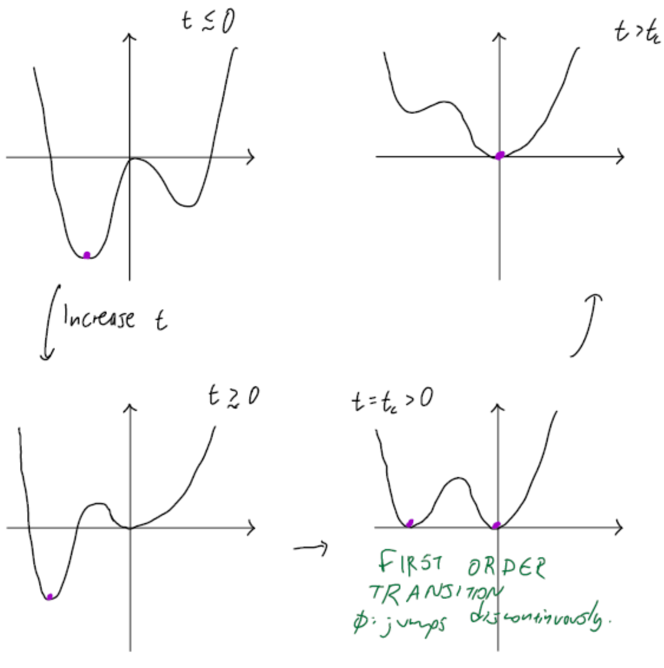
$$V(\phi) = t\phi^2 + u\phi^6 + v\phi^6$$

so we retain the $\mathbb{Z}_2$ symmetry, and we can vary 2 parameters. This microscopic model could correspond to an Ising model with vacancies, so the "spin" can take values $s = -1, 0, 1$, and the Hamiltonian is,

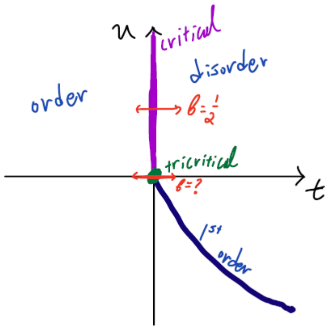
$$-\mathcal{H} = J \sum_{\langle ij \rangle} s_i s_j + \Delta \sum_i s_i^2$$

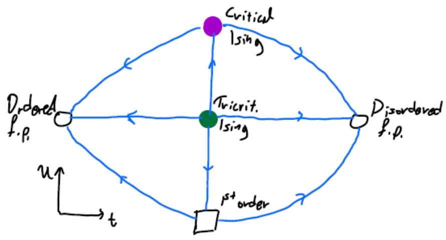
With two parameters to control, the phase diagram is indeed richer, and we can understand with mean field theory. We will not show how to translate between the microscopic model and the Landau-Ginzburg action but we can assume that by using the two parameters in $\mathcal{H}$ we can control the signs of t, u in $V(\phi)$. We further assume $v > 0$.

Previously, we had $u > 0$ so that the $v\phi^6$ term can be neglected. In that case, tuning t from positive to negative gives the same behaviour as before, with critical point at $t = 0$. When $u < 0$ we have a first order transition: at $t = 0$, there are two minima, and as we increase t, the values of the minima are pulled up until we have three degenerate minima.



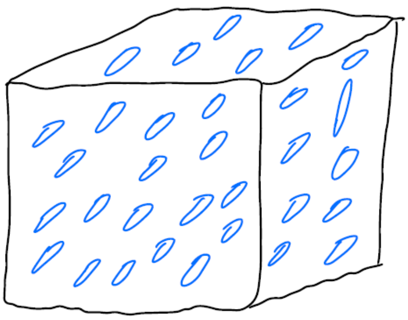
It turns out that $t_c = \frac{u^2}{4v}$, which is the point where the first order transition between disorder phase ($\phi = 0$) and order $\phi = \pm\phi_0$. It is a first order transition because even at t_c, there is finite jump of the order parameter. We can thus draw a phase diagram (for small t, u) and it will look like the one below. At $t = u = 0$, the potential is $\sim \phi^6$ with a single minimum at $\phi = 0$. This point corresponds to the *tricritical Ising* point, with the terminology originating from the fact that there are 3 relevant couplings, t, u, h. At the tricritical point, there is a different critical exponent. This suggests an RG flow diagram as in the figure below.

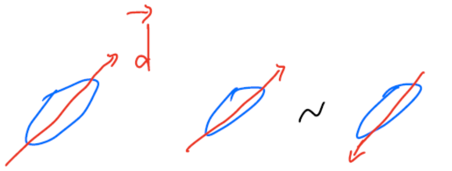




Nemantic liquid crystals

A liquid crystal is a state of matter with more ordered than a liquid but less order than a crystal. A nematic liquid crystal can be thought of as a dense fluid of rod-like molecules, with a preferred axis.





It can have spontaneous long range order of axes. In order to define an order parameter, we can define a vector $\vec{d}$ parallel to the axis of the molecule, such that $|\vec{d}|^2 = 1$. This is similar to the $O(3)$ model but it is not the same because here two vectors are identified, $\vec{d} \sim -\vec{d}$. For example, if we write a Landau theory with a term like $(\nabla \vec{d})^2$, this would artificially give energy to configuration which are microscopically the same, parallel and antiparallel rods for example. Essentially, the sign of $\vec{d}$ is a gauge choice, and the physical order parameter should be gauge invariant. Thus, a good order parameter should be quadratic in $\vec{d}$, like

$$Q_{ab} = d_a d_b - \frac{d^2}{3}\delta_{ab}$$

which contains all information in $\vec{d}$ except its sign, and $\mathrm{Tr}\{Q\} = 0$. It is symmetric under $d_a \to R_{ab} d_b$ which for the tensor Q reads, $\underline{Q} \to \underline{R}\underline{Q}\underline{R}^T$.

A Landau Ginzburg functional should then contain terms which are invariant under the symmetries of Q,

$$\mathcal{H} = \frac{1}{2}\operatorname{Tr}(\nabla Q)^2 + \frac{t}{2}\operatorname{Tr} Q^2 + \frac{g}{3}\operatorname{Tr} Q^3 + \frac{u}{4}\operatorname{Tr} Q^4 + \frac{u'}{4}(\operatorname{Tr} Q^2)^2$$

where t corresponds to the mass term, and the presence of the cubic term makes $\mathcal{H}$ unbounded (meaning arbitrarily large free energy) and therefore the addition of the quartic terms was necessary to make the theory stable. Note that $V(Q)$ is only a function of the eigenvalues of Q, which by the traceless condition must be $(\lambda_1, \lambda_2, -\lambda_1 - \lambda_2)$. Before estamblishing long range order all directions of $\vec{d}$ are equivalent, but after long range order gets singled out, so the ordering pattern which we are interested in is really $\left(\lambda, -\frac{\lambda}{2}, -\frac{\lambda}{2}\right)$. This is because the two perpendicular axes are equivalent and the eigenvalues corresponding to those are degenerate.

Thus, for a Landau theory to be able to describe this transition the coefficients g, u, u' should be such that as t is varied, we reach a minimum with eigenvalues as we described and a minimum where rotational symmetry is unbroken (all eigenvalues are equivalent), and thus $Q = 0$, corresponding to a disorder state. This transition is first order because of the cubic term and the minimum jumps discontinuously as t is varied.

The nematic liquid is interesting and actually different from the $O(3)$ model because it has "topological defects", singularities of line in space where $Q = 0$ (in the continuum limit) and $\vec{d}$ is undefined, and there is a "winding" of the order parameter as we go around the defect. Looking at a slice of the 3D block, the orientation of the rods would look like the figure below. Such a configuration is not possible in the $O(3)$ model: if we are to assign $\vec{d}$ vectors for the configuration, there would be a discontinuity which would cost a lot of energy. But in the nematic liquid this doesn't cost any energy since $\vec{d} \sim -\vec{d}$. This configuration is a topologically stable vortex: it cannot disappear by local deformations of the configuration.

Note. The $O(3)$ symmetry here is not an "internal" symmetry. It is about spatial rotations, $\underline{Q} \to \underline{R}\underline{Q}\underline{R}^T$ and $\vec{r} \to \underline{R}\vec{r}$. Thus any term that is invariant under the combined transformation is allowed. A good exercise would be to figure out the simplest additional terms which are allowed by this combined symmetries.

4.4 Free Field Theory: RG

Recall that perturbing the free field theory by including a mass term,

$$\mathcal{H} = \frac{1}{2}(\nabla\phi)^2 + \frac{t}{2}\phi^2$$

and by dimensional analysis we can define a length scale $\xi^{-2} = t$ which is the correlation length. If $t = 0$ ($\xi = \infty$) or if $r \ll \xi$ then $\langle\phi(0)\phi(r)\rangle \sim A/r^{d-2}$. If $t > 0$ and $r \gg \xi$, $\langle\phi(0)\phi(r)\rangle \sim r^{\frac{d-1}{2}} e^{-r/\xi}$.

Note. Note that for $d = 2$,

$$\langle\phi(0)\phi(r)\rangle \sim \frac{1}{\sqrt{r}} e^{-r/\xi}$$

> This formula is familiar from the high temperature expansion of the Ising model (corresponding to positive t here), which we computed as a partition function by summing over directed paths. Indeed we can think of the 2-point function of free field theory as a sum over Browinian paths.

To compute the above we refer to Tong's lecture notes in *Statistical Field Theory* (p.42). But as a sketch of the calculation, it will reduce to the computation of the integral,

$$\int \frac{d^dk}{(2\pi)^d} \frac{e^{ik\cdot r}}{k^2+\xi^{-2}}$$

where the limits of the integral are from 0 to ∞. This is allowed in $d>2$ where we do not have an IR divergence. For large r the upper limit is also justified. The next step is using the identity,

$$\frac{1}{\alpha} = \int_0^\infty dt\; e^{-\alpha t}$$

This reduces the integral over k to just a Gaussian integral, and the t integral can be computed using the saddle point approximation, justified by the large $|r|$.

Recall now how we defined the block Hamiltonian on the lattice,

$$e^{-\mathcal{H}'[s']} = \sum_{\substack{s \\ (\mapsto s')}} e^{-\mathcal{H}[s]}$$

with s' the coare grained spins on scale $b\times a$, for a the lattice spacing. Also, recall that $s(r)\sim b^{-X_h}s'(R)$, with $R=r/b$. The continuum analogue is as follows. We write the field,

$$\phi(r) = \int \frac{d^dk}{(2\pi)^d}\; e^{i\underline{k}\cdot\underline{r}}\phi(\underline{k})$$

with $\int = \int_{|k|\leq\frac{2\pi}{a}}$, the momenta are bounded by the UV cutoff given by the lattice spacing. We will split the Fourier integral into two terms, depending on the magnitude of the momenta,

$$\phi(r) = \phi_<(r) + \phi_>(r)$$

where $\phi_<(r)$ contains the momenta which are small than the new UV cutoff (corresponding to a larger length scale), $|k|\leq\frac{2\pi}{ab}$, and $\phi_>(r)$ contains the remaining modes, $|k|\in\left(\frac{2\pi}{ab},\frac{2\pi}{a}\right)$

In analogy with the lattice model, integrating over the short wavelength modes with the long wavelength modes remaining fixed, we replace,

$$\sum_{\substack{s \\ (\mapsto s')}} \mapsto \int \mathcal{D}\phi_>(r)$$

The analogue of the coarse grained spin s', it is not simply $\phi_<$. It has to be rescaled by a rescaling a rescaling factor since $R=r/b$,

$$\phi'(R) = b^{X_\phi}\phi_<(r).$$

This agrees with what we had on the lattice when $s(r)\sim b^{-X}s'(R)$ (an expression for the long distance correlation functions). Considering long distance correlation functon for the free field theory, only low momentum part is important, so $\phi(r) \sim \phi_<(r) \sim b^{-X_\phi}\phi'(R)$. X_ϕ is determined by the requirement that the RG transformation leaves $\mathcal{H} = \frac{1}{2}\int(\nabla\phi)^2$ invariant. We will see this explicitly later.

The steps for performing Wilsonian RG in analogy with the lattice RSRG are as follows.

1. Integrate over $\phi_>$ to get an effective $\mathcal{H}$ for $\phi_<$ only. This is only trivial for the free field theory.

2. Rescale lengths by a factor of b so that the UV cutoff is restored to a, so we can compare "like with like".

Implementation for the FFT

The key property of the free field theory is that different momenta decouple. This allows us to write the partition function as,

$$Z = \int \mathcal{D}\phi_> \mathcal{D}\phi_< \; e^{-\mathcal{H}_>[\phi_>]} e^{-\mathcal{H}_<[\phi_<]}$$

This makes step 1 trivial. A generalised version of the Hamiltonian is,

$$\begin{aligned}\mathcal{H} &= \int d^d r \; \left(\frac{1}{2}(\nabla\phi)^2 + t\phi^2 + h\phi\right) \\ (\text{FFT}) \to &\int \frac{d^d k}{(2\pi)^d} \; \left(\frac{1}{2}(k^2+t)|\phi(k)|^2 + h\delta(k)\phi(k)\right)\end{aligned}$$

Now, according to the first step of the Wilsonian RG and using Eq. 4.4, we will integrate out the large k modes. Since the momenta are decoupled, this is simple, and in analogy with the lattice,

$$\begin{aligned}e^{-\mathcal{H}'_>[\phi_<]} = &= \int \mathcal{D}\phi_> \; e^{-\mathcal{H}[\phi_< + \phi_>]} \\ &= \int \mathcal{D}\phi_> \; e^{-\mathcal{H}_>[\phi_>]} e^{-\mathcal{H}_<[\phi_<]}\end{aligned}$$

which means that $\mathcal{H}'$ is just a constant multple of $\mathcal{H}$,

$$\mathcal{H}' = \int d^d r \; \left(\frac{1}{2}(\nabla\phi_<)^2 + t\phi_<^2 + h\phi_<\right)$$

The next step is to restore the UV cutoff since when we integrate the high frequency part, it got rescaled. As usual, $R = r/b$, thus,

$$\mathcal{H}' = \int d^d R \; b^d \left(\frac{1}{2}(\nabla_R\phi_<)^2 \cdot b^{-2} + t\phi_<^2 + h\phi_<\right)$$

This tells us that the free theory is a fixed point. To see this, we need to choose the "right" normalisation for the field ϕ. Therefore we define the coarse grain field as $\phi' = b^{\frac{d-2}{2}}\phi_< = b^{X_\phi}$[3], chosen such that the coefficient of the kinetic term is restored to its previous value,

$$\mathcal{H}'[\phi'] = \int d^d r \; \left(\frac{1}{2}(\nabla\phi')^2 + tb^2\phi'^2 + hb^{\frac{d+2}{2}}\phi'\right)$$

[3]Note that the exponent X_ϕ is the same as what we encountered before, $\langle\phi\cdot\phi\rangle \sim r^{-2X_\phi}$

so we see that the other coefficients have picked up extra factors of b. More explicitly, the RG eigenvalues of t and h are b^2 and $b^{\frac{d+2}{2}}$ respectively. In terms of the partition function,

$$Z[t, h; a, L] = Z[tb^{y_t}, hb^{y_h}; a, L/b]e^{-f_{UV}L^d}$$

where f_{UV} is the constant contribution coming from performing the integral above.

We see that $t = 0, h = 0$ is a fixed point, with relevant eigenvalues. We note that the values we obtained are consistent with general relation,

$$y_h = d - X_\phi = d - \frac{d-2}{2} = \frac{d+2}{2}$$

Similarly for y_t, but for that we need the scaling dimension of ϕ^2, X_{ϕ^2}.

Infinitesimal RG

So far, all the RG transformations we considered were done with some finite rescaling factor. In momentum space, we can perform a rescaling with an infinitesimal RG "step". We take $b = e^{d\tau}$ with the terminology that τ is the *RG time*, so that τ parametrises the amount of rescaling done. After an RG time τ, the total rescaled length is $\ell = e^\tau a$. Then,

$$\begin{aligned} u' &= b^{y_u}u + O(u^2) \\ u + du &= e^{yd\tau}u + O(u^2) = u + uyd\tau + O(u^2), \end{aligned}$$

so that,

$$\frac{du}{d\tau} = yu + O(u^2)$$

Generally, with more couplings, the RG equations take the form,

$$\frac{dK_\alpha}{d\tau} = -\beta_\alpha(\underline{K})$$

with β_α being the *beta function.* Near the fixed point $\underline{K}^*$, we linearise to get,

$$\frac{d\delta K_\alpha}{d\tau} = T_{\alpha\beta}\delta K_\beta(\underline{K})$$

with $T_{\alpha\beta} = -\frac{d\beta_\alpha}{dK_\beta}\Big|_{K^*}$. As usual, we diagonalise T to get the scaling variables and the eigenvalues of T are the y_α. This explains why we initially called $y's$ the RG eigenvalues.

4.5 Scaling Operators at the Gaussian fixed point

We focus here on $d > 2$ because $d = 2$ is special and requires a different treatment. Recall that the idea of a scaling operator is that they have a pure power law decay. An example is the field itself in free field theory. We normalise the field so that

$$\langle\phi(r)\phi(r')\rangle = \frac{1}{|r - r'|^{d-2}}$$

We say that ϕ is a scaling operator with scaling dimension $X_\phi = \frac{d-2}{2}$. Using ϕ we can build more scaling operators by taking powers of ϕ.

Note. For any Gaussian random variable ϕ,

$$\langle \phi_i \phi_j \phi_k \dots \rangle = \sum_{\text{pairing}} \langle \phi\phi \rangle \langle \phi\phi \rangle \dots$$

which means that we group the variables in $\langle \cdot \rangle$ into pairs. We also assumed $\langle \phi \rangle = 0$ for simplicity. For example, if x is a Gaussian random variable,

$$\begin{aligned} \langle x^4 \rangle &= \langle xxxx \rangle \qquad \text{(3 ways to pair them)} \\ &= \overbracket{xx}\,\overbracket{xx} + xxxx + xxxx \\ &= 3 \langle x^2 \rangle^2 \end{aligned}$$

In our case, the contractions for the two point function of ϕ give,

$$\overbracket{\phi(r)\phi(r')} = \langle \phi(r)\phi(r') \rangle = G(r - r') = \frac{1}{|r - r'|^2}$$

Consider $\langle \phi^2(0)\phi^2(r) \rangle$.

$$\begin{aligned} \langle \phi^2(0)\phi^2(r) \rangle &= \overbracket{\phi(0)\phi(0)}\,\overbracket{\phi(r)\phi(r)} + 2\phi(0)\phi(0)\phi(r)\phi(r) \\ &= G(0)^2 + 2G(r)^2 \end{aligned}$$

This is not a scaling operator since its two point function is not a pure power law and it contains a constant piece. We can fix this by *normal ordering*, which we define by

$$:\!\phi^2(r)\!: = \phi^2(r) - \overbracket{\phi(r)\phi(r)}$$

with the idea being that if we compute a correlation function using the normal ordered $:\!\phi^2\!:$ and something else, we get,

$$\begin{aligned} \langle :\!\phi^2\!:(\dots) \rangle &= \langle \phi^2(\dots) \rangle - \overbracket{\phi\phi} \langle (\dots) \rangle \\ &= \sum \text{Contractions where the two } \phi\text{'s are paired with stuff in}(\dots) \end{aligned}$$

Thus, normal ordering eliminates self-contractions. Considering now the two-point function,

$$\langle :\!\phi^2(0)\!: :\!\phi^2(r)\!: \rangle = \frac{2}{|r - r'|^{2X_{\phi^2}}}$$

where $X_{\phi^2} = d - 2$, which is consistent with the RG eigenvalue of the conjugate coupling t, $y_t = d - X_{\phi^2} = 2$.

Similarly, we can normal order higher powers, eg

$$\begin{aligned} :\!\phi^3\!: &= \phi^3 - 3\overbracket{\phi\phi}\phi = \phi^3 - 3G(0)\phi \\ :\!\phi^4\!: &= \phi^4 - \binom{4}{2}\overbracket{\phi\phi}\phi^2 - 3\overbracket{\phi\phi}\,\overbracket{\phi\phi} \end{aligned}$$

The normal ordered powers of ϕ provide a basis for scaling operators. More precisely, this is true for *some* of the scaling operators since there exist scaling operators which are derivatives. The basis $\{:\phi^k:\}$ is related to the "naive" operator basis $\{\phi^k\}$ by a triangular matrix. We can write,

$$\boxed{O_k(r) =: \phi(r)^k : \text{ has dimension } X_k = \frac{d-2}{2}k}$$

The free field theory is a rare example where we can write the full spectrum of scaling operators explicitly.

4.6 Perturbation of Free Field Theory

We will now think about the stability of the Gaussian fixed point. For the Ising case, as we claimed above, the corresponding field theory is ϕ^4 which is relevant for $d < 4$ and irrelevant for $d > 4$. The Hamiltonian rewritten in the scaling operator basis is,

$$\mathcal{H}_* + \frac{t}{2}\phi^2 + \frac{u}{4}\phi^4 = \mathcal{H}_* + \left(\frac{t}{2} + \frac{6u}{4}G(0)\right) :\phi: + \frac{u}{4} :\phi^4: + \text{ const.}$$

and from $y_k = d - k \cdot \frac{d-2}{2}$, so that $y_4 = 4 - d$, which justifies the comment above. For ϕ^6, we find that $y_6 = 6 - 2d$, so that it is irrelevant above $d = 6$. Both of these are *marginally irrelevant* in the critical value of the dimensionality. For example, in ϕ^4, the RG equation is,

$$\frac{du}{d\tau} = (4-d)u - (\dots)u^2$$

where $(\dots)$ is a positive number. Thus, although it's a marginal eigenvalue, it still tends to zero under RG. We will justify this later.

Generalisation to $O(n)$ model

To generalise to the $O(n)$ model, we need to take symmetry into account. For $n = 1$, we just needed to take only one power of ϕ into account (in the absence of derivatives), but there are more terms for general n. We will classify the terms under their $O(n)$ symmetry, or in other words, what the representation of the symmetry is.

- Singlets: $:\vec{\phi}^2:, :(\vec{\phi}^2)^2:, \dots$
- Vector: $:\vec{\phi}:, \dots$
- Symmetric traceless matrix: $M_{ab} = :\phi_a\phi_b - \frac{1}{n}\delta_{ab}\phi^2:$

and so on. In free field theory, many of these operators have the same scaling dimension, since in that case the scaling dimension only depends on powers of ϕ. For example, $:\vec{\phi}:$ and M_{ab} have the same dimension. This is not true in an interacting field theory. In general at an interacting fixed point they can have diferent scaling dimensions.

Operators can also contain derivatives. For example, in the Ising model had the simple emergent symmetry of continuous spatial symmetries, in the continuum. To check that this is true, we need to see if the corresponding perturbation is irrelevant or relevant. The leading perturbation allowed by the presence of the hypercubic lattice (assuming $d > 4$ so that ϕ^4 is also irrelevant) is,

$$\sum_{\mu=1}^{d} (\partial_\mu^2 \phi)^2$$

which is invariant only under discrete spatial rotations. This can now be decomposed into representations of the spatial symmetries. Every derivative increases the dimension X by 1 so that this operator is highly irrelevant. This is also true at the interacting Ising fixed point. Since it is irrelevant, we get a continuous rotational symmetry in the IR limit.

4.7 Validity of MFT for $O(n)$ model

We haven't yet justified the usage of the saddle point approximation in the context of MFT. Mathematically, we know that MFT is justified when there is a large parameter that suppresses the fluctuations around the saddle point and the integral gets all the contributions from around the saddle point. We will see here that for high enough dimensions, under the RG we will acquire such a large parameter. The main requirements that we expect from a MFT is that,

1. Is the phase diagram valid, i.e. does MFT predict the existence of an ordered phase?
2. Are the prediction for the exponents valid?

We start with the second point. In $d \geq 4$ MFT is valid because $u \to 0$ under the RG. We first look at the correlators at the critical point. In saddle point treatment we computed $\langle \phi\phi \rangle$ using the Gaussian action, $\mathcal{H} = \int (\nabla\phi)^2$, and this is justified by the fact that $u \to 0$. What about the magnetisation? We compute the magnetisation by minimising V. We cannot just set $u = 0$ in MFT since this would lead to a contradiction in the ordered phase, when $t < 0$, and thus the potential would be unbounded. We refer to u as a *dangerously irrelevant* variable. Although at some very large length scale, u has become very small (but non zero), so it does not necessarily make sense to set it to zero.

We start with $\mathcal{H}_* + t :\phi^2: + u :\phi^4:$, where $\mathcal{H}_* = (\nabla\phi)^2$ and we perform the RG by some factor $b \gg 1$ to get,

$$\mathcal{H} = \int d^d R \; \left[(\nabla\phi')^2 + tb^2 :\phi'^2: + ub^{4-d} :\phi'^4: \right]$$

With a change of variable to make the use of saddle point obvious, $\phi' = \psi b^{\frac{d-4}{2}}$, so that the kinetic and quartic terms end up with the same power of b,

$$\mathcal{H} = b^{d-4} \int d^d R \; \left[(\nabla\psi)^2 + tb^2 :\psi^2: + u :\psi^4: \right]$$

Now we choose b so that $|tb^2| = \mathcal{O}(1)$. Since $t \ll 1$ then $b \gg 1$, and we note that $b = \xi = |t|^{-\nu_{MF}}$, and $\nu_{MF} = 1/2$. Finally the Hamiltonian takes the form,

$$\mathcal{H} = \xi^{d-4} \int d^d R \; [\dots\dots\dots]$$

where the prefactor is large when $d > 4$ and $[\dots]$ represents some $\mathcal{O}(1)$ parameters when t is small. This is precisely the form that justifies the saddle point approximation: computing $\langle \phi \rangle$ (or $\langle \psi \rangle$) by minimising $V(\phi)$. In the ordered phase, we find

$$|\psi| = \sqrt{\frac{|tb^2|}{u}} \sim \frac{1}{\sqrt{u}}$$

and we can relate this to ϕ by the definition above.

Tricritical Ising model

Recall now the phase diagram of the tricritical Ising model with potential,

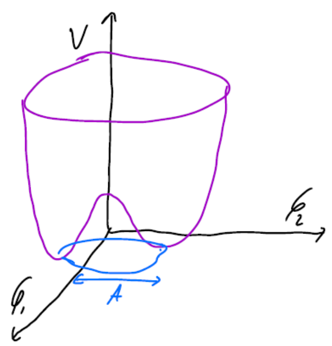
$$V(\phi) = t\phi^2 + u\phi^4 + v\phi^6, (v > 0)$$

The $(0, u), u > 0$ line is Ising criticality given by the ϕ^4 theory, and there is another first order transition in the $t > 0, u < 0$ section. These lines separate order from disorder and there is tricritical point sitting at $(0, 0)$ described by ϕ^6 theory. The Ising critical line is MF-like for $d \geq 4$. The tricritical point if MF-like for $d \geq 3$ with logarithmic divergences in $d = 3$.

4.8 Stability of ordered phase for $n > 2$

When $n > 2$ there is a continuous symmetry. The MFT gives the same phase diagram structure for all n. But of course $n \geq 2$ is fundamentally different due to the continuous symmetry. What's more is that $n = 2$ is different from $n > 2$ as the symmetry is then abelian, but we ignore this for now. The question we want to understand is when the ordered phase is stable.

For $n = 2$ the potential $V(\phi)$ has a circle of minima. We write $\vec{\phi} = (A + \delta A)(\cos\theta, \sin\theta)$, where A is the magnitude of the minimum.



We find that,

$$(\nabla\vec{\phi})^2 = (\nabla\delta A)^2 + (A + \delta A)^2(\nabla\theta)^2$$

and the Hamiltonian becomes,

$$\mathcal{H} = \frac{A^2}{2}(\nabla\theta)^2 + \left[\frac{(\nabla\delta A)^2}{2} + \frac{\mu}{2}(\delta A)^2 + \dots\right] + \left[\sim (\delta A)(\nabla\theta)^2 + \dots\right]$$

so that the Hamiltonian is interacting. Neglectic the interactions, we see that δA is a massive field since its mass which is relevant will become more important under RG. The expectation is that we can integrate out the amplitude field, in the cost of introducing effective interactions for θ. Since δA is a massive field, it has short ranged correlators which are exponentially decaying for fixed θ, and thus the effective interactions in θ are also short ranged, which will lead to renormalisation of the coupling. The effective Hamiltonian for θ is,

$$\boxed{\mathcal{H}_{\text{eff}} = \frac{K}{2}\int (\nabla\theta)^2 \; d^d r}$$

with K the *stiffness*.

This is *almost* a free field theory. The difference is that $\theta = 0$ and $\theta = 2\pi$ are identified. This means that there are configurations which are allowed here but not allowed in free field theory. These are called "vortices" and they look like the picture below. These are singularities of θ where $\phi = 0$ and as we go around them θ acquires a winding of 2π. This is not allowed in free field theory since this would be a discontinuity between $0 \to 2\pi$. If these vortices can be neglected, $\mathcal{H}$ is basically a free field theory. The neglect will be okay if the stiffness $K \to \infty$ under RG since the fluctuations are then suppressed and the vortices becomes very rare.

Recall that under the RG, if we define $\theta' = b^X \theta_<$, with $X = \frac{d-2}{2}$, then $\mathcal{H}_{\text{eff}}$ is invariant under the transformation. However, the range of θ changes from $[0, 2\pi] \longrightarrow [0, 2\pi b^X]$. It is thus more convenient to change the convention we are using here. To keep the range fixed, we do not rescale θ and use a domain $[-\pi, \pi]$. What happens then is that the stiffness flows to a different value,

$$\boxed{\mathcal{H}_{\text{eff}} = K b^{2X}\int (\nabla\theta)^2 \; d^d r}$$

If $d > 2$ then $Kb^{d-2} \to \infty$ as $b \to \infty$, and thus the neglect of vortices is justified and the field is ordered. The fluctuations of the renormalised field θ' don't explore the full compact space $[-\pi, \pi]$. The order parameter gets better and better localised at one value of θ as we do RG. This means that, there is order and thus θ is a Goldstone mode whose correlation function can be calculated using free field theory.

For $d < 2$ there is no long range order since the coupling gets weaker under RG. For $d = 2$ we need to analyse separately, and what we'll find is that for $n = 2$ there is a phase where FFT correctly gives correlation functions, even though there is no long range order.

Long Range Order and Correlation functions

Assume θ is a goldstone mode in the ordered phase, governed by FFT. To check if this is consistent we calculate the fluctuations,

$$\begin{aligned}\langle[\theta(0) - \theta(r)]^2\rangle &= 2G(0) - 2G(r) \\ &= (\text{const.}) - \frac{\text{const.}}{r^{d-2}}\end{aligned}$$

and find that this they are bounded for $r \to \infty$ when $d > 2$. This is consistent with long range order (LRO). For $d \leq 2$ we find this this quantity diverges.

4.9 Upper and lower critical dimension

We define the upper and lower critical dimension by the example of the Ising and $O(n)$ models as below. At the lower critical dimension and below, there is no ordered phase. In the $O(n)$ model, the LCD is $d = 2$, whereas in the Ising model it is $d = 1$. The order in the Ising model is more stable by the existence of domain walls. This is not possible in $O(n)$ models where there can be smooth twists of the spins. The upper critical dimension is the dimension above which MFT exponents are valid. For both $O(n)$ and Ising, the UCD is $d = 4$.

Before moving on, we summarise what we found so far. So saw that MFT gives correct exponents for $d \geq 4$ but not $d < 4$, and that the ordered phase is destroyed by Goldstone modes when $d \leq 2$ if $n \geq 1$. The large n limit will give us a regime where we can go beyond MFT for critical exponents in $2 < d < 4$ and it will also show us qualitatively what happens in $d = 2$ for $n \geq 3$ where the symmetry is non-abelian. In the latter case the Goldstone modes are interacting.

4.10 Operator Product Expansion (OPE)

For a general fixed point (not just the Gaussian fixed point), we have a list of scaling operators $O_\alpha(r)$. In free field theory this list included the normal ordered powers of ϕ. We consider scalar operators (operators that have no spatial index),

$$\langle O_\alpha(0) O_\alpha(r) \rangle \sim r^{-2X_\alpha}$$

An example of a non-scalar operator is $\partial_\mu \phi$ in free field theory. More generally we can consider correlation functions between different operators,

$$\langle O_\alpha(r) O_\beta(r') \Phi \rangle$$

where Φ denotes other operators at distance much larger than $|r - r'|$. The idea of the OPE is that in such correlation functions we can expand the product $O_\alpha(r) O_\beta(r')$ as a sum of single operators,

$$O_\alpha(r) O_\beta(r') \sim \sum_\gamma C_{\alpha\beta\gamma}(r - r') O_\gamma \left(\frac{r + r'}{2} \right)$$

The point is that $\{O_\gamma\}$ is by definition a basis for all *local* operators. Local operators are operators that depend on sites on a finite patch of a system. The product on the LHS in the equation above is also a local operator since $|r - r'|$ is finite. Another way of thinking about it is that after coarse-graining to a scale of the order $\sim |r - r'|$, $O_\alpha O_\beta$ looks local on the scale of the UV cutoff. The coefficients $C_{\alpha\beta\gamma}(r - r')$ are independent of Φ.

Note. $\sum_\gamma$ will always include derivative operators that don't contribute to $\mathcal{H}$. To see this we note that we can expand the operator around r so that,

$$O_\gamma \left(\frac{r + r'}{2} \right) = O_\gamma(r) + \left(\frac{r - r'}{2} \right)_\mu \partial_\mu O_\gamma(r) + \dots$$

We can fix the form of the constants $C_{\alpha\beta\gamma}(r-r')$ by require the OPE to be consistent with the behaviour of O_α under RG. For simplicity we consider only scalar operators and work with $O_\alpha(0)O_\beta(r)$. We will first apply the OPE and coarse grain and then rederive this by coarse graining and then apply the OPE. We start by coarse-graining by a scaling factor $b=r$ and set the UV cutoff $a=1$,

$$O_\alpha(0)O_\beta(r) \xrightarrow{\text{RG}} r^{-X_\alpha-X_\beta}O_\alpha(0)O_\beta(1) \xrightarrow{\text{OPE}} r^{-X_\alpha-X_\beta}\sum_\gamma O_\gamma\left(\frac{1}{2}\right)C_{\alpha\beta\gamma}(1)$$

Performing the operations in the opposite way,

$$O_\alpha(0)O_\beta(r) \xrightarrow{\text{OPE}} \sum_\gamma C_{\alpha\beta\gamma}(r)O_\gamma\left(\frac{r}{2}\right) \xrightarrow{\text{RG}} \sum_\gamma C_{\alpha\beta\gamma}(r)r^{-X_\gamma}O_\gamma\left(\frac{1}{2}\right)$$

Comparing the two final expressions, we can define $C_{\alpha\beta\gamma}(1)=c_{\alpha\beta\gamma}$ to be the *OPE coefficients*, to get,

$$\boxed{C_{\alpha\beta\gamma}(r) = \frac{c_{\alpha\beta\gamma}}{r^{X_\alpha+X_\beta-X_\gamma}}}$$

Conventionally, we fix the normalisation of the operators O_α by requiring that $\langle O_\alpha(0)O_\alpha(r)\rangle \sim 1/r^{2X_\alpha}$, so the set of numbers $\{c_{\alpha\beta\gamma}\}$ is a set of universal constants.

RG equations

The point now is that the OPE coefficients determine the RG equations to order u_α^2. With $b=e^\tau$ we have,

$$\frac{du_\alpha}{d\tau} = y_\alpha u_\alpha - \frac{1}{2}S_d\sum_{\beta\gamma}c_{\beta\gamma\alpha}u_\beta u_\gamma + O(u^3)$$

with S_d the volume of a $d-1$ sphere, $S_d=\frac{2\pi^{d/2}}{\Gamma(d/2)}$ (although this is not very important here). In a certain RG scheme, we write the partition function,

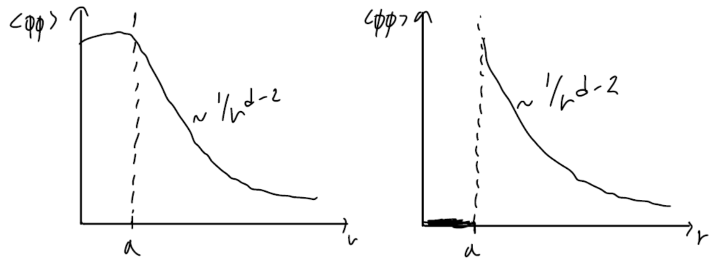
$$Z = \int \mathcal{D}\phi e^{-\mathcal{H}_*[\phi]-\sum_\alpha u_\alpha\int O_\alpha(r)d^dr}$$

and we expand in $\{u_\alpha\}$,

$$Z = \int \mathcal{D}\phi e^{-\mathcal{H}_*[\phi]}\left(1-\sum_\alpha u_\alpha\int O_\alpha(r)\ d^dr + \frac{1}{2}\sum_{\alpha\beta}u_\alpha u_\beta\int O_\alpha(r)\ d^dr\int O_\beta(r')\ d^dr' + \dots\right)$$

We can think of the partition function as an infinite sum of the correlation functions in the fixed point theory $\mathcal{H}_*$. Each term is an integral of a correlation function. We should also think about the UV cutoff. The effect of the cutoff is captured entirely by the two-point function. This is true and known of course in free field theory where all correlation functions can be expressed in terms of the two-point function. What the cutoff is doing is to regulate a divergence in the two point function.

In Cardy's book what is done is to implement the UV cutoff in an ad hoc way by restricting the integrals in the infinite expansion to separations which are $\geq a$. Correlation functions $\langle\phi\phi\rangle$ for $r<a$ are set to 0. At the M^{th} order, there are M coordinates and we impose the rule that they



cannot get closer than a. This is just regularisation and what Cardy claims is that since it doesn't matter how it is done, we choose the simpler way.

We now perform the RG, so we change the UV cutoff from a to ba. This means that certain configurations which were allowed previously (eg at a distance a) are not allowed now since their separation is smaller than $ae^{\delta\tau}$, and thus these configurations must be removed. To do this we will use the OPE,

$$O_\alpha(\underline{r})O_\beta(\underline{r}') = \sum_\gamma \frac{c_{\alpha\beta\gamma}}{a^{X_\alpha+X_\beta-X_\gamma}} O_\gamma\left(\frac{\underline{r}+\underline{r}'}{2}\right)$$

We will apply this idea in the $4-\epsilon$ expansion later in the notes.

Let us do this more explicitly. We consider a Hamiltonian which is of the form,

$$\mathcal{H} = \mathcal{H}_* + \sum_\alpha u_\alpha \int d^d r \; O_\alpha(r)$$

where O_α is a scaling operator and $\mathcal{H}_*$ is a fixed point Hamiltonian. We can normalise the scaling operators so that,

$$\langle O_\alpha(0) O_\alpha(r)\rangle = \frac{1}{r^{2X_\alpha}}$$

and with this normalisation, the dimension of O_α is $[O_\alpha] = (\text{length})^{-X_\alpha}$ and the $\{u_\alpha\}$ are dimensionless. The OPE coefficients are defined by the relation,

$$O_\alpha(0)O_\beta(r) \sim \sum_\gamma \frac{c_{\alpha\beta\gamma}}{r^{X_\alpha+X_\beta-X_\gamma}} O_\gamma\left(\frac{r}{2}\right)$$

which holds in correlation functions. This is consistent with the dimensional analysis above. The numbers $c_{\alpha\beta\gamma}$ are universal constnat that depends on the normalisation convention. This is not how we will normalise the operators we will use below.

We restrict to the case of a single coupling for simplicity, and the partition function can then be expanded as,

$$Z = \int \mathcal{D}\phi \; e^{-\mathcal{H}_*}\left[1 - u\int \frac{d^d r}{a^{d-X}} O(r) + \frac{1}{2}u^2 \int \frac{d^d r \; d^d r'}{a^{2(d-X)}} \; O(r)O(r') + \dots\right]$$

To implement the UV cutoff we impose $|r-r'| \geq a$ and similarly for the higher terms. For the IR cutoff, we take the system to finite, of size L, to make sure that the perturbative series we have is convergent. This allows us to write the partition function as a sum of correlation functions in the fixed point theory,

$$Z = Z_*\left(1 - u\int \frac{d^d r}{a^{d-X}} \langle O(r)\rangle_* + \frac{1}{2}u^2 \int \frac{d^d r \; d^d r'}{a^{2(d-X)}} \; \langle O(r)O(r')\rangle_* + \dots\right)$$

To implement the RG, we increase the UV cutoff and rescale the coordinate to restore it to its original value. We do it in the opposite order here, first rescaling the length and then coarse grain. Let $R = r/b$ and $b = e^{d\tau}$ for an infinitesimal increment. The rescaling will reduce the UV cutoff, $a \mapsto a/b$, so we restore it to a by integrating out some modes. Here it amounts to eliminating some configurations where the operators are very close to each other. We use the defining property of the scaling operator, $\langle O_\alpha(r) O_\alpha(r') \dots \rangle_* = b^{-X} b^{-X} \langle O_\alpha(r/b) O_\alpha(r'/b) \dots \rangle_*$. Thus,

$$\begin{aligned} Z &= Z_* \left(1 - u \int \frac{d^d R b^d}{a^{d-X}} \langle O(bR) \rangle_* + \frac{1}{2} u^2 \int \frac{d^d R b^d \; d^d R'}{a^{2(d-X)}} \; \langle O(bR) O(bR') \rangle_* + \dots \right) \\ &= Z_* \left(1 - u \int \frac{d^d R b^d}{a^{d-X}} b^{-X} \langle O(R) \rangle_* + \frac{1}{2} u^2 \int \frac{d^d R b^d \; d^d R'}{a^{2(d-X)}} \; b^{-2X} \langle O(R) O(R') \rangle_* + \dots \right) \\ &= Z_* \left(1 - u b^y \int \frac{d^d R}{a^{d-X}} \langle O(R) \rangle_* + \frac{1}{2} (u b^y)^2 \int \frac{d^d R b^d \; d^d R'}{a^{2(d-X)}} \; \langle O(R) O(R') \rangle_* + \dots \right) \end{aligned}$$

where we used the definition of the RG eigenvalue $y = d - X$, and it is clear that the appearance of b^y will happen at all orders.

For the second step, we note that the condition $|r - r'| \geq a$ becomes $|R - R'| \geq a/b$ in the rescaled coordinate. Thus we need to restore the UV cutoff to its original value. To do this, we need to perform the integral over all configurations where $a/b \leq |R - R'| < a$, which are not allowed when the UV cutoff is a (but they are allowed in the coordinate used in the integral above). Schematically we can draw something like the figure below. There is a shell between the two cutoffs whose thickness is $a d\tau$ since $b = e^{d\tau}$. The volume of the shell in d dimensions is $a^d d\tau S_d$, with S_d the area of the $(d-1)$-sphere, eg for $d = 2$ the volume is $2\pi a^2 d\tau$. The integral is thus,

$$\begin{aligned} \int_{|R-R'| \geq a/b} \int d^d R \; d^d R' \; \langle O(R) O(R') \rangle &= \int_{|R-R'| \geq a} \int d^d R \; d^d R' \; \langle O(R) O(R') \rangle \\ &\quad - \int_{R, R' \in \text{shell}} \int d^d R \; d^d R' \; \langle O(R) O(R') \rangle \end{aligned}$$

and we will apply the OPE in the last integral so that we will replace the two operators which are too close by a single operator,

$$O(R) O(R') = \sum_\alpha \frac{c_{00\alpha}}{a^{2X - X_\alpha}} O_\alpha \left(\frac{R + R'}{2} \right)$$

The subscripts 0 are because we did not put any subscript to the operator O, and translational invariance ensures that the fact that O_α lives in the midpoint of R and R' is not actually significant here. We then have a single integral to perform for the third term above,

$$\int_{R, R' \in \text{shell}} \int d^d R \; d^d R' \; \langle O(R) O(R') \rangle = \sum_\alpha c_{00\alpha} \int d^d R \; \langle O_\alpha(R) \rangle \frac{a^d d\tau S_d}{a^{2X - X_\alpha}}.$$

With $b = e^{d\tau}$ and $d\tau$ infinitesimal, collecting all the above, we find the expansion to leading order in $d\tau$,

$$\begin{aligned}
Z = Z_* \Bigg(& 1 - (1+yd\tau)u \int \frac{d^dR}{a^{d-X}} \langle O(R)\rangle_* \\
& + \frac{1}{2}(u(1+yd\tau))^2 \int_{|R-R'|\geq a} \frac{d^dRb^d \; d^dR'}{a^{2(d-X)}} \; \langle O(R)O(R')\rangle_* \\
& + \frac{(ub^y)^2}{2} S_d d\tau c_{000} \frac{a^d}{a^x} \int \frac{d^dR}{a^{2(d-X)}} \; \langle O(R)\rangle + \dots \Bigg) \\
= Z_* \Bigg(& 1 - [u + yd\tau u - d\tau \frac{S_d}{2} c_{000} u^2] \int \frac{d^dR}{a^{d-X}} \langle O(R)\rangle_* + \dots \Bigg)
\end{aligned}$$

For simplicity here, we kept only the term in the OPE where $O_\alpha = O$, i.e. the RG equation involving only u, and kept only terms up to $O(d\tau)$. Terms appearing in the higher order integral will be consistent with the renormalised value of u,

$$u \to u + d\tau[yu - c_{000}\frac{S_d}{2}u^2 + \mathcal{O}(u^3)]$$

leading to an RG equation,

$$\frac{\mathrm{d}u_\alpha}{\mathrm{d}\tau} = y_\alpha u_\alpha - \frac{S_d}{2}\sum_{\beta\gamma} c_{\beta\gamma\alpha} u_\beta u_\gamma + O(u^3)$$

We can now aborb the factor of $S_d/2$ by rescaling all the couplings $\{u_\alpha\}$ by the same constant. And as we noted before, the constants $\{c_{\beta\gamma\alpha}\}$ depend on our normalisation convention. In the standard normalisation where the coefficient of the power law is 1, conformal invariance implies a total symmetry of the indices,

$$c_{\alpha\beta\gamma} = c_{\beta\alpha\gamma} = c_{\gamma\alpha\beta} = \dots$$

Finally, $\{c_{\alpha\beta\gamma}\}$ also sets the normalisation of the 3-point function.

5 O(n) model

Recall from before the effective Hamiltonian of the $O(n)$ model,

$$\mathcal{H} = \frac{K}{2} \int d^d r \, (\nabla h)^2$$

which gives a two point function,

$$G(r) = \langle h(0) h(r) \rangle = \frac{1}{K} \int \frac{d^d k}{(2\pi)^d} \frac{e^{i\underline{k}\cdot\underline{r}}}{k^2}$$

In $d = 2$, $G(r)$ has an IR divergence ($k \to 0$) corresponding to an infinite size limit divergence. However, $G(r) - G(0)$ has a finite limit,

$$G(r) - G(0) = \frac{1}{K} \int \frac{d^d k}{(2\pi)^d} \frac{e^{i\underline{k}\cdot\underline{r}} - 1}{k^2} = -\frac{\ln(r/\alpha)}{2\pi K} + (\dots)$$

for large r.

Writing $\vec{\phi} = (\cos\theta, \sin\theta)$ for θ a gaussian variable, we consider the two point function,

$$\begin{aligned} \langle \vec{\phi}(0) \cdot \vec{\phi}(r) \rangle &= \langle \cos\left(\theta(r) - \theta(0)\right) \rangle \\ &= \langle e^{i[\theta(r) - \theta(0)]} \rangle \\ &= e^{-\frac{1}{2}\langle [\theta(r) - \theta(0)]^2 \rangle} \\ &= e^{G(r) - G(0)} \end{aligned}$$

In $d > 2$, $G(r) \sim \text{const.} + \frac{A}{r^{d-2}}$, so that,

$$\langle \vec{\phi}(0) \cdot \vec{\phi}(r) \rangle \sim \left(1 + \frac{A}{r^{d-2}} + \dots\right)$$

for large r. This proves that there is LRO. For $d = 2$, according to this treatment, ie assuming that θ is a free field, we get

$$\langle \vec{\phi}(0) \cdot \vec{\phi}(r) \rangle \overset{???}{=} e^{-\frac{1}{2\pi K} \ln(r/a) + \dots} \sim \left(\frac{r}{a}\right)^{-1/2\pi K}$$

which vanishes for $r \to 0$. So as we see there is no long range order in that case.

Naively, this gives power law decay at large r for the case $n = 2$ (this is called the XY model). However, in that case, it is not clear yet whether FFT is justified since θ is a periodic variable in the range $[-\pi, \pi]$. Another point is that for $n > 2$ there are multiple Goldstone modes that interact, and we need to understand the effect of these interactions. The large n treatment is the simplest way to see what happens in $d = 2, n > 2$ but it also gives us results for $d > 2$.

5.1 Large n expansion

We will now obtain a direct solution for the $O(n)$ model for large n, without using any RG. The critical exponents we will obtain are exact as $n \to \infty$ and the qualitative picture is similar for $n \geq 3$

(and $n \geq 2$ for $d > 2$). This will be a starting point so we can go beyond the leading order result with a system expansion in $1/n$. Recall the partition function,

$$Z = \int \mathcal{D}\vec{\phi}\, e^{-\mathcal{H}}, \qquad \mathcal{H} = \int d^d r \, \left[\frac{1}{2}(\nabla\vec{\phi})^2 + \frac{t}{2}\vec{\phi}^2 + \frac{u}{4n}(\vec{\phi}^2)^2\right]$$

where we have chosen to normalise the field so that the $1/n$ appears in front of the quartic term.

We start by considering the terms in the Hamiltonian which contain only ϕ_1 (without loss of generality) collected as an effective Hamiltonian,

$$\mathcal{H}_1 = \int d^d r \left[\frac{1}{2}(\nabla\phi_1)^2 + \frac{1}{2}\left(t + \frac{u}{n}\vec{\phi}^2\right)\phi_1^2 + \mathcal{O}\left(\frac{1}{n}\right)\right]$$

What we did was to write the quartic term in the following form,

$$\begin{aligned}\frac{1}{n}(\vec{\phi}^2)^2 &= \frac{1}{n}(\phi_1^2 + \sum_{a>1}^{n}\phi_a^2)^2 \\ &= \frac{2}{n}\phi_1^2\sum_{a>1}\phi_a^2 + \frac{1}{n}\phi_1^4 + (\dots) \\ &= \frac{2}{n}\phi_1^2\vec{\phi}^2\left(\vec{\phi}^2 - \phi_1^2\right) + + \frac{1}{n}\phi_1^4 + (\dots) \\ &= \frac{2}{n}\phi_1^2\vec{\phi}^2 - \frac{1}{n}\phi_1^4 + (\dots)\end{aligned}$$

where $(\dots)$ denotes terms which do not include ϕ_1 and they are therefore not part of $\mathcal{H}_1$. The idea now is to use n as a large parameter. In the second line we notice that the second term is of order $\mathcal{O}(1/n)$ while the first term is actually $\mathcal{O}(n^0)$ since it is a sum of $\sim n$ terms. We then reorganise the sum in order to keep only the $\mathcal{O}(1)$ terms and drop the rest.

The Hamiltonian shows that the field ϕ_1 is interacting with all the other components so we need to find a way to justify separating it from all the others. The field $\vec{\phi}^2$ is a random fluctuating quantity which is a sum of $n \gg 1$ random variables. Using the central limit theorem, we conjecture that the fluctuations in $\frac{1}{n}\vec{\phi}^2$ become much smaller than the average $\langle\frac{1}{n}\vec{\phi}^2\rangle$ in the limit $n \to \infty$. More precisely, we restrict the conjecture to the case where there is no LRO (disorder phase or at the critical point). To extend the argument for the case of an ordered phase, we need to single out the ordering direction and treat that separately. This is the core idea behind the large n approximation.

The effective Hamiltonian for ϕ_1 is thus a Gaussian Hamiltonian which we know how to treat. To rewrite it, we absorb the expectation value of $\vec{\phi}^2$ into a new coupling $\tilde{t}$, representing an effective mass squared.

$$\mathcal{H}_1 = \int d^d r \frac{1}{2}\left[(\nabla\phi_1)^2 + \tilde{t}\phi_1^2\right], \qquad \tilde{t} = t + \frac{u}{n}\langle\sum_a \phi_a^2\rangle = t + \langle\phi_1^2\rangle$$

where to rewrite $\tilde{t}$ we have used the symmetry between components of $\vec{\phi}$. Also note that $\frac{1}{n}\langle\vec{\phi}^2\rangle$ is **not** an order parameter since it is always positive. $\tilde{t}$ or equivalently $\langle\phi_1^2\rangle$ are determined self-consistently. The two point function of ϕ_1 using $\mathcal{H}_1$ is,

$$\langle\phi_1(0)\phi_1(r)\rangle = \int \frac{d^d k}{(2\pi)^d}\,\frac{e^{ik\cdot r}}{k^2 + \tilde{t}} \sim r^{-\frac{d-1}{2}} e^{-r/\xi}, \qquad \tilde{t} = 1/\xi^2$$

so that at $r = 0$,

$$\langle \phi_1^2 \rangle = \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2 + \tilde{t}}$$

Physically, in a lattice $O(n)$ model, $\tilde{t}$ is the correlation length of the two point function. Thus, the consistency relation becomes,

$$\frac{1}{\xi^2} = t + u \int^{|k| < \frac{2\pi}{a}} \frac{d^d k}{(2\pi)^d} \frac{1}{k^2 + \xi^{-2}}$$

This equation determines the correlation length $\xi = \xi(t)$ as a function of the bare coupling t. Depending on the value of the spatial dimension, we need to treat different cases separately: $d > 2$ and $d = 2$, with the difference being the existence of a critical point separating the ordered from the disorded phase, since in $d = 2$ there is no LRO.

Case $d > 2$

There exists a critical point t_* where $\xi = \infty$. Using the self consistency equation and solving for t_*,

$$-t_* = u \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2}.$$

Because of the UV cutoff there is no UV divergence of the integral on the RHS so we do not need to worry about large k. For $d > 2$ by dimensional analysis we find that there is no IR divergence either, so that we end up with a negative finite value for t_*. For $d \leq 2$, the RHS diverges as $L \to \infty$ and therefore there is no phase transition at any finite t_*. Contrasting the negative value of t_* with MFT we see that there is a disagreement, essentially because of fluctuations. The quadratic term in the effective Hamiltonian is different than the quadratic term in the bare Hamiltonian. The bare potential at t_* still has two minima.

Scaling of $\xi(t)$ for $t \gtrsim t_*$

We take here $(t - t_*) \ll 1$ and $\xi \gg a$. We use the equation above for t_* and write it in terms of $t - t_*$ by subtracting the equations at t and at t_*,

$$\begin{aligned} \frac{1}{\xi^2} - (t - t_*) &= u \int \frac{d^d k}{(2\pi)^d} \left(\frac{1}{k^2 + \xi^{-2}} - \frac{1}{k^2} \right) \\ \iff t - t_* &= \frac{1}{\xi^2} + \frac{u}{\xi^2} \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2 (k^2 + \xi^{-2})} \end{aligned}$$

We rewrite the last equation in terms of a dimensionless parameter q given by $k = \frac{1}{\xi} q$,

$$t - t_* = \frac{1}{\xi^2} + \frac{1}{\xi^{d-2}} \int^{|q| < 2\pi \frac{\xi}{a}} \frac{d^d q}{(2\pi)^d} \frac{1}{q^2 (q^2 + 1)}$$

and the problem is reduced to finding the scaling of the integral as $\xi \to \infty$.

Case $2 < d < 4$

The integral is finite as $\xi \to \infty$,

$$t - t_* = \frac{1}{\xi^2} + \frac{1}{\xi^{d-2}}(A + \dots)$$

for some constant A. Since $d < 2$, the dominant term is the second one, and therefore $\xi \sim (t - t_*)^{-1/(d-2)}$ from which we read the critical exponent $\nu = \frac{1}{d-2}$. This is our first exact result for a critical exponent beyond MFT. In RG language, we expect a flow diagram with a fixed point separating a disordered and ordered phase. In the ordered phase (negative values of $(t - t_*)$), and close to the critical point we find $\delta t' = b^y \delta t$ with $y = 1/\nu$. For the other coupling u, we find that it is irrelevant.

As an example, in $d = 3$, $y_{:\phi^2:} = \frac{1}{\nu} = 1$, which gives the scaling dimension, $X_{:\phi^2:} = d - y_{:\phi^2:} = 2$. Note that this is different from what we find in free field theory where $X^{\text{FREE}}_{:\phi^2:} = 2X^{\text{FREE}}_{\phi} = 2 \cdot (d-2)/2 = 1$. And although the two-point function here and in the free field case agree to leading order, the two-point function of ϕ^2 does not agree.

Case $d > 4$

We use the same equation as above, but now the integral behaves differently. For large ξ now the integral is unbounded, and by dimensional analysis we find that the integral scales as $\sim \xi^{d-4}$. Therefore both terms in RHS scale as $1/\xi^2$ and thus, now $\nu = \frac{1}{2}$. This agrees with the MFT prediction and it is consistent with the RG argument that MFT is valid for $d \geq 4$.

Case $d = 4$

The integral now has a logarithmic divergence for large q, and thus the second term dominates in the equation,

$$(t - t_*) \sim \frac{\log \xi}{\xi^2} \iff \xi \sim (t - t_*)^{-1/2} \sqrt{\log \xi}$$

Taking the logarithm of the last equation we end up with,

$$\xi \sim (t - t_*)^{-1/2} \sqrt{\log \frac{1}{t - t_*}}$$

We conclude that in $d = 4$ there are logarithmic correction to MFT. This is a general lesson to be understood later in the RG language.

Exercise 2. *What happens for $t < t_*$? So far we implicitly assumed that all the components of $\vec{\phi}$ are of the same size by using the central limit theorem argument. In an ordered phase, this is not necessarily true.*

> **Note.** Aside: Scaling operators at large n fixed point The effective Hamiltonian for ϕ_1, $\mathcal{H}_1$, gives the scaling of the two-point function as in FFT, $\langle \phi_1 \phi_1 \rangle \sim 1/r^{d-2}$. From that we can extrapolate that by symmetry, the off diagonal elements vanish, so that $\langle \phi_a(0)\phi_b(r) \rangle \sim \delta_{ab}/r^{d-2}$, and thus the scaling dimension of ϕ is $X_\phi = \frac{d-2}{2}$. We also know that to leading order in $1/n$, higher correlation functions in ϕ_1 are given by Wick's theorem, since $\mathcal{H}_1$ is a Gaussian

Hamiltonian. For example,

$$\begin{aligned}\langle :\phi_1^2(0): :\phi_1^2(r): \rangle &= \phi_1(0)\phi_1(0)\phi_1(r)\phi_1(r) + \phi_1(0)\phi_1(0)\phi_1(r)\phi_1(r) \\ &= 2\left(\frac{1}{r^{d-2}}\right)^2\end{aligned}$$

Naively we might expect all operators with quadratic order in ϕ to have dimension $2X_\phi$. This is not the case here since we know the scaling dimension of the two point function of $\vec{\phi}^2$, $\langle :\vec{\phi}^2: :\vec{\phi}^2: \rangle$. The scaling dimension of this is $X_{:\vec{\phi}^2:=d-y_{:\vec{\phi}^2:}} = d-(d-2) = 2 \neq 2X_\phi$. From that we have,

$$\langle :\vec{\phi}^2: :\vec{\phi}^2: \rangle \sim \frac{1}{r^4}$$

whereas for ϕ_1 and at leading order,

$$\langle :\phi_1^2: :\phi_1^2: \rangle \sim \frac{1}{r^{2(d-2)}} \overset{\text{d=3}}{=} \frac{1}{r^2}$$

So what is going on here? We first note that this is not a contradiction because we can distinguish different scaling operators by symmetry. The first one is the $O(n)$ singlet $:\vec{\phi}^2:$ with scaling dimension $X_{\phi^2} = 2$, consistent with the first formula. The second one is, $M_{ab} = \phi_a\phi_b - \delta_{ab}\vec{\phi}^2/n$, an $O(n)$ symmetric matrix, has scaling dimension $X_M = d-2$, consistent with the second formula. In free field theory, these operators had the same dimension, but in an interacting theory they do not need to have the same one. This is precisely what the formula above is saying. Computing the correlation function of ϕ_1^2, it turns out that in the large n limit, it is dominated by just the M_{ab} operator,

$$:\phi_1^2: = M_{11} + \frac{1}{n} :\vec{\phi}^2:$$

We should thus be careful to distinguish operators with different symmetry.

<u>Case $d = 2$</u>

The case $d = 2$ there is no LRO. We will use the Landau-Ginzburg machinery to investigate the low temperature regime of the theory. The self consistency equation for $d = 2$ is,

$$\frac{1}{\xi^2} - t = u\int \frac{d^2k}{(2\pi)^2}\,\frac{1}{k^2+\xi^{-2}}$$

The correlation length ξ remains finite for all t but diverges fast as $t \to -\infty$ (corresponding to $T \to 0$ as we will see later). The integral is easy to analyse,

$$\begin{aligned}\frac{1}{\xi^2} - t = u\int \frac{d(k^2)}{4\pi}\,\frac{1}{k^2+\xi^{-2}} &= \frac{u}{4\pi}\left[\ln\left(k^2+\frac{1}{\xi^2}\right)\right]_0^{(2\pi/a)^2} \\ &= \frac{u}{4\pi}\ln\left(\frac{\left(\frac{2\pi}{a}\right)^2+\frac{1}{\xi^2}}{\frac{1}{\xi^2}}\right)\end{aligned}$$

We are interested in the case where $t \to -\infty$ and $\xi \to +\infty$ so that we can drop the $1/\xi^2$ on the

LHS and in the numerator inside the logarithm. Thus, to leading order,

$$-t = \frac{u}{4\pi} \ln \left(\frac{2\pi \xi}{a} \right)^2 \iff \boxed{\xi = \frac{a}{2\pi} \exp \left[\frac{2\pi |t|}{u} \right]}$$

The last result is a known result from classical statistical mechanics for 2D systems with an order parameter with a continuous non abelian symmetry. This holds for all $n > 2$ and we will revisit this in the context of the RG.

> **Note.** Aside: $T \to 0$
> Consider a microscopic model with Hamiltonian,
>
> $$\mathcal{H} = \frac{1}{T} \int d^2 r \; \left[\frac{1}{2} (\nabla \vec{\phi})^2 + \frac{t}{2} \vec{\phi}^2 + \frac{u}{4n} (\vec{\phi}^2)^2 \right]$$
>
> This could be an approximate coarse grained description of some lattice model with $t < 0$ and fixed, and the variable we can vary is T, the temprature. We make a change of variables, $\phi \to \sqrt{T} \tilde{\phi}$, and following that, rescale the spatial coordinate by $r \to \tilde{r} / \sqrt{T}$. The kinetic term (for $d = 2$) is invariant under the rescaling, so the Hamiltonian becomes,
>
> $$\mathcal{H} = \int d^2 \tilde{r} \; \left[\frac{1}{2} (\nabla \vec{\tilde{\phi}})^2 + \frac{t}{2T} \vec{\tilde{\phi}}^2 + \frac{u}{4n} (\vec{\tilde{\phi}}^2)^2 \right]$$
>
> It is now clear that $T \to 0$ is equivalent to $t \to -\infty$ for the rescaled coordinate. To leading order, the exponential dependence found above is then,
>
> $$\xi \sim \exp \left(\frac{\text{const.}}{T} \right)$$
>
> for small T.

Aside: CLT conjecture and formal treatment

We will briefly discuss how we can do a more formal treatment without invoking the assumption of the CLT about weak fluctuations of ϕ in the large n limit. One way is to do it diagrammatically with Feynman diagrams, but here we will do it via the saddle point method to perform the path integral,

$$Z = \int \mathcal{D}\vec{\phi} \; e^{-\int d^d r \; \left[\frac{1}{2} (\nabla \vec{\phi})^2 + \frac{t}{2} \vec{\phi}^2 + \frac{u}{4n} (\vec{\phi}^2)^2 \right]}$$

We will use the "Hubbard-Stratonovich" transformation by introducing another field λ so that the integral over $\vec{\phi}$ is Gaussian,

$$Z' = \int \mathcal{D}\vec{\phi} \; \mathcal{D}\lambda \; e^{-\int d^d r \; \left[\frac{1}{2} (\nabla \vec{\phi})^2 + \frac{t}{2} \vec{\phi}^2 + + i\lambda \vec{\phi}^2 - \frac{1}{2} \left(\frac{2n}{u} \right) \lambda^2 \right]}$$

Performing the integral over λ gives Z up to a constant. We first Z' over $\vec{\phi}$ using the fact that it is an independent integral for each ϕ_a, $a = 1, \ldots, n$,

$$Z' = \int \mathcal{D}\lambda e^{-\frac{1}{2} \int \left(\frac{2n}{u} \right) \lambda^2 - \frac{n}{2} \int \frac{d^d k}{(2\pi)^d} \; \ln \left(k^2 + t + 2i\lambda \right)}$$

This is now in the form that allows us to use the saddle point method for large n, since n sites outside the effective action for λ, say S_λ. We calculate

$$\frac{\partial S_\lambda}{\partial \lambda} = 0 \iff -\frac{2n}{u}\lambda - \frac{1}{2}\int \frac{d^d k}{(2\pi)^d}\,\frac{2i}{k^2+t+2i\lambda} = 0$$

which is the self consistency equation from before, with $2i\lambda \propto \langle \phi_1^2 \rangle$.

5.2 $4-\epsilon$ expansion

In the previous subsection we have performed a large n expansion and found that this gave us the mean field exponents for $d > 4$ but non-trivial exponents for $d < 4$. We now take a different approach by treating d as a continuous parameter. The RG equations derived in the previous section can be applied to the $O(n)$ model. There are two simplifying limits for the RG of the $O(n)$ model:

- $d = 2+\epsilon$, in which case the RG equations are simple because the Goldstone modes are weakly interacting at the fixed point when $\epsilon \ll 1$
- $d = 4-\epsilon$, in which case $\vec{\phi}$ is weakly interacting at the non-trivial fixed point when $\epsilon \ll 1$.

As we will see, both expansions give results consistent with the previous large n calculation.

We start with the latter case of the two above. The $4-\epsilon$ expansion dates back to Wilson and Fisher (1972) and the basic idea is that when $\epsilon = 4-d \ll 1$, the Ising fixed point (of ϕ^4 theory) is close to the free fixed point in the space of coupling constants. We can then use the perturbative RG equations. We will use the OPE method which will give us the lowest non-trivial order for the equations. The Hamiltonian is,

$$\mathcal{H} = \int d^d r\, \left[\frac{1}{2}(\nabla\phi)^2 + \frac{t}{a^2}\phi_2 + \frac{u}{a^{4-d}}\phi_4\right]$$

where $\phi_2 \equiv {:}\phi^2{:}$ and $\phi_4 \equiv {:}\phi^4{:}$. The normalisation we will choose is $\langle\phi\phi\rangle = \frac{1}{r^{d-2}}$. From the general formula of the RG equations,

$$\frac{\mathrm{d}u_\alpha}{\mathrm{d}\tau} = y_\alpha u_\alpha - \sum_{\beta\gamma} c_{\beta\gamma\alpha} u_\beta u_\gamma,$$

we see that we need to compute the OPE coefficients that include ϕ_2 and ϕ_4 eg $c_{222}, c_{224}, \ldots$.

Thinking about the correlation function of ϕ_2 for example, we see that there are terms which are contracted with terms in the bracket $(\ldots)$ and terms which are contracted between themselves, and we can organise the product as,

$$\begin{aligned}
\phi_2(0)\phi_2(r) &= {:}\phi(0)\phi(0){:}\,{:}\phi(r)\phi(r){:} \\
&= {:}\phi(0)\phi(0)\phi(r)\phi(r){:} + 4\overbrace{\phi(0)\phi(r)}\,{:}\phi(0)\phi(r){:} + 2(\overbrace{\phi(0)\phi(r)})^2 \\
&= [\phi_4\left(\frac{r}{2}\right) + \text{derivative ops}] \\
&\quad + \frac{4}{r^{d-2}}[\phi_2\left(\frac{r}{2}\right) + \text{derivative ops}] \\
&\quad + \frac{2}{r^{2(d-2)}}\mathbb{1}
\end{aligned}$$

where we used the property of the OPE, $O_\alpha(0)O_\beta(r) = \sum_\gamma \frac{c_{\alpha\beta\gamma}}{r^{X_\alpha+X_\beta-X_\gamma}} O_\gamma\left(\frac{r}{2}\right)$. From this, we immediately read off the OPE coefficients,

$$c_{224} = 1, c_{222} = 4$$

The problem has thus become an exercise in combinatorics and we count the contractions possible. We can even simplify this be ignoring the r argument since it is consistent with the OPE by construction. For example, the coefficient c_{242} gives the number of ways to contract $\phi_2 \cdot \phi_4$ so that the end result is ϕ_2. We calculate it as follows,

$$\phi_2 \cdot \phi_4 = {:}\phi\phi{:}\,{:}\phi\phi\phi\phi{:} = c_{242}\,{:}\phi\phi{:}$$

and $c_{242} = 4 \cdot 3 = 12$ ways to contract. Similarly, for c_{244},

$$\phi_2 \cdot \phi_4 = {:}\phi\phi{:}\,{:}\phi\phi\phi\phi{:} = c_{244}\,{:}\phi\phi\phi\phi{:}$$

and $c_{244} = 8$. Finally, for c_{444},

$$\phi_4 \cdot \phi_4 = {:}\phi\phi\phi\phi{:}\,{:}\phi\phi\phi\phi{:} = c_{444}\,{:}\phi\phi\phi\phi{:}$$

and $c_{444} = \binom{4}{2} \cdot \binom{4}{2} \cdot 2 = 72$.

Using the OPE coefficients we write down the RG equations with $u_1 \to t$ and $u_2 \to u$,

$$\frac{\mathrm{d}u}{\mathrm{d}\tau} = (4-d)u - 72u^2 - 16ut - t^2$$
$$\frac{\mathrm{d}t}{\mathrm{d}\tau} = (2-24u)t - 4t^2 - 96u^2$$

Some of the terms are subleading, meaning that they can be ignored because they give the same fixed point to $O(\epsilon)$. We find the trivial fixed point $t = 0, u = 0$, which is the fixed point we are perturbing around, and additionally a non-trivial one $t = 0, u_* = \frac{\epsilon}{72}$. Note that for $\epsilon \ll 1, u_* \ll 1$. The non-trivial fixed point is referred to as the *Wilson-Fisher* fixed point.

Flows in $d < 4$

At the free fixed point $(0,0)$ both of the RG eigenvalues are positive thus the outside flow lines. The Wilson-Fisher fixed point has a irrelevant direction (the u variation).

We linearise around the Wilson-Fisher fixed point by writing $u = u_* + \delta u$ to get,

$$\frac{\mathrm{d}\delta u}{\mathrm{d}\tau} = -\epsilon\delta u + O(\delta u^2, \delta t^2, \delta u \delta t)$$
$$\frac{\mathrm{d}\delta t}{\mathrm{d}\tau} = \left(2 - \frac{\epsilon}{3}\right)\delta t + \ldots$$

from which we can read the critical exponents. From the second equation,

$$\frac{d\delta t}{\delta t} = \left(2 - \frac{\epsilon}{3}\right) d\tau \iff \delta t \sim e^{\left(2-\frac{\epsilon}{3}\right)\tau} \equiv (e^\tau)^{1/\nu}$$

so that the thermal RG eigenvalue is $y_t = 2 - \frac{\epsilon}{12}$ and the correlation length exponent is thus,

$$\nu = \frac{1}{y} = \frac{1}{2} + \frac{\epsilon}{12} + O(\epsilon^2)$$

so that there is an $\mathcal{O}(\epsilon)$ deviation from the MF exponent, when $d > 4$. By contrast, when $\epsilon < 0$ ($d > 4$) then the free fixed point is stable.

The lowest order ϵ expansion is useful because often it correctly captures the topology of the flows or the existence of the fixed point even when d is physical (ϵ an integer). It is also a starting point for a higher order expansion in ϵ which is quantitatively accurate even when $\epsilon = 1$.

5.2.1 Extension to $O(n)$ model

For the $O(n)$ model, the corresponding operators are $\sum_{a=1}^{n} {:}\phi_a\phi_a{:}$ and $\sum_{a,b=1}^{n} {:}\phi_a\phi_a\phi_b\phi_b{:}$. The OPE coefficients which we need for the RG equations are $c_{444} = 8(n+8)$ and $c_{242} = 4(n+2)$. The fixed point now is at,

$$u_* = \frac{\epsilon}{8(n+8)} + \mathcal{O}\Big(\epsilon^2\Big)$$

with an RG eigenvalue, $y_t = 1/\nu = 2 - \frac{n+2}{n+8}\epsilon + \mathcal{O}(\epsilon^2)$. Note that as $n \to \infty$, u_* is small even if ϵ is not small. As a result, the above gives the exact critical exponent ν when $n \to \infty$ for any $0 < \epsilon < 2$,

$$\nu = \frac{1}{2-\epsilon} = \frac{1}{d-2}$$

which matches the large n calculation! For n large **and** ϵ small, we are permitted to use either expansion method.

5.3 $2+\epsilon$ expansion

We now turn our attention to understanding the $O(n)$ model close to $d = 2$. We expect this to be qualitatively different because in $2D$ the Ising model has a nonzero critical temperature T_c, and on the other hand the $O(\infty)$ model has no phase transition with correlation length $\xi \sim e^{-A/T}$ as $T \to 0$. The RG treatment shows that this is also true for any $n \geq 3$ in 2D. Nonlinear interactions between "Goldstone modes"[4] drive the theory away from the ordered fixed point in 2D. The key idea is due to Polyakov (1975) in a paper called "Interaction of Goldstone particles in two dimensions".

The Hamiltonian we will be using is known as the *nonlinear sigma model*,

$$\mathcal{H} = \frac{K}{2}\int \frac{d^dx}{a^{d-2}}\,(\nabla\vec{\phi})^2$$

with the constraint $\vec{\phi}^2 = 1$. This is different from the Landau Ginzburg theory which was constraint free but included a potential. However, they can plausibly access the same universal physics. The contraint on the field $\vec{\phi}$ means that the Hamiltonian is not the free field theory anymore. The coupling K as written above is dimensionless and we will write $K = \frac{1}{T}$ with T a dimensionless temperature parameter, so that $K \gg 1$ corresponds to the regime of well ordering in short scales.

[4] in quotation marks because they are the degrees of freedom which we would call Goldstone modes had there been long range order, in the absence of interactions.

For a direct approach, we essentially solve the constraint explicitly by writing

$$\vec{\phi} = (\sqrt{1-\vec{\psi}^2}, \vec{\psi})$$

and now $\vec{\phi}$ has $(n-1)$ constraint free components. In terms of $\vec{\psi}$, the Hamiltonian is,

$$\mathcal{H} = \frac{K}{2}\int \frac{d^d x}{a^{d-2}} \left[(\nabla\vec{\psi})^2 + (\vec{\psi}\cdot\nabla\vec{\psi})^2 + \dots\right]$$

We can write this in terms of a rescaled parameter $\tilde{\psi} = \sqrt{K}\vec{\psi}$,

$$\mathcal{H} = \frac{1}{2}\int \frac{d^d x}{a^{d-2}} \left[(\nabla\tilde{\psi})^2 + \frac{1}{K}(\tilde{\psi}\cdot\nabla\tilde{\psi})^2 + \dots\right]$$

which is in a form that allows us to perform a perturbative RG treatment in large K, keeping only the lowest order term in $\mathcal{H}$. In doing that we will find the RG equations,

$$\frac{\mathrm{d}T}{\mathrm{d}\tau} = -\epsilon T + \left(\frac{n-2}{2\pi}\right)T^2 + \mathcal{O}\left(T^3\right)$$
$$\implies \frac{\mathrm{d}K}{\mathrm{d}\tau} = \epsilon K - \left(\frac{n-2}{2\pi}\right) + \mathcal{O}(1/K)$$

and we will derive these later. The first term in K is what we had before: the flow of the stiffness K in free field theory if we have a fixed normalisation of the field (here it is $\vec{\phi}^2 = 1$). The non-trivial term is an effect of the interactions.

For the flows in 2D, $\epsilon = 0$ in the equations above. At low temperatures, $K(a) \gg 1$, that is the stiffness measured at the scale of the UV cutoff, the "microscopic stiffness". At a different scale ab with $b > 1$, integrating the RG equation we get,

$$K(ba) - K(a) = -\left(\frac{n-2}{2\pi}\right)\ln b$$

In terms of an effective temperature in the renormalised model, $T(ba)$, and the physical temperature T, we have

$$\frac{1}{T(ba)} = \frac{1}{T} - \left(\frac{n-2}{2\pi}\right)\ln b$$

When $T \ll 1$, the renormalised temperature will eventually become $\mathcal{O}(1)$ when the terms in the RHS are of the same order. This happens when $b = \exp\left(\frac{2\pi}{(n-2)T}\right)$ which is the scale where we have escaped the vicinity of the ordered fixed point ($T = 0$). We expect that this is the scale at which the model becomes disorded so that the correlation length is,

$$\xi \sim a\exp\left(\frac{2\pi}{(n-2)T}\right)$$

up to some multiplicative $\mathcal{O}(1)$ factor. Even if the field is well ordered on short scales, it becomes "floppy" on this large scale.

Let us now derive the RG equations stated above following Cardy (sec. 6.5). It is sufficient to do the calculation in $d = 2$ to get the $\mathcal{O}(1)$ coefficient of the T^2 term. The linear term in the RG equation is just the scaling dimension of the stiffness in FFT. In $2D$ the Hamiltonian is,

$$\mathcal{H} = \frac{K}{2}\int d^2x\ (\nabla\vec{\phi})^2, \qquad \vec{\phi}^2 = 1$$

Instead of writing $\vec{\phi}$ as before, we write

$$\vec{\phi} = \left(\cos\theta\sqrt{1-\vec{t}^2}, \sin\theta\sqrt{1-\vec{t}^2}, \vec{t}\right)$$

singling out two components instead of one. The Hamiltonian is then,

$$\mathcal{H} = \frac{K}{2}\int d^2x \ \left[(1-t^2)(\nabla\theta)^2 + \frac{(\vec{t}\cdot\nabla\vec{t})^2}{\sqrt{1-t^2}} + (\nabla\vec{t})^2\right]$$

For convenience, we can rescale $t = \sqrt{T}\tilde{t}$ and $\theta = \sqrt{T}\tilde{\theta}$, and split the Hamiltonian in the convenient form,

$$\mathcal{H} = \mathcal{H}_{\tilde{\theta}} + \mathcal{H}_{\tilde{t}} + \mathcal{H}_{\tilde{t},\tilde{\theta}}$$

where,

$$\begin{aligned} \mathcal{H}_{\tilde{\theta}} &= \frac{1}{2}\int d^2x \ (\nabla\tilde{\theta})^2 \\ \mathcal{H}_{\tilde{t}} &= \frac{1}{2}\int d^2x \ (\nabla\tilde{t})^2 + T(\tilde{t}\nabla\tilde{t})^2 + \dots \\ \mathcal{H}_{\tilde{t},\tilde{\theta}} &= -\frac{T}{2}\int d^2x \ \tilde{t}^2(\nabla\tilde{\theta})^2 \end{aligned}$$

We will use momentum shell RG and integrate out the higher momentum modes by writing $\tilde{\theta} = \tilde{\theta}' + \tilde{\theta}_>$ and $\tilde{t} = \tilde{t}' + \tilde{t}_>$[5]. The normalisation for the angle is $\theta \in [-\pi, \pi]$ and we will return to this in the end.

The trick we will use now is that we don't need to compute all the terms in the renormalised Hamiltonian. We will only compute the change in the coefficient of $(\nabla\theta)^2$ and then use symmetry to fix the rest. At $\mathcal{O}(\nabla^2)$ the functional form of the Hamiltonian is the initial one, with $\vec{\phi}^2 = 1$. The coefficient of $\tilde{\theta}'$ appears in $\mathcal{H}$ in the kinetic term $\mathcal{H}_{\tilde{\theta}}$ which is free and thus the high and low momentum modes decouple, and also in $\mathcal{H}_{\tilde{t},\tilde{\theta}}$. In the second term, the low momentum modes of $\tilde{\theta}$ are coupled to both the high and low modes of $\tilde{t}$. Thus,

$$\mathcal{H}_{\text{eff}} = \frac{1}{2}(\nabla\tilde{\theta}')^2\left[1 - T(\tilde{t}')^2 - T(\tilde{t}_>)^2\right]$$

and the $\tilde{t}'$ term will be kept in the RG Hamiltonian and we need to integrate out the last term. To lowest order in T, the new "Boltzmann weight" for $\tilde{\theta}'$ is given by,

$$\langle e^{\frac{1}{2}(\nabla\tilde{\theta}')^2\left[1-T(\tilde{t}_>)^2\right]}\rangle_{\tilde{t}_>}$$

with the notation meaning that this is computed in $\mathcal{H}_{\tilde{t}_>} = \frac{1}{2}\int d^2x \ (\nabla\tilde{t}_>)^2$ because the other terms that involve $\tilde{t}_>$ are $O(T)$ relative to using the free field. Since the renormalization of $\tilde{\theta}'$ is already suppressed by a factor of T it enough to use the free part of $\mathcal{H}_{\tilde{t}_>}$.

We use the cumulant expansion, $\langle e^A\rangle = e^{\langle A\rangle} + \dots$ controlled by $T \ll 1$, to get,

$$\langle e^{\frac{1}{2}\int d^2x \ (\nabla\tilde{\theta}')^2\left[1-T(\tilde{t}_>)^2\right]}\rangle_{\tilde{t}_>} = e^{\frac{1}{2}\int d^2x \ (\nabla\tilde{\theta}')^2\left[1-T\langle\tilde{t}^2_>\rangle\right]}$$

[5]Using primes instead of $\tilde{\theta}_<$ is the same notation as the spins in RSRG

It remains to calculate the fluctuations in $\langle\tilde{t}^2_>\rangle$ which is simple enough since they are given by the massless free field theory.

$$\langle\tilde{t}^2_>\rangle = \int_{2\pi/ab}^{2\pi/a} \frac{d^2k}{(2\pi)^2}\frac{1}{k^2} = \frac{1}{2\pi}\ln b\times(n-2) = \frac{n-2}{2\pi}d\tau$$

since $\tilde{t}$ has $(n-2)$ components and we used an infinitesimal RG step $b=e^{d\tau}$. Thus,

$$\begin{aligned}\langle e^{\frac{1}{2}\int d^2x\ (\nabla\tilde{\theta}')^2\left[1-T(\tilde{t}_>)^2\right]}\rangle_{\tilde{t}_>} &= e^{\frac{1}{2}\int d^2x\ (\nabla\tilde{\theta}')^2\left[1-\frac{n-2}{2\pi}Td\tau\right]}\\ &= e^{\frac{1}{T}\left(1-\frac{n-2}{2\pi}Td\tau\right)\int d^2x\ (\nabla\theta')^2}\end{aligned}$$

having restored the original coupling variable θ. The RG equation to order $d\tau$ is then,

$$\frac{1}{2T'} = \frac{1}{2T\left(1-\frac{n-2}{2\pi}Td\tau\right)}$$

and with $T'=T+dT$, we get,

$$\frac{\mathrm{d}T}{\mathrm{d}\tau} = \frac{n-2}{2\pi}T^2$$

and we note again that this calculation has been performed in $d=2$. To go to $d=2+\epsilon$, we simply restore the $\mathcal{O}(\epsilon)$ term we found using the Wilsonian RG in free field theory. Even though the renormalised could pick up an extra $\mathcal{O}(\epsilon)$ correction in $(\cdots+\mathcal{O}(\epsilon))T^2$, this won't matter in the fixed point where we are after T_* to $\mathcal{O}(\epsilon)$. Thus, we can write,

$$\boxed{\frac{\mathrm{d}T}{\mathrm{d}\tau} = -\epsilon T + \left(\frac{n-2}{2\pi}\right)T^2}$$

RG flows

In $d=2$ there are two fixed points: $T=0$ corresponding to an ordered phase, and $T=\infty$ corresponding to disorder. The flow is from the ordered to the disorded fixed point. However the flow near $T=0$ is very slow due to the quadratic term, leading to a very rapid correlation length $\xi\sim\exp(1/T)$.

In $d=2+\epsilon$ with $\epsilon>0$, there is a non trivial fixed point at,

$$T_* = \frac{2\pi\epsilon}{n-2}$$

and the correlation length exponent is $\nu=\frac{1}{\epsilon}=\frac{1}{d-2}$. At higher orders in ϵ there will be an n dependence. Is this consistent with the large n expansion? For $n\to\infty$ the correlation length exponent is exactly the same, so that for large enough n this supports the idea that we are finding the same fixed points in these different approaches. There is a difference though: in $4-\epsilon$ we considered a Hamiltonian $\mathcal{H}$ with couplings u,t in the RG equations whereas in the $2+\epsilon$ we only had one coupling, T. There is no contradiction here since in both cases there is only one relevant coupling at the fixed point, t in $4-\epsilon$ and $(T-T_*)$ in $2+\epsilon$.

6 2D models: XY and Sine-Gordon

We now turn our attention to some specific two dimensional models: the XY and the Sine-Gordon model.

6.1 XY model

The XY model can be represented in various ways in general dimensions. We studied it's Landau-Ginzburgh theory for a field $\vec{\phi} = (\phi_1, \phi_2)$ with the $\vec{\phi}^4$ Hamiltonian,

$$\mathcal{H} = \int d^d x \, \left[\frac{1}{2}(\nabla\vec{\phi})^2 + \frac{t}{2}\vec{\phi}^2 + \frac{u}{4}(\vec{\phi}^2)^2 \right]$$

and we vary t to reach the critical point. This is a useful representation in "high" dimensions where mean field is exact ($d > 4$), but it acn also be used to access the WIlson-Fisher fixed point at $u_* = \mathcal{O}(\epsilon)$ in $d = 4 - \epsilon$. It is however less useful in 2D.

An alternative representation is to retain only the "would-be Goldstone modes, θ". We write the field $\vec{\phi} = (\cos\theta, \sin\theta)$ in which the Hamiltonian is,

$$\mathcal{H} = \frac{K}{2} \int d^d x \, (\nabla\theta)^2$$

We can access the same universal behaviour, meaning we can flow to the same RG fixed point, using either Hamiltonian. However there is a subltelty in the phase representation: we must properly treat the topology of θ. The equivalence of the two representations is more plausible when we think of the lattice XY model with $\phi_i = (\cos\theta_i, \sin\theta_i)$. The two field theories then corresponding to different protocols for coarse-graining ϕ_i or θ_i.

Phase diagram

We focus on $d = 2$ here and try to find the phase diagram. We consider a 2D lattice on which there is a two component vector of spins which interact ferromagnetically, and thus have the tendency to align at low enough temperature, while at the same there is competition with entropic fluctuations. Recall that for the $O(n)$ model in 2D, at large K, we found the RG equation,

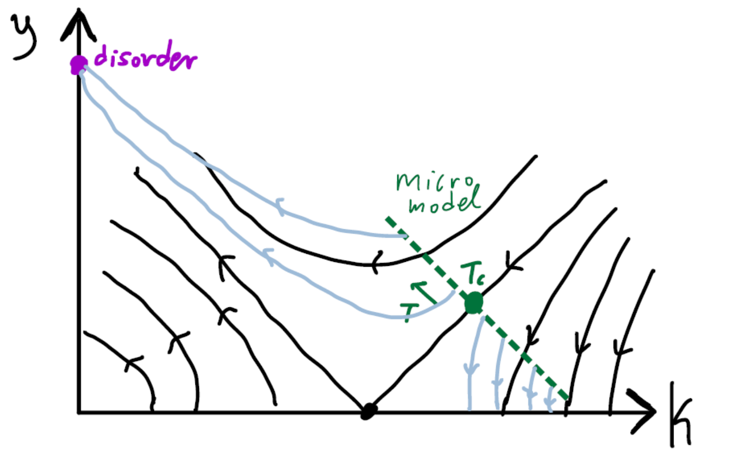
$$\frac{\mathrm{d}K}{\mathrm{d}\tau} \propto (2 - n) \iff dvT\tau \propto (n-2)T^2$$

The $(n - 2)$ terms originate from the interacting terms present in $\mathcal{H}$ when $n > 2$. To see this, recall that the vector $\vec{t}$ we used and found that was interacting contains no components for $n = 2$. Thus, in this perturbative calculation,

$$\frac{\mathrm{d}T}{\mathrm{d}\tau} = 0 \qquad \text{at all orders in } T$$

This might look like a paradox when we think about the RG flows: all the points seem to be fixed points. We call this a *fixed line*. However this cannot be the case since we know that the lattice XY model certainly has disorder in high enough temperatures which is stable. This can be seen both by a high-temperature expansion on the lattice and by the Ginzburg-Landau theory where the disorder phase is a massive phase, which is certainly possible to achieve for any n and d. In fact it seems to be impossible to write an RG equation with analytic terms which give fixed points on a finite part of the line, but on a different part, there is flow towards the disorder phase.

The perturbative treatment is in fact valid but only for large enough values of K. To understand the full range of temperatures we must consider two couplings, with the new coupling y denoting the topological defects. What we will find is that the structure of the RG flows is as in the figure below.



There is still a fixed line for any value of K, but it can be either stable or unstable in the y direction. We can imagine varying the microscopic model with a bare value of $y \neq 0$ and a large value of stiffness K. As the temperature is increase (K is decreased) and y also increases which means that it is easier to excite topological defects. There is a critical value of T, T_c, below which the model flows towards $y = 0$ and some asymptotic value of K. Above T_c, the model flows away to large values of y and small values of K, corresponding to the disorder fixed point $K = 0, y = \infty$. The paradox above was then that we were projecting a two-parameter flow diagram to only one direction. To understand this flow diagram, we need to understand an apparently unrelated model, and we will return to it shortly.

6.2 Sine-Gordon model

The 2D Sine-Gordon model is given by the Hamiltonian,

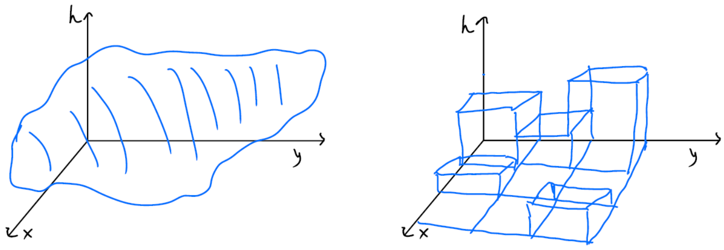
$$\mathcal{H} = \int d^2r \ \left[\frac{K}{2}(\nabla h)^2 - g\cos(2\pi h)\right]$$

for h a real scalar field, $h \in \mathbb{R}$, and the potential favours integer values of h. Tha Hamiltonian has the global symmetry described by the group $\mathbb{Z}$,

$$h \to h + (\text{integer})$$

We can think of h as a height of a surface above the $x - y$ plane. A question we can ask about the surface is whether or not it is "rough" or "smooth". Suppose that there is factor of temperature in the coupling constants so that low temperature corresponds to large K and large g, corresponding to the "smooth" phase. On the other hand, for small K and small g the surface is in the "rough" phase. The Sine-Gordon theory is related to the XY-model, Dirac fermions in $1 + 1D$ through "bosonization" and other physical systems of interest.

We can define the smooth phase by the spontaneous breaking of the global $\mathbb{Z}$ symmetry where the field h picks one of the minima of the cosine, and the field fluctuates around the chosen value, say $h = P$. In the rough phase, the $\mathbb{Z}$ symmetry is unbroken. The phases by distinguished by looking at the fluctuations of h, $\langle (h(0) - h(r))^2 \rangle$. In the smooth phase, the fluctuation in the height between two points is finite as $r \to \infty$ while in the rough phase, the fluctuations grow logarithmically as $r \to \infty$.



We can think of the field theory as the continuum analog of the lattice model known as the "solid-on-solid" model, on the right of the figure above. On a 2D lattice with square plaquettes we have integers h_p sitting inside the plaquettes. The partition function is then given by,

$$Z = \sum_{\{h\}} e^{-\frac{1}{2T} \sum_{\langle p,p' \rangle} (h_p - h_{p'})^2}$$

which preserves the $\mathbb{Z}$ symmetry of the continuum model. After coarse-graining by averaging over larger blocks, h will not be an integer in general. As in the Landau-Ginzburg arguments, in the continuum theory, we include the lowest harmonics of the field h which are compatible with the $\mathbb{Z}$ symmetry. We will see that higher harmonics like $\sim \cos(4\pi h)$ become less relevant when the potential is weak. The lattice model is then a discrete surface of square plaquettes sitting at heights h_p and we can think of it as the surface of a crystal, having the thermodynamic phase transition between the two phases explained above. This is what we will try to understand using the RG.

We want to to understand the thermodynamic properties of the fluctuating surface $h(x, y)$ using the RG. An approach commonly used in textbooks is momentum shell RG, by writing $h = h_> + h_<$. There are a few subtleties involving divergent integrals when using this method, so it is perhaps simpler to use the OPE method used and developed before.

6.3 RG for Sine-Gordon via the OPE method

We write the Hamiltonian as a fixed point Hamiltonian plus a perturbation. The Hamiltonian is,

$$\mathcal{H} = \int d^2r \, \left[\frac{K}{2} (\nabla h)^2 - g \cos(2\pi h) \right] = \mathcal{H}_* + \delta\mathcal{H}$$

The free part of the Hamiltonian is a fixed point for any value of K. Thus we will choose an arbitrary value of K as a reference and write the fixed point Hamiltonian,

$$\mathcal{H}_* = \frac{K_*}{2}\int d^2r\ (\nabla h)^2, \qquad \delta\mathcal{H} = \int d^2r\ \left[\frac{\delta K}{2} :(\nabla h)^2: -g\cos(2\pi h)\right]$$

Even though the splitting is artificial, it allows us to apply the general result of the OPE method, but we should do a consistency check at the end of the calculation to make sure that the specific value of the reference point K_* does not show up in the RG equations. We use the notation $u_1 = \frac{\delta K}{2}$ and $u_2 = -g$ to match the RG equation notation.

Aside: Two notes on normal ordering

Note. It will be convenient to be able to go between normal and non-normal order expressions, which we will do by using the handy identity,

$$e^{i\alpha h(r)} = e^{-\frac{\alpha^2}{2}G(0)} :e^{i\alpha h(r)}: \qquad \text{with } G(0) = \contraction{}{h}{(r)}{h}h(r)h(r)$$

To show this, we use the Taylor series explicitly, and use the fact that normal ordering is defined by,

$$h^k = :h^k: + (\text{terms with self contractions})$$

$$\begin{aligned} e^{i\alpha h(r)} &= \sum_{k=0}^{\infty}\frac{(i\alpha)^k}{k!}h^k \\ &= \sum_{k=0}^{\infty}\frac{(i\alpha)^k}{k!}\sum_{l\le k/2} :h^{k-2l}: G(0)^l \binom{k}{2l}\frac{(2l)!}{l!\cdot(2l)!} \end{aligned}$$

where on the second line we considered contracting l pairs by choosing $2l$ $h's$ out of the k by $\binom{k}{2l}$ to contract and the final factor is the number of ways to pairing the $2l$ objects. Canceling factors reduces to,

$$\begin{aligned} e^{i\alpha h(r)} &= \sum_{k=0}^{\infty}(i\alpha)^k\sum_{l\le k/2}\frac{:h^{k-2l}:}{(k-2l)!}\frac{[G(0)/2]^l}{l!} \\ &= \sum_{k=0}^{\infty}\sum_{l\le k/2}(i\alpha)^{k-2l}(i\alpha)^{2l}\frac{:h^{k-2l}:}{(k-2l)!}\frac{[G(0)/2]^l}{l!} \\ &= e^{-\frac{\alpha^2}{2}G(0)} :e^{i\alpha h(r)}: \end{aligned}$$

We can check that this is true by taking expectation values on both sides,

$$\begin{aligned} \langle e^{i\alpha h(r)}\rangle &= e^{-\frac{\alpha^2}{2}\langle h(r)h(r)\rangle} = e^{-\frac{\alpha^2}{2}G(0)} \\ \langle :e^{i\alpha h(r)}:\rangle &= 1, \text{since } \langle :h^k:\rangle = 0 \end{aligned}$$

Note also that $G(0)$ is IR divergent in $d=2$,

$$G(0) = \langle h(r)^2\rangle = \int_{2\pi/L}^{2\pi/a}\frac{d^2k}{(2\pi)^2}\frac{1}{k^2} \sim \frac{1}{2\pi}\log\frac{L}{a}$$

so it is sensitive to the UV cutoff and diverges with L. However, it drops out from all the results, and thus $e^{-\frac{\alpha^2}{2}G(0)}$ becomes infinitesimally small at large L, which is the reason it is not included in the Hamiltonian density in $\delta\mathcal{H}$.

Note. To elaborate on the above calcuation we make some more remarks on normal ordering. We previously defined $:h^k:$ via the evaluation in correlators by saying that no contractions are allowed within $:\ldots:$. Equivalently, we can say that the un-normal ordered operator is,

$$h^k = :(\text{Sum of all numbers and patterns of self-contractions}):$$

which includes the term with no contractions. For example,

$$\begin{aligned} h^2 &= :h^2: + \wick{\c h \c h} \\ h^4 &= :h^4: + 6\wick{\c h \c h}\, :hh: + 3(\wick{\c h \c h})^2 \end{aligned}$$

In free field theory, if we compute the correlation function involving $:h^2:$, we have,

$$\langle :h^2:(\ldots)\rangle = \langle \wick{\c1 h \c2 h (\c1 . \c2 . .)}\rangle$$

so that the effect of the normal ordering is to only allow contractions of the field h with stuff in $(\ldots)$. For example,

$$\begin{aligned} \langle h^2(\ldots)\rangle &= \langle \wick{\c1 h \c2 h (\c1 . \c2 . .)}\rangle + \langle \wick{\c h \c h}(\ldots)\rangle \\ &= \langle :h^2:(\ldots)\rangle + \langle \wick{\c h \c h}(\ldots)\rangle \end{aligned}$$

Using the above we can also solve for $:h^4:$, giving,

$$:h^4: = h^4 - 6\wick{\c h \c h} h^2 + 3(\wick{\c h \c h})^2$$

Alternatively, we can define normal ordering not by contractions but by the operators themselves. For simplicit we consider operators h^k at a single point. We claim that it is defined by,

$$:h^k: = e^{-\wick{\c h \c h}(\frac{1}{2}\frac{d^2}{dh^2})} h^k$$

where as usual, the exponential is defined by its power series. For example,

$$\begin{aligned} :h^2: &= \left(1 - \wick{\c h \c h}\frac{1}{2}\frac{d}{dh}\frac{d}{dh}\right) h^2 \\ &= h^2 - \wick{\c h \c h} \end{aligned}$$

This definition holds for a general function $f(h)$, as long as it admits an analytic expansion in powers of h,

$$:f(h): = e^{-\wick{\c h \c h}(\frac{1}{2}\frac{d^2}{dh^2})} f(h)$$

This can also be inverted to give a definition of h^k in terms of the normal ordered powers,

$$\boxed{h^k = :e^{-\wick{\c h \c h}(\frac{1}{2}\frac{d^2}{dh^2})} h^k:}$$

This agrees with our definition above as can be verified by direct computation.

$$\begin{aligned} :e^{i\alpha h}: &= e^{-\frac{1}{2}\wick{\c h \c h}\frac{d^2}{dh^2}} e^{i\alpha h} \\ \left(\frac{d}{dh} \leftrightarrow i\alpha\right) \qquad &= e^{-\frac{1}{2}G(0)(i\alpha)^2} e^{i\alpha h} \\ &= e^{\frac{\alpha^2}{2}G(0)} e^{i\alpha h} \end{aligned}$$

OPE coefficients

We now compute the RG eigenvalues and the OPE coefficients in order to set up the RG equations for the Sine-Gordon model. Let the operators be given by,

$$O_1 = :(\nabla h)^2:, \quad O_2 = \frac{1}{2}\left(e^{2\pi i h} + e^{-2\pi i h}\right)$$

We first consider two point functions,

$$\begin{aligned} G(r) - G(0) &= \langle h(0)h(r)\rangle - \langle h(0)^2\rangle \\ &= \frac{1}{K_*}\int_{|k|\leq \frac{2\pi}{a}} \frac{d^2k}{(2\pi)^2}\,\frac{e^{ik\cdot r}-1}{k^2} \\ &= -\frac{1}{2\pi K_*}\left(\ln\frac{\pi r}{a} + \gamma + \dots\right) \end{aligned}$$

and thus it is IR finite as $L \to \infty$. The constant here is called the "Euler-Mascheroni constant" and has value $\gamma = 0.5772\dots$. It is non-universal and depends on the normalisation of the length. It is convenient to measure lengths in units such that the constant term vanishes, and the correlation function is simply,

$$G(r) - G(0) = -\frac{1}{2\pi K_*}\ln r$$

which in turn gives a simple expression for the exponential operator,

$$\begin{aligned} \langle e^{2\pi i h(0)} e^{-2\pi i h(r)}\rangle &= \langle e^{2\pi i[h(0)-h(r)]}\rangle \\ &= e^{\frac{1}{2}(2\pi i)^2\langle [h(0)-h(r)]^2\rangle} \\ &= e^{-2\pi^2[2G(0)-2G(r)]} \\ &= r^{-2\pi/K_*} = r^{-2X_2} \end{aligned}$$

That tells us also the scaling dimension of the cosine operator,

$$\begin{aligned} \langle \cos(2\pi h(0))\cos(2\pi h(r))\rangle = \frac{1}{4}\Big(&\langle e^{2\pi i h(0)} e^{-2\pi i h(r)}\rangle + \langle e^{-2\pi i h(0)} e^{2\pi i h(r)}\rangle \\ &+ \cancel{\langle e^{2\pi i h(0)} e^{2\pi i h(r)}\rangle} + \cancel{\langle e^{-2\pi i h(0)} e^{-2\pi i h(r)}\rangle}\Big) \end{aligned}$$

where the last two terms vanish in the thermodynamic limit. We can show this by computing an integral like the one above.

We can immediately read off the scaling dimension of O_2 from the two point function computed above, $X_2 = \frac{\pi}{K_*}$. The scaling dimension of O_1 is simply, $X_1 = 2$, which we find by counting derivatives. The two-point function of h is a logarithm and thus the scaling dimension of h is 0. By differentiating we increase the dimension to 2. More formally, we need to compute, $\langle :(\nabla h)^2::(\nabla h)^2:\rangle$ using Wick's theorem and find that it is written as 4 derivatives of $G(r)^2$, so that it becomes $\sim \frac{1}{r^4}$. From $y = d - X$, we get the RG eigenvalues, $y_1 = 0$ and $y_2 = 2 - \frac{\pi}{K_*}$. y_1 is expected to be zero because the stiffness is exactly marginal: the theory is a fixed point for any value of K, and thus its RG eigenvalue has to be 0.

For clarity, we define the *vertex operator*, $V(r) = e^{2\pi i h(r)}$. We have the shift symmetry of the field here, $h \to h + \chi$, which is a symmetry for any $\chi \in \mathbb{R}$ in the fixed point theory $\mathcal{H}_*$, but only for $\chi \in \mathbb{Z}$ in the interacting theory. We can define an operator of charge q under the symmetry,

$$V_q \to e^{2\pi i q\chi} V_q$$

For example, V has charge 1, V^* has charge -1 and $:(\nabla h)^2:$ has charge 0. In writing the OPE, the charges on both sides of the equations must match, which immediately gives us,

$$c_{112} = c_{121} = c_{222} = 0$$

Let us see this explicitly. For c_{112} we have,

$$:(\nabla h)^2::(\nabla h)^2: = \sum (\text{charge 0 operators})$$

and since

$$O_2 = (\text{charge +1 operator}) + (\text{charge -1 operator})$$

we see that O_2 can't appear on the RHS. For c_{121},

$$:(\nabla h)^2:\frac{1}{2}(V + V^*) = \sum (\text{charge +1 operators}) + \sum (\text{charge -1 operators})$$

so that no charge 0 operators appear on the RHS, and O_1 can't appear. For c_{222},

$$\frac{1}{2}(V + V^*)\frac{1}{2}(V + V^*) = (\text{charge +2 operators}) + (\text{charge -2 operators}) + (\text{charge 0 operators})$$

so that V and V^* can't appear on their own on the RHS. However, $c_{221} \neq 0$ since $:(\nabla h)^2:$ can appear. Additionally, $c_{111} = 0$, which follows from the fact that for $g = 0$ the theory is Gaussian and thus a fixed point for any K. This can also be checked explicitly,

$$:(\nabla h)^2::(\nabla h)^2: = :\nabla h(0)\nabla h(0)::\nabla h(r)\nabla h(r):$$

and for c_{111} we look at terms on RHS which are quadratic in h. We have to contract on pair of $h's$ to get a term $\propto \nabla_\mu \overbracket{h(0)\nabla_\mu h}(r) = -\nabla_\mu\nabla_\mu G(r) = 0$ since $G(r)$ in FFT satisfies the equation of motion for $r \neq 0$.

Given the above, there are only two OPE coefficients to calculate: c_{122} and c_{221}.
$\underline{c_{122}}$: In looking at the product of the two operators, it is convenient to take the normal ordering of O_2 since it corresponds only to a multiplication by the constant $e^{\frac{1}{2}G(0)(2\pi)^2}$, and remove the constant in the end,

$$:\nabla h(0)\nabla h(0)::e^{2\pi i h(r)}: = :\nabla h(0)\nabla h(0):\sum_k :h(r)\ldots h(r):\frac{(2\pi i)^k}{k!}$$

We are after the coefficient of $:e^{2\pi i h}:$, an operator with no derivatives. We thus need to get rid of ∇h operators by contractng both of them. We think of the contraction of the operators under the derivative as,

$$\nabla_{r'}\overbracket{h(r')\nabla_r h}(r)\big|_{r'=r} = (\nabla_\mu G(r))^2$$

so that the end result is,

$$(\nabla_\mu G(r))^2 \sum_{k\geq 2} \frac{(2\pi i)^k}{k!} :h(r)^{k-2}: \cdot k \cdot (k-1) = (\nabla_\mu G(r))^2(-4\pi^2) :e^{2\pi i h(r)}:$$

with,

$$(\nabla_\mu G(r))^2 = (\nabla_\mu G(r)) \cdot (\nabla_\mu G(r)) = \frac{1}{(2\pi K_*)^2} \sum_\mu (\partial_\mu \ln r)^2$$
$$= \frac{1}{(2\pi K_*)^2} \sum_\mu \left(\ln \frac{r_\mu}{r^2}\right)^2$$
$$= \frac{1}{(2\pi K_*)^2 r^2}$$

Putting everything together, we find,

$$:\nabla h(0)\nabla h(0)::e^{2\pi i h(r)}: = -\frac{1}{K_*^2} \cdot \frac{1}{r^2} :e^{2\pi i h(r)}: + \dots$$

with $\dots$ including other operators from the OPE that we are not interested here. Recall that the OPE formula is $O_1(0)O_2(r) = \frac{c_{122}}{r^{X_1+X_2-X_2}} O_2(r/2) + \dots$, and here $:e^{2\pi i h(r)}: = O_2(r/2) + (\text{deriv. ops})$. Since $X_1 = 2$, the power of r is as expected, and we can read off the OPE coefficient,

$$c_{122} = -\frac{1}{K_*^2}$$

As we see, $:e^{2\pi i h(r)}:$ appears on both sides of the equation, and since it corresponds to multiplying with a constant, what we found is actually the OPE coefficient for the un-normal ordered operator.

$\underline{c_{221}}$: The last OPE coefficient we need comes from the operator product,

$$\cos(2\pi h(0))\cos(2\pi h(r)) \sim \frac{e^{-4\pi^2 G(0)}}{4}\left(\underbrace{:e^{2\pi i h(0)}::e^{-2\pi i h(r)}:}_{\text{Charge } 0} + \underbrace{:e^{-2\pi i h(0)}::e^{2\pi i h(r)}:}_{\text{Charge } 0} + \underbrace{\dots\dots}_{\text{Charge } \pm 2}\right)$$

and we have converted again the RHS to normal ordered form by just multiplying with the factor $e^{\frac{1}{2}G(0)(2\pi)^2}$ for each exponential. The two terms with charge 0 contribute in the same way to the OPE, so we just multiply this by a factor of 2. We thus need to extract the coefficient of $:(\nabla h(r/2))^2:$ in the first term above. As before, we first Taylor expand and then contract all but two of the h's.

$$\frac{e^{-4\pi^2 G(0)}}{2} :e^{2\pi i h(0)}::e^{-2\pi i h(r)}: = \frac{e^{-4\pi^2 G(0)}}{2} \sum_{m,n} \frac{(2\pi i)^m(-2\pi i)^m}{m!n!} :h(0)\dots h(0)::h(r)\dots h(r):$$

There are now three cases,

1. $m = n$ and contract $(n-1)$ pairs $\implies \propto h(0)h(r)$
2. $m = n+2$ and contract n pairs $\implies \propto h(0)^2$
3. $n = m+2$ and contract m pairs $\implies \propto h(r)^2$

We consider case (1) explicitly,

$$
\begin{aligned}
\frac{e^{-4\pi^2 G(0)}}{2} :e^{2\pi i h(0)}::e^{-2\pi i h(r)}: &= \frac{e^{-4\pi^2 G(0)}}{2} \sum_m \frac{(-1)^m(-4\pi^2)^m}{(m!)^2} G(r)^{m-1} :h(0)h(r): \cdot m \cdot m \cdot \underbrace{(m-1)!}_{\text{pairings of contr. terms}} \\
&= \frac{e^{-4\pi^2 G(0)}}{2} \sum_m \frac{(4\pi^2)^m}{(m-1)!} G(r)^{m-1} :h(0)h(r): \\
&= 2\pi^2 e^{-4\pi^2 G(0)} e^{4\pi^2 G(r)} :h(0)h(r): \\
&= 2\pi^2 e^{-4\pi^2 [G(0)-G(r)]} :h(0)h(r): \\
&= \frac{2\pi^2}{r^{2\pi/K_*}} :h(0)h(r):
\end{aligned}
$$

All together, we find that theproduct of cosines is,

$$
\begin{aligned}
\cos(2\pi h(0))\cos(2\pi h(r)) &= \cdots + \frac{2\pi^2}{r^{2\pi/K_*}} \left[:h(0)h(r): - \frac{1}{2} :h(0)^2: - \frac{1}{2} :h(r)^2: \right] \\
&= \cdots + \frac{2\pi^2}{r^{2\pi/K_*}} \left(\frac{1}{2}\right) :(h(0)-h(r))^2:
\end{aligned}
$$

where we used the fact that the LHS is invariant under $h \to h+1$ enforcing the RHS to be as well. This fixes the coefficients for cases (2) and (3) above to be a $-1/2$. We can now Taylor expand the term in the normal ordering which to leading order gives,

$$
:(h(0)-h(r))^2: = :(\nabla h)^2: + \dots
$$

To check for consistency the power of r, we expect its power to be $X_2 + X_2 - X_1 = 2X_2 = 2\pi/K_*$. Therefore,

$$
c_{221} = -\pi^2
$$

RG equations

We are now in position to write the RG equations for the couplings u_1, u_2.

$$
\begin{aligned}
\frac{\mathrm{d}u_1}{\mathrm{d}\tau} &= y_1 u_1 - \pi c_{221} u_2^2 = \pi^3 u_2^2 \\
\frac{\mathrm{d}u_2}{\mathrm{d}\tau} &= y_2 u_2 - 2\pi c_{122} u_1 u_2 = \left(2 - \frac{\pi}{K_*}\right) u_2 + \frac{2\pi}{K_*^2} u_1 u_2
\end{aligned}
$$

It more natural to write them in terms of the original couplings for which $u_1 = \frac{\delta K}{2}$ and $u_2 = g$.

$$
\begin{aligned}
\frac{\mathrm{d}\delta K}{\mathrm{d}\tau} &= 2\pi^3 g^2 \\
\frac{\mathrm{d}g}{\mathrm{d}\tau} &= \left(2 - \frac{\pi}{K_*}\right) g + \frac{\pi}{K_*^2} \delta K g
\end{aligned}
$$

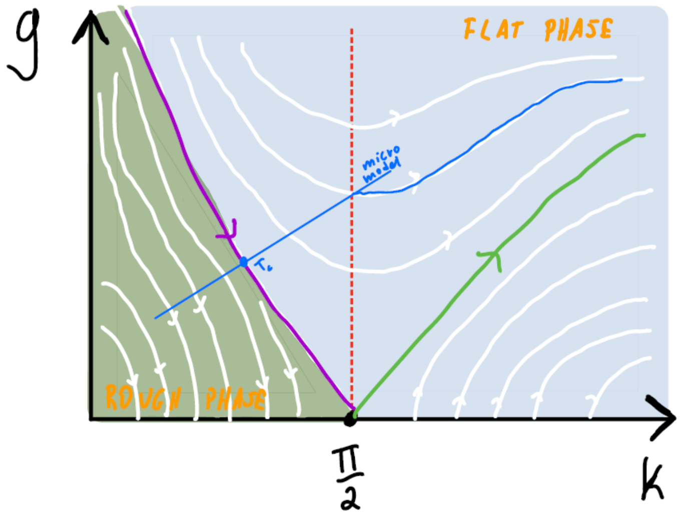
As a consistency check we note that we have chosen to expand around K_* but this was an arbitrary choice. The RG equation for g should depend only on $K = K_* + \delta K$ and not on K_* and δK separately. The physical Hamiltonian only *sees* K and the splitting was just a book-keeping exercise.

The flows should then be independent on it, at least to the order that we did the calculation. To see that this is the case we write the RG equations in terms of g and K,

$$\begin{aligned}\frac{\mathrm{d}K}{\mathrm{d}\tau} &= 2\pi^3 g + \mathcal{O}\left(g^4\right)\\ \frac{\mathrm{d}g}{\mathrm{d}\tau} &= \left(2 - \frac{\pi}{K}\right) g + \mathcal{O}\left(g^3\right)\end{aligned}$$

and we have used the symmetry of the RG equations $g \to -g$, which holds because the theories with $\pm g$ are related by a change of variable $h \to h + \frac{1}{2}$. The first term in the RG equation for g is exact since it is the RG eigenvalue of g.

We first give some qualitative remarks on the structure of the flows. There is a critical value of $K_c = \frac{\pi}{2}$ where the *RG* eigenvalue changes sign. For small stiffness $K < K_*$, the field is floppy with lots of fluctuations and the potential is weak under the RG. In the limit of $g \to 0$ the flow line is vertical since the RHS for the g-equation is linear in g whereas the flow in K is quadratic for small g. The the flow in the vertical direction is parametrically faster than the flow in the horizontal direction. Similarly, if $K > K_*$, the field is stiff and it is more easily "trapped" by the potential. The flows here are in the opposite direction. At $K = K_c$, g is not flowing, and K flows horizontally since $\frac{\mathrm{d}K}{\mathrm{d}\tau}$ is positive. With a more careful treatment we will find a separatrix which flows in towards K_c.



What the flow diagram shows is essentially two different phases. On the right of the separatrix there is a region where the flows will end up to large K and large g. The field here is in a strong potential so we expect that the symmetry is broken and the field spontaneously chooses one of the minima of cosine. Expanding around the minimum we find massive (quadratic) fluctuations. This phase is called the "flat phase" or "smooth phase". The other phase, called the "rough phase" is the one where the flows end up to $g = 0$.

To understand the difference between the two phases, we consider two points on the surface at 0 and r and ask what is the difference in height $|h(0) - h(r)|$ between these two points. We look at the two-point function,

$$\langle (h(0) - h(r))^2 \rangle_{\text{rough}} \propto \frac{1}{\tilde{K}} \ln r$$

since in the rough phase we essentially have a free theory, and here $\tilde{K}$ is the renormalised value for K. The result is then that the size of fluctuations diverge with the separation r. In the flat phase, we look at the fluctuations within the picked minimum,

$$\langle (h(0) - h(r))^2 \rangle_{\text{flat}} \propto \mathcal{O}(1)$$

even when r is large.

The rough phase is not really a phase since it is not given by a single fixed point. It is universal and governed by an RG fixed line. The point on the fixed line that the microscopic model will flow to, depends on the initial conditions. But since K is continuoysly varying on the fixed line, this means that we have continuoysly varying scaling exponents. We consider vertex operators of charge q and find,

$$e^{2\pi i q h(0)} e^{-2\pi i q h(r)} \sim r^{-2\pi q^2/K}$$

giving $X_q = \frac{\pi q^2}{K}$. To get a fixed line we need to have an exactly marginal coupling constant, as is K in this case, and $\frac{\mathrm{d}K}{\mathrm{d}\tau} = 0$ when $g = 0$.

Let us look at the separatrix in a little more detail by studying the flows near $(K, g) = (\pi/2, 0)$. We write, $K = \frac{4}{\pi}(k - \frac{\pi}{2})$ and absorb the $2\pi^2$ constant into g, so that the RG equations for k, g are,

$$\frac{\mathrm{d}k}{\mathrm{d}\tau} = g^2 \qquad \frac{\mathrm{d}g}{\mathrm{d}\tau} = kg$$

to quadratic order in k, g. The separatrix is now the line $k = -g$ in which case the solution for g is,

$$\boxed{g(\tau) = \frac{g(0)}{1 + g(0)\tau}}$$

We can now connect the Sine-Gordon model with the microscopic model of the discrete theory which had only integer translational symmetry of the lattice, the “solid on solid model”, controlled by a parameter T. Heuristically, as we vary T in the SOS model, we move along a trajectory in the theory space which crosses the separatrix at T_c (high T would correspond to low stiffness and low g). At low temperatures the model flows to the flat phase while for low T the flow is towards the fixed line. This is only heuristic though because we only used two couplings while more couplings would be necessary to describe the long wavelength properties of the microscopic limit.

Exercise 3. *In the rough phase, what is the probability that two points at distance r lie in the same minimum of the cosine potential?*

6.4 2D XY-model and KT-transition

We have seen so far that among the $O(n)$ models with continuous symmetry ($n \geq 2$) only the case $n = 2$ (XY model) has a phase transition in 2D. For $n > 2$ we must go to $2 + \epsilon$ as we saw. The key difference is that the Hamiltonian of the non-linear sigma model formulation becomes quadratic if we parametrise the field in terms of an angle,

$$\mathcal{H} = \frac{J}{2} \int d^2 r \, (\nabla\theta)^2$$

In general for large J and arbitrary n, the RG equation is,

$$\frac{\mathrm{d}J}{\mathrm{d}\tau} = -\left(\frac{n-2}{2\pi}\right) + \mathcal{O}(1/J)$$

For $n > 2$, there is flow towards a disorded fixed point. But for $n = 2$ the RG equation vanishes to all orders in $1/J$ simply because of the free form of J. The Hamiltonian then describes a fixed line parametrised by J and governed by free field theory. We will see that we indeed have a fixed line, which is not obvious since θ is a compact variable. However this fixed line is only stable for $J \geq J_c$ with $J_c = \frac{2}{\pi}$.

For $J < J_c$ ($T > T_c$), the fixed line is destablised by non perturbative effects that are not captured by simply perturbation theory in $1/J$. In fact, perturbation theory in that case is trivial since there are no interactions. The non-perturbative interactions are the *vortices*. For $J \geq J_c$, on the fixed line we have continuously varying exponents,

$$\langle \vec{S}(0) \cdot \vec{S}(r) \rangle = \langle \cos(\theta(r) - \theta(0)) \rangle = \langle e^{i(\theta(r)-\theta(0))} \rangle = e^{-\frac{1}{2}\langle(\theta(r)-\theta(0))^2\rangle} = \frac{1}{r^{\frac{1}{2\pi J}}}$$

This is called "quasi-long range order" which differs from true long-range order in that the two point function $\langle \vec{S} \cdot \vec{S} \rangle$ does not approach a positive constant at large distances. At the same time however, it is also different from a trivial disordered phase where $\langle \vec{S} \cdot \vec{S} \rangle \sim e^{-r/\xi}$.

Vortices and Kosterlitz-Thelouss

We will first try to understand what happens if we decrease J by thinking in terms of the original lattice formulation. At some point, we expect that the system must enter some convential disorder phase, since $J = 0$ is a stable fixed point. This might seem contradictory when looking at Hamiltonian which is a free fixed line for any J. The key idea to remember though is that the angle is a compact variable,

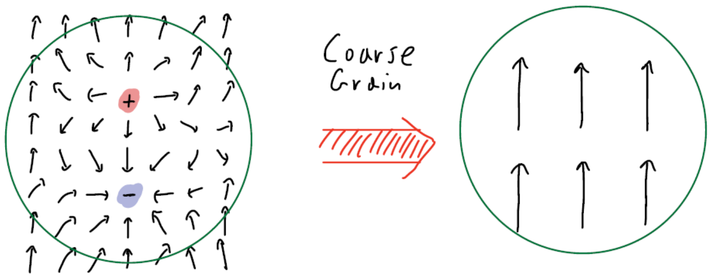
$$\theta \equiv \theta + 2\pi$$

This means that there are additional configurations which are not allowed in the free case when $\theta \in \mathbb{R}$. Here there are configurations where θ jumps by 2π, known as *vortices*. Vortices are point-like singularities characterised by a charge, positive or negative, which specifies the winding of the phase θ as we go around the centre. Mathematically we have,

$$\oint d\theta = \oint (\nabla\theta) \cdot d\vec{r} = 2\pi \cdot m$$

for m an integer. Such configurations are neglected when θ is a free field but in a lattice model, they are perfectly valid configurations.

Let us first try to understand when the neglect of these configurations from free field treatment can be consistent. The free field description will be valid if there are no vortices. This can be the case for example when the energetic cost of having them is very large. More generally though, the configurations with vortices are valid if the vortices disappear under coarse-graining. This can be the case when the vortices appear in bound pairs of positive and negative charges in some microscopic region, and by integrating over the bound pair will give zero, $\oint d\theta = 0$. An example of such a configuration which disapperas under coarse-graining is shown below.



So when is it consistent to assume that these vortices disappear under coarse-graining? To answer the question we will run a consistency check by assuming that the vortices disappear under RG and check that this is a self-consistent assumption, free of contradictions. Thus we assume that after coarse-graining to some scale L_0 the vortices are absent. We then compute the free energy cost ΔF of introducing a dilute gas of vortices with typical separation between vortices $L_* > L_0$. If $\Delta F > 0$, then the assumption is self-consistent, where is $\Delta F < 0$, it isn't. The state that we assumed to be the equilibrium state is not the state of lowest free energy. The change in the free energy is given by, $\beta \Delta F = \beta E - S/k_B$ where E is the energy of the configuration and S its entropy. We compute E and S per vortex.

Entropy: We will do a crude estimate of the entropy by considering a dilute gas of vortices with mean separation L_*. In a box of size L_* there will be on average a single vortex. Imagining that th box is sitting on the lattice of spacing a, the box contains $(L_*/a)^2$ plaquettes which corresponds to the total number of locations that the vortex can be placed. Thus a crude estimate for the entropy is simply,

$$S = \ln\left(\frac{L_*}{a}\right)^2 = 2\ln\frac{L_*}{a} = 2\ln\left(\text{No. of states for vortex within box}\right)$$

The reason that this estimate is okay to use is the logarithmic form S. Even though the UV cutoff suffers from some $\mathcal{O}(1)$ ambiguity, the large L_* behaviour will be independent of that cutoff, at least to leading order.

Energy: We first consider the energy of a rotationally symmetric vortex within a disc of size L_*. We can take $\theta = \tan^{-1}(y/x)$, so that $|\nabla\theta = 1/r$. To see this, we basically integrate around a circular path of radius r and the total change should be 2π, $\int dl|\nabla\theta| = 2\pi = (2\pi r)|\nabla\theta|$. The energy is then given by,

$$\beta E = \frac{J}{2}\int d^2r\,(\nabla\theta)^2 = \frac{J}{2}\cdot 2\pi\int_a^{L_*}\frac{dr}{r} = \pi J\ln\left(\frac{L_*}{a}\right)$$

The calculation is also valid to leading order at large L_* even for disorded configurations without rotational invariance. This is essentially because on scales $\gg L_*$ and $\ll a$, the configuration still looks rotationally symmetric. You can try to compute the integral in different geometries to see that the leading is the same. This calculation misses the $\mathcal{O}(1)$ terms from contributions very close to the vortex. Let us call this core energy cost, E_{UV}, although its effect is subleading.

The cost of the free energy per vortex is thus,

$$\beta f = \beta E_{UV} + (\pi J - 2)\ln\left(\frac{L_*}{a}\right)$$

There is a critical value $J_c = \frac{2}{\pi}$ so that,

- $J > J_c$: Δf is positive which is expected since it these vortices cost energy to "excite"
- $J < J_c$: Δf may be positive for small L_* if E_{UV} large enough. Eventually however, since E_{UV} is fixed, for sufficiently enough L_*, Δf will become negative.

This shows that there is a scale ξ beyond which vortices proliferate. Once the vortices proliferate because it is thermodynamically favourable to do so, the system is essentially in the disorder phase.

Note. In this treatment we implicitly used the quadratic nature of the Hamiltonian to assume that fluctuations around classical solutions with fixed vortices give a contribution to the free energy which is independent of the classical solution. This means that in computing the energy cost of adding a vortex, we didn't care about how the vortex change the free energy of the quadratic fluctuations around a given configuration.

We can justify the above more regorously. We know that J_c is the location of the phase transition between QLRO and disorder. We also note that this is the first example of a phase transition without symmetry breaking, and it is known as the "Kosterlitz-Thouless transition", who first understood it in the context of the RG. This transition is relevant in the case of superfluid in thin films.

RG treatment: Initial Remarks

We will show now that the XY model is dual to Sine-Gordon with,

$$J_{XY} = \frac{1}{K_{SG}}$$

with the subscript put explicitly here to denote the coupling in each model, and we will drop it from now on. The core energy of a vortex, E_{UV} is related to the coupling constant in the Sine-Gordon model through,

$$e^{-\beta E_{UV}} \propto g$$

We will show that this mapping holds but for now we will think about the RG flows in the XY-model, assuming that these relations hold. The separatrix is now on the right of the critical point, and on that side, flows end up on the fixed line which is the semi-infinite line from J_c to ∞. This side describes the QLRO. On the left of the separatrix, we have the disorded phase and large vortex fugacity, g.

The fixed line exists for all $g > 0$ but it becomes unstable for $J < J_c$. We can see this explicitly by the RG equations (to be derived later),

$$\frac{\mathrm{d}g}{\mathrm{d}\tau} = (2 - \pi J)g$$
$$\frac{\mathrm{d}J}{\mathrm{d}\tau} = -g^2$$

In the Sine-Gordon model, g is the coefficient of the potential which renormalises the Sine-Gordon stiffness K upwards. This made sense in the physical model with the height by thinking about the potential as trying to pin the field by making it stiffer. On the other hand, in the XY model, vortices renormalise J downwards, as can be seen by the minus sign in the RG equation of J. The factor of g^2 comes out by integrating out pairs of vortices. Additionally, in the Sine-Gordon model, the fixed line is stable when the field h has a small stiffness: a floppy system doesn't *see* the potential, and it avoids pinning to $\cos(2\pi h)$ so that it renormalises to zero and at large length scales, free theory is a good description. In the XY model, the fixed line is stable when the field θ has a large stiffness. This is because a large stiffness allows the vortices in the system to be suppressed at large scales.

Duality between XY and Sine-Gordon

The duality between the models is an important example of the fact that two very different looking field theorys can have a hidden equivalence. We will now prove that the two models are dual to each other. We start by describing an abstract way of thinking about the vortices in the XY model, by introducing "vortex operators", $V_+(r), V_-(r)$, defined formally as follows. We define a "primed" functional integral " $\int'$, to mean the fact that we should only integrate over vortex free configurations,

$$\int' \mathcal{D}\theta \; e^{-\frac{J}{2}\int(\nabla\theta)^2}$$

In this way, we can treat θ like an ordinary free field, since there are no locations where θ is not defined or discontinuously jumps by 2π.

We then define the path integral with vortex operators,

$$\int' \mathcal{D}\theta \; e^{-\frac{J}{2}\int(\nabla\theta)^2} V_+(r_1)V_+(r_2)\ldots V_-(r_{m+1})\ldots V_-(r_{2m})$$

defined as a functional integral over field configurations θ with vortices at $r_1,\ldots r_m$ and antivortices at $r_{m+1}\ldots r_{2m}$. We can thus rewrite the above as a correlation function multiplying the vortex free partition function,

$$Z' \left\langle V_+(r_1)V_+(r_2)\ldots V_-(r_{m+1})\ldots V_-(r_{2m})\right\rangle'$$

Here we assumed the same number of V_+ and V_-. As an exercise you can show that on closed manifolds, the number of vortices is the same as the number of antivortices. The correlation function $\langle V_+\ldots V_-\ldots\rangle$ is then defined as the number of V_+ different from V_-. The interpretation of the vortex operators are that $V_+(r)$ inserts a vortex at the point r.

We now use the vortex operators to write the partition function of the XY model, Z_{XY},

$$\begin{aligned}
Z[J,g] &= \int' \mathcal{D}\theta \; e^{-\frac{J}{2}\int(\nabla\theta)^2+\frac{g}{2}\int[V_+(r)+V_-(r)]} \\
&= \int' \mathcal{D}\theta \; e^{-\frac{J}{2}\int(\nabla\theta)^2} \exp\left[\frac{g}{2}\int d^2r \sum_{\sigma'=\pm} V_{\sigma'}(r)\right] \\
&= \sum_{N=0}^{\infty}\left(\frac{g}{2}\right)^N \int \frac{d^2r_1\ldots d^2r_N}{N!} \sum_{\sigma_1\ldots\sigma_N} \int' \mathcal{D}\theta e^{-\frac{J}{2}\int\,(\nabla\theta)^2} V_{\sigma_1}(r_1)\ldots V_{\sigma_N}(r_N)
\end{aligned}$$

and the last steps shows that this is precisely the partition function of the XY-model including the vortices. The sum over N is the sum over all possible numbers of vortices and antivortices.

We also integrate over all possible positions of the vortices, and the $1/N!$ factor compensates for overcounting due to relabelling $r_i \leftrightarrow r_j$. The sum $\sum_{\{\sigma_i\}}$ is over all the possible assignments of vortices, $\sigma = \pm$ for each vortex. Finally, there is a "fugacity", $\propto \frac{g}{2}$ for each vortex, where the $\propto$ is to amount to the effect of the UV regularisation on the normalisation of V. More precisely, the fugacity carries a core energy, with Boltzmann weight $e^{-\beta E_{UV}}$ which is proportional to $\frac{g}{2}$ up to some $\mathcal{O}(1)$ constant.

Formally this now looks like the problems of the type that we have considered before, with Hamiltonian of the form,

$$\mathcal{H} = \mathcal{H}_* + g\mathcal{O}$$

where $\mathcal{O}$ is some perturbation to the fixed point Hamiltonian $\mathcal{H}_*$. We could apply now the RG directly to compute the scaling dimensions of the operators, the OPE coefficients etc. However, we take a different approach and instead we will exhibit the duality in which $V_\pm(r)$ will become ordinary local operators written in terms of a field that is integrated over in a functional integral.

Exercise 4. *More generally we can define an operator that inserts a vortex of strength q, V_q, with $q \in \mathbb{Z}$. Then the two point function,*

$$\langle V_q(0) V_{-q}(r)\rangle$$

is equal to a ratio of partition functions with and without vortex insertions. Show that this ratio can be expressed in terms of the energy of a the saddle point configuration of θ for a vortex pair. Then you can show using this that,

$$\langle V_q(0) V_{-q}(r)\rangle \sim r^{-2X_q}$$

with $X_q = \pi J q^2$.

Let us now go back to seeking the realtion with the Sine-Gordon model. We first explore the duality in the absence of vortices. We start with a primed functional integral,

$$\int' \mathcal{D}\theta \; e^{-\frac{J}{2}\int(\nabla\theta)^2} \mapsto \int \mathcal{D}h \; e^{-\frac{K}{2}\int(\nabla\theta)^2}$$

mapped to an ordinary functional integral over the dual field h, with $K = 1/J$. We will then extend this to include vortices, and find the duality,

$$V_\pm(r) \mapsto e^{\pm 2\pi i h(r)}$$

The first claim to be proved is thus that the partition function of the vortex free XY model, *schematically* written as "$\int \mathcal{D}\theta \; e^{-\frac{J}{2}\int(\nabla\theta)^2}$", is such that,

$$\begin{aligned} Z_{XY} &= \int \mathcal{D}\theta \; e^{-\frac{J}{2}\int(\nabla\theta)^2} \\ &\equiv \int' \mathcal{D}\theta \; e^{-\frac{J}{2}\int(\nabla\theta)^2 + \frac{g}{2}\int[V_+(r)+V_-(r)]} \\ &= \int' \mathcal{D}h \; e^{-\frac{K}{2}\int(\nabla h)^2 + \frac{g}{2}\int\cos(2\pi h)} \end{aligned}$$

This is easy in the vortex free case. We will use the Hubbard-Stratonovich transformation with new field s_μ which couples to $\nabla_\mu\theta$. Thus,

$$\begin{aligned} Z' &= \int' \mathcal{D}\theta\ e^{-\frac{J}{2}\int(\nabla\theta)^2} \propto \int' \mathcal{D}\theta \int \mathcal{D}s_\mu\ e^{-\frac{1}{2J}\int s_\mu s_\mu + i\int s_\mu(\nabla_\mu\theta)} \\ &\to \int' \mathcal{D}\theta\ e^{-i\int \theta\cdot(\nabla s)} \end{aligned}$$

having integrated by parts. This is now a functional version of the delta function representation. To see this, we can imagine defining it on a lattice, and there is then an "ordinary" delta function at each site. Integrating now over θ gives the constraint,

$$\nabla\cdot s = 0$$

and the partition function takes the form,

$$Z' = \int \mathcal{D}s_\mu e^{-\frac{1}{2J}\int s^2}\delta(\nabla\cdot s)$$

We can now solve for the contraint explicitly by taking $s_\mu = \epsilon_{\mu\nu}\nabla_\nu h$, so that

$$\nabla_\mu s_\mu = \epsilon_{\mu\nu}\nabla_\mu\nabla_\nu h = 0$$

which vanishes by the antisymmetry of $\epsilon_{\mu\nu}$. Plugging it back in and defining $J = \frac{1}{K}$,

$$Z' = \int \mathcal{D}h e^{-\frac{K}{2}\int(\nabla h)^2}$$

and we used the identity $(\epsilon\nabla h)^2 = (\nabla h)^2$.

We now repeat the calculation by including vortex operators. We have,

$$\begin{aligned} \tilde{Z} &= \int' \mathcal{D}\theta\ e^{-\frac{J}{2}\int(\nabla\theta)^2} V_\pm(r_1)\ldots \\ &\propto \int' \mathcal{D}\theta \int \mathcal{D}s_\mu\ e^{-\frac{1}{2J}\int s_\mu s_\mu + i\int s_\mu(\nabla_\mu\theta)} V_{\sigma_1}(r_1)\ldots \end{aligned}$$

To perform the integral, it is easiest to extract all the singular terms arising from the vortex operators explicitly. We write $\theta(r) = \theta_{\text{vortex}}(r) + \tilde{\theta}(r)$, with $\theta_{\text{vortex}}(r)$ an arbitrary configuration with vortices at the desired locations $\{r\}$. It is fixed, so it is not to be integrated over in the functional integral. The remaining part, $\tilde{\theta}(r)$ is then non-singular, and it is the degree of freedom to be integrated over. Thus,

$$\begin{aligned} \tilde{Z} &= \int' \mathcal{D}\tilde{\theta}\ \mathcal{D}s_\mu\ e^{-\frac{K}{2}\int s^2 + i\int s_\mu(\nabla_\mu\theta_{\text{vortex}} + \nabla_\mu\tilde{\theta})} \\ &= \int' \mathcal{D}\tilde{\theta}\ e^{-i\int\tilde{\theta}(\nabla\cdot s)} \int \mathcal{D}s_\mu\ e^{-\frac{K}{2}\int s^2 + i\int s_\mu(\nabla_\mu\theta_{\text{vortex}})} \end{aligned}$$

where $\tilde{\theta}$ was integrated by parts, as before, and we get the same constraint,

$$\nabla\cdot s = 0 \iff s_\mu = \epsilon_{\mu\nu}\nabla_\nu h$$

Plugging above, we get,

$$\begin{aligned} \tilde{Z} &= \int \mathcal{D}s_\mu\ e^{-\frac{K}{2}\int s^2 + i\int \epsilon_{\mu\nu}(\nabla_\nu h)(\nabla_\mu\theta_{\text{vortex}})} \\ &= \int \mathcal{D}s_\mu\ e^{-\frac{K}{2}\int s^2 + i\int\ h[\epsilon_{\mu\nu}(\nabla_\nu(\nabla_\mu\theta_{\text{vortex}}))]} \end{aligned}$$

We note now that the term in the exponent above,

$$\epsilon_{\mu\nu}\nabla_\nu(\nabla_\mu\theta_{\text{vortex}}) \neq 0$$

since θ_{vortex} is a singular function at the locations of the vortices and therefore the derivatives do not commute.

Exercise 5. *Use Stoke's theorem to show that*

$$\epsilon_{\mu\nu}\nabla_\nu(\nabla_\mu\theta_{vortex}) = 2\pi\sum_a \sigma_a\delta(r - r_a)$$

where a runs over the vortices and σ_a is the sign of the vorticity. To see this, note that close to the vortex, we have $\nabla_\mu\theta \sim \frac{\epsilon_{\mu\nu}r_\nu}{r^2}$, so that integrating around the loop gives the factor of 2π.

Substituting the above from the above exercise in the partition function $\tilde{Z}$, we have

$$\begin{aligned}\tilde{Z} &= \int \mathcal{D}s_\mu\ e^{-\frac{K}{2}\int(\nabla h)^2 + 2\pi i\sum_a \sigma_a h(r_a)} \\ &= \int \mathcal{D}s_\mu\ e^{-\frac{K}{2}\int(\nabla h)^2}\prod_a e^{2\pi i\sigma_a h(r_a)}\end{aligned}$$

which shows that the mapping that confirms the duality is indeed,

$$V_{\sigma_a}(r_a) \to e^{2\pi i\sigma_a h(r_a)}$$

By trivially extending this argument, the more general mapping for a vortex operator of strength q is,

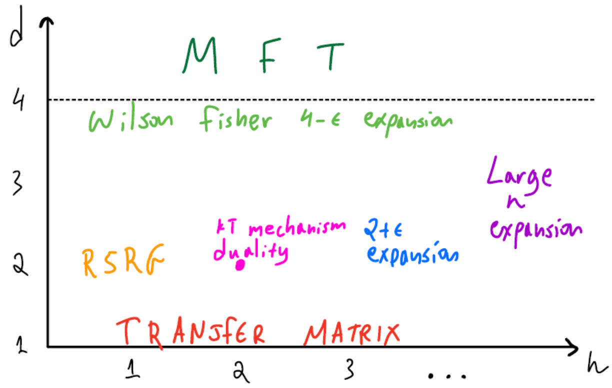
$$V_q(r) \to e^{2\pi i q h(r)}$$

and the scaling dimension of the operator on the LHS has been computed before.

The argument we have just developed shoes the equivalence between the XY and the Sine-Gordon model, which gives the RG equations that we have stated in the start of this discussion for the XY model. QLRO is now the phase where vortex fugacity flows to zero under coarse graining, whereas disorder is the phase where vortices drive the field towards disorder by allowing the exponential decay of correlations. As we stated in the start, the XY model is unique in that the phase transitions we have looked so far were all related to spontaneus symmetry breaking. There are more models which show phase transitions without SSB like for example gauge theories – the simplest being the $\mathbb{Z}_2$ or Ising gauge theory on the 3D cubic lattice.

This concludes our puzzle for the different methods and models with various RG methods. The puzzle is shown schematically below,

Fin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M F T
4
Wilson fisher 4-ϵ expansion
3
Large n expansion
KT mechanism
duality
2+ϵ expansion
2
RSRG
TRANSFER MATRIX
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1
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